

编号： SQ2023RA2C200024

科技创新领军人才 推 荐 表

推 荐 人 选： 张帆

前 沿 领 域： 集成电路

技 术 领 域： 信息领域

申 报 单 位： 浙江大学

推 荐 单 位： 浙江省科学技术厅

芯 片 人 才： 是☒ 否☐

填 报 日 期： 2023 年 02 月 20 日

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中华人民共和国科学技术部

二〇二三年制

填写说明

一、填写内容应实事求是、内容翔实、文字精炼。

二、“技术领域”请从“农业领域”、“能源领域”、“信息领域”、“资源领域”、“环境领域”、“人口与健康领域”、“材料领域”、“先进制造领域”、“生物领域”、“海洋领域”、“交通领域”、“公共安全领域”、“其他领域”中选择填写（“其他领域”指前十二个领域不能覆盖的领域）。

三、“基本信息”中，“学习经历”从大学填起。

四、“海归人才”是指在国外（及香港、澳门地区）正规大学、学院、科研机构求学、攻读学位、进修业务或从事科学研究及进行学术交流时限不短于一年后回国定居工作的人才。

五、申报单位银行账号（零余额账号除外）及财务联系人信息，请如实填写，入选后将通过此账号拨付经费。

六、“相关材料”按提纲提供齐全，不得缺项漏项。

七、“申报单位承诺和意见”中要对公示时间、范围和结果情况进行说明。

八、表中栏目没有内容的一律填“无”。

九、涉密内容不得在推荐材料中体现。

十、在线打印《科技创新领军人才推荐表》，签字盖章后报科技部。



一、基本信息

推 荐 人 选	姓 名	张 帆	性 别	男	国 籍	中国	
	出生日期	1978-10-20	政治面貌	中共党员	民 族	汉族	
	最高学历	研究生		最高学位		博士	
	是否高等学校、科研院所等法人单位的主要负责人	否	是否为海归人才	是	最近一次回国工作起始年月	2014-02-01	
	专业技术职务	教授	证件类型	身份证	证件号码	330125197810201818	
	所从事专业或方向	网络空间安全	所属主要学科 1	计算机科学技术—计算机科学技术其他学科	所属主要学科 2		
	所属战略性新兴产业领域	其他—其他			其他产业领域	网 络 空 间 安 全	
	研究工作所服务的行业	信息传输、软件和信息技术服务业—软件和信息技术服务业—集成电路设计					
	主要研发类别	☐ 基础研究 ☒ 应用基础研究 ☒ 技术开发 ☒ 成果产业化 ☐ 其他__					
	电话/传真	0571-87951111/0571-88981358	手 机	13777356409	电子邮箱	fanzhang@zj u. edu. cn	
	通讯地址	杭州市西湖区浙大路 38 号曹光彪主楼 526-3			邮 编	310027	
	学习经历	国家	院校	专业	学历/学位	起始时间	结束时间
		美国	康涅狄格大学（University of Connecticut）	计算机科学与工程	研究生/博士	2005 年 07 月	2011 年 12 月
		中国	上海交通大学	计算机应用技术	研究生/硕士	2001 年 09 月	2004 年 03 月
中国		西安交通大学	计算机科学与技术	大本/学士	1997 年 09 月	2001 年 07 月	
工作经历	国家	单位		职务	起始时间	结束时间	
	中国	浙江大学计算机科学与技术学院/网络空间安全研究中心		教授	2021 年 01 月	至今	

		中国	浙江大学计算机科学与技术学院/网络空间安全研究中心		副教授	2019年07月	2020年12月
		新加坡	新加坡科技设计大学 (Singapore University of Technology and Design)		访问教授	2018年06月	至今
		中国	浙江大学信息与电子工程学院		副教授	2017年01月	2019年06月
		新加坡	新加坡国立大学 (National University of Singapore)		访问学者	2016年12月	2018年01月
		中国	浙江大学信息与电子工程学院		讲师	2014年02月	2016年12月
申报单位	单位名称	浙江大学		统一社会信用代码 (或组织机构代码)		12100000470095016Q	
	单位类别	大专院校		主管部门		教育部	
	法定代表人	杜江峰		所在地区		浙江省杭州市西湖区	
	单位地址	浙江省杭州市西湖区余杭塘路866号		邮 编		310058	
	单位联系人	覃莉茜	手 机	13575700696	电 话	0571-88981080	
			电子邮箱	qinlixizju.edu.cn	传 真	0571-88981080	
	开户名称	浙江大学	开户行	工行杭州市浙大支行	银行账号	1202024609908808891	
	银行机构代码(12位)	102331002462	单位财务联系人	胡素英	电 话	座机	0571-88981163
						手机	13606618899
传 真	0571-88981163		电子邮箱		syhu@zju.edu.cn		

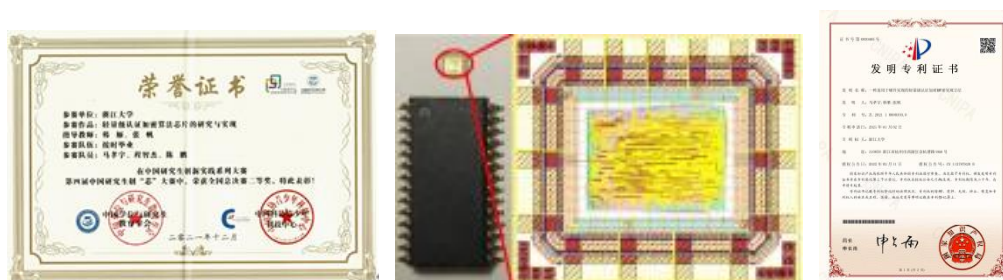


具体如下:

① 首次提出密码芯片的持久性故障分析方法并随后提出其增强型改进和代数优化方法，获 ChinaCrypt2018、CryptoTE2019 和 SAFE2021 最佳论文奖；



② 完成国际标准算法 ASCON 安全芯片的研究与设计，并实现中芯国际 180 纳米工艺流片，获华为杯创“芯”大赛全国二等奖。



③ 编著《密码故障分析与防护》一书，在华为海思 Kirin 解决方案平台软件开发部形成了故障分析检测工具套件，赋能华为针对企业级安全芯片的密码故障分析能力，直接应用于海思麒麟系列芯片的产品测试。



3. 产出效益

推荐人前期针对密码芯片的硬件安全工作，转化成为一套旁路信息硬件采集和分析平台，得到包括郑建华院士在内的 4 位专家推荐和包括中电科 30 研究所 在内 3 家企事业单位的应用证明，于 2020 年 12 月获得军队科学技术进步二等奖。



推荐人针对 AI 芯片的硬件安全工作，已形成专利“一种基于模版构建的网络流信道化时序筛分方法和装置”，融入到国家重大科研仪器研制项目“面向威胁发现的网络流谱分析仪研制”中并获得国家基金委立项。



4. 团队建设

推荐人所在的硬件安全与密码学团队成员 8 人(均为正高),其中教授 2 人、百人计划研究员 6 人。其中国家青千 2 人、海外优青 1 人、浙江省千人 1 人、杭州市 521 人才 2 人。团队成员拥有扎实的基础理论功底和丰富的科研实践经验,他们结合国家重大战略需求,开展原创性研究以及应用集成开发研究,已逐渐成为位于世界科技前沿,代表国内顶尖水平的硬件安全方向实力最强团队之一。

**密码与硬件安全
教师队伍**

主力成员平均年龄35岁左右深入安全学术前沿



- 张帆,教授,博导
- 浙江省千人
- 美国康涅狄格大学博士
- 硬件安全,密码学,系统安全



- 许海海,百人计划研究员,博导
- 曾任教于美国亚利桑那州立大学
- 网络安全,隐私保护,数据分析



- 沈浩宇,百人计划研究员,博导
- 海外优青
- 曾任职于内华达州立大学
- 美国宾夕法尼亚州立大学博士
- 硬件安全,物联网安全



- 杨坤,百人计划研究员,博导
- 曾任职于美国英特尔总部
- 美国佛罗里达大学博士
- 硬件安全,芯片安全架构,平台安全



- 王小航,教授,博导
- 浙江大学博士
- 曾任职于华南理工大学
- 硬件安全,众核芯片



- 张懿,百人计划研究员
- 曾任职于美国马里兰州大学
- 密码学,应用密码学



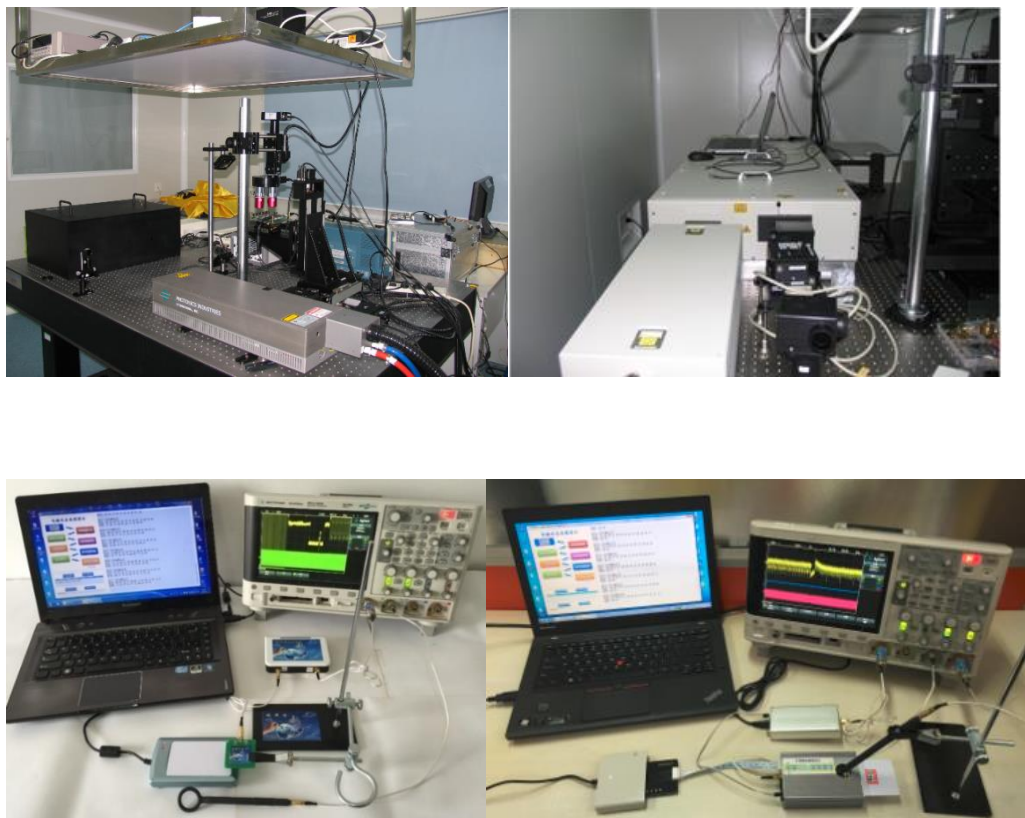
- 张秉墨,百人计划研究员,博导
- 青千
- 曾任教于美国兰卡斯特大学
- 密码学,区块链技术,移动安全



- 秦湛,百人计划研究员,博导
- 青千
- 曾任教于美国德克萨斯大学圣安东尼奥分校
- 差分隐私,云计算安全,人工智能安全

5. 科研条件

浙江大学网安中心硬件安全实验室面积约 300 平方米，包含 50 平米集成电路安全评测实验室，拥有多套密码芯片故障分析和旁路分析的商用测评平台，主要包括：利用智能卡/随机函数发生器/PC 机搭建的故障注入平台、自主开发的典型密码算法故障分析软件平台、自主开发的 ARM 系列微架构泄露分析检测平台、利用智能卡/FPGA/ASIC/示波器/PC 机搭建的功耗/电磁辐射旁路检测平台等。



2. 研究的主要内容及创新点（500 字以内）;

硬件安全技术是当前国内外芯片企业高度关注的信息安全保障手段之一，重点是基于各类密码算法，利用硬件资源以及硬件特性，从产品端保障用户信息安全。难点在于，既要在产品有限的硬件资源中进行密码算法的部署，对密码算法实现高效的硬件加速，又要支持多种不同类别的密码算法，包括传统算法和后量子密码算法。此外，后量子硬件必须要提前考虑旁路和故障分析等硬件安全威胁。推荐人拟围绕“后量子密码安全芯片的设计与评测”展开研究。

主要研究内容包括：

- ③ 后量子密码算法资源性能极致优化研究;
- ④ 后量子密码算法指令集扩展研究;
- ⑤ 后量子密码旁路/故障测评和防护技术研究。

主要创新点包括：

- ① 设计一套有限资源的硬件架构以支持不同的密码算法，重点研究资源性能极致优化的密码架构技术，探索如何以可复用的底层算子增加运算模块的灵活性以及重配置性，同时增加硬件资源能效比；
- ② 针对支持后量子密码算法的硬件模块，设计与企业 and 国家需求所适配的 CPU 扩展指令集，以支持更加高效的软硬协同后量子密码算法实现方案；
- ③ 将对设计的支持后量子密码的硬件安全模块进行完整的后量子密码旁路/故障测评技术评估并且设置对应的防护技术，以确保该模块在部署到产品中时，可以真正保障信息安全，从而延长产品的安全生命周期。

3. 科研组织形式、国内外合作设想 (200 字以内):

推荐人将围绕浙大网安中心密码硬件安全核心团队，按芯片硬件安全首席科学家负责制组织科研创新团队。团队将发表高水平的国际会议学术论文，组织“后量子密码工程和硬件安全”系列国内和国际会议，通过学术交流扩大影响力。推荐人将推动团队积极参与 ISO 等国际标准组织的行业标准制定，抢夺行业的国际话语权。推荐人还将举办公开的硬件安全夺旗对抗赛，吸引国内外知名企业和团队共同参与硬件安全评测，构建更加深厚的竞争壁垒。

4. 个人能力提升、人才培养和团队建设 (200 字以内):

在完善专业知识体系的基础上，加强后量子密码实现关键技术学习，强化企业级产品的研发能力，加速构建密码、芯片和安全相关的技术体系。落实立德树人任务、培养工程复合人才，鼓励团队创新实践，攻克卡脖子难题，勇攀科学技术前沿。依托华为-浙大联合实验室、区块链与数据安全国家重点实验室，组建硬件安全创新研究团队，核心教师编制约 10 人；依托工程师学院组建专项导师组约 40 人。打造一支老中青、校内外结合的一流协同创新研究团队。

5. 支撑保障条件需求 (200 字以内)。

要实现上述研究计划，需要四方面的支撑和保障条件，一是经费支撑保障。据初步测算，推荐人选所在科研团队年均需要科研经费支持 300 万元；二是科研团队支撑。要在稳定现有科研团队的基础上，根据创新研究需求进一步优化和适度扩充科研队伍，需要依托单位给予适当的倾斜；三是信息主管部门的支持，芯片硬件安全研究需要跨部门、跨学科的协作和典型试点应用的支持；四是开放交流平台条件支撑，包括国内国际广泛的交流和信息共享。

四、推荐人选自我评价

主要包括：主要研究方向的科研成果水平，科技创新能力和组织管理效果，职业发展目标和动力等方面的自我评价（500 字以内）

围绕芯片硬件安全方向，发表论文 58 篇（含网安 A 类论文 31 篇），5 次获最佳论文奖，在硬件安全顶级国际会议 CHES 上发表论文 7 篇，是以第一或通讯作者发表 CHES 论文最多的华人；主持国家级项目 5 项、省部级项目 5 项，总经费 2122.83 万元；授权发明专利 14 项。

研制一套自动化的旁路信息硬件采集和分析平台，突破异构密码芯片接入、功耗信息动态预处理等关键技术，在中电科 30 所、信工大、上海交大得到应用，获 2020 年军队科学技术进步二等奖，应用于 USIM 卡自动功耗采集，获亚洲硬件安全年会 AsianHOST2019 最佳海报论文奖；提出基于多推断的密码芯片代数旁路容错分析技术，获 COSADE2012 最佳论文奖；首次提出密码芯片的持久性故障分析方法并随后提出其增强型改进和代数优化方法，获 ChinaCrypt2018、CryptoTE2019 和 SAFE2021 最佳论文奖；完成轻量级认证加密算法 ASCON 安全芯片的研究与设计，并实现中芯国际 180 纳米工艺流片，参加第四届中国研究生创“芯”大赛并获全国总决赛二等奖，得到包含华为、新思科技、日月光等集成电路行业龙头企业的高度肯定，授权专利 1 项。



五、近五年相关情况列表

1.承担任务情况（不超过5项，主持或作为主要完成人参与的任务）

承担主要科研任务情况						
序号	项目（课题）名称	任务来源*	科研任务编号	经费（万元）	起止年月	担任角色
1	XYY-124002-E52 202	国家科技重大专项,军委科技委-创新特区项目	KM20220658	360.00	2022-12-15 至 2023-12-15	主持
2	XYY-110202-E62 202	国家科技重大专项,军委科技委-173 基础加强计划	KM20220484	1935.0 0	2022-08-01 至 2026-12-01	课题负责人
3	人工智能安全理论及验证平台-AI 系统硬件架构安全适配与优化	国家科技重大专项,科技部-科技创新 2030	2020AAA010 7700	5823.0 0	2020-11-01 至 2023-10-31	参与
4	基于持久性故障的密码芯片和安全系统的分析机理和方法研究	国家自然科学基金,面上项目	62072398	69.60	2021-01-01 至 2024-12-31	主持
5	后量子密码硬件安全测评算法技术合作项目等	其他,校企联合横向项目-华为技术有限公司	K 横 20210034	468.76	2019-12-01 至 2024-12-31	主持

* 任务来源包括任务委托方及所属计划等的名称

2. 成果产出和贡献

代表性论著（“第一作者”或“通讯作者”的论文、专著）（不超过 5 项）

序号	论著题目	所有作者（通讯作者 请标注*）	期刊（或出版 社）名称	出版日期	卷期及页码
1	Persistent Fault Analysis on Block Ciphers	Fan ZHANG*, Xiaoxuan LOU, Xinjie ZHAO, Shivam BHASIN, Wei HE, Ruyi DING, Samiya QURESHI, Kui REN	IACR Transactions on Cryptographi c Hardware and Embedded Systems	2018	2018. 2018 (3) , 150-172
2	Free Fault Leakages for Deep Exploitation: Algebraic Persistent Fault Analysis on Lightweight Block Ciphers	Fan ZHANG*, Tianxiang FENG, Zhiqi LI, Kui REN, Xinjie ZHAO	IACR Transactions on Cryptographi c Hardware and Embedded Systems	2022	2022. 2022 (2) , 289-311
3	Side-Channel Analysis and Countermeasure Design on ARM-Based Quantum-Resist ant SIKE	Fan ZHANG, Bolin YANG, Xiaofei DONG, Sylvain GUILLEY, Zhe LIU*, Wei HE, Fangguo ZHANG, Kui REN	IEEE Transactions on Computers	2020	2020. 69 (11) , 1681-1693
4	DARPT: Defense Against Remote Physical Attack Based on TDC in Multi-tenant Scenario	Fan ZHANG, Zhiyong WANG, Haoting SHEN*, Bolin YANG, Qianmei WU, Kui REN	Proceedings - Design Automation Conference	2022	2022. 559-56 4



5	Pushing the Limit of PFA: Enhanced Persistent Fault Analysis on Block Ciphers	Guorui XU, Fan ZHANG*, Bolin YANG, Xinjie ZHAO, Wei HE, Kui REN	IEEE Transactions on Computer-Aided Design of Integrated Circuit and Systems	2021	2021. 40 (6), 1102-1116	
发明专利授权情况（不超过 5 项）						
序号	名称	授权号	类别	排序	授权时间	授权国别或组织
1	基于 SIKE 算法的旁路防御方法装置电子设备和可读介质	ZL 2020 1 0553818.1	产品发明	1	2021/11/23	中国专利
2	基于 TDC 模块的硬件攻击的防御方法及装置、电子设备	ZL 2022 1 0035523.4	产品发明	1	2022/6/24	中国专利
3	一种基于稳态故障的芯片安全性的评估方法	ZL 2019 1 0789976.4	方法发明	1	2021/3/19	中国专利
4	基于线性码掩码和位切片技术的抗旁路攻击的防御方法	ZL 2022 1 0047078.3	方法发明	1	2022/6/24	中国专利
5	一种适用于硬件实现的轻量级认证加密解密实现方法	ZL 2021 1 0000330.0	方法发明	3	2022/2/11	中国专利
软件著作权或动植物新品种（不超过 5 项）						
序号	名称		登记号		开发完成时间	
/	/		/		/	
标准制定情况（不超过 5 项）						



序号	标准号	标准名称	类别	颁布/修订时间	本人排序
1	待定（依密标委发[2021]14号）制定	汽车智能处理器硬件可信根和安全启动密码应用技术研究	国家	2023年08月01日	待定
2	待定（依密标委发[2021]14号）制定	车载领域 SoC 密码安全子系统保护轮廓与测评要求研究	国家	2023年08月01日	待定
主要新产品（含新品种）/新装置（装备）/新工艺/新材料开发/创新服务情况*（不超过5项）					
序号	名称	创新性	开发阶段	功能、应用领域（限50字）	经济及社会效益（限50字）
1	国家重大科研仪器研制项目-面向威胁发现的网络流谱分析仪研制	首创	原型	应用于网络安全领域，研制网络流谱分析仪，实现网络空间中网络行为的即测、即用、即分析	为网络威胁分析研究提供小型化、智能化、通用化的基础性和创新性专用科研仪器
2	NIST 认证加密算法 ASCON 芯片-SMIC 180nm 工艺	重大改进	原型	具有抗旁路分析安全属性的低功耗、高吞吐实现，兼具认证和加密功能。可广泛应用于医疗、物联网等领域。	符合美国 NIST 标准，可成为未来相关设备的基础安全核心芯片。

*此栏主要反映本人研发成果及主要作用的相关开发产品、装备、工艺、材料和服务等

3. 其他



国内外科研组织及重要学术期刊任职情况			
序号	组织或期刊名称	职务	任期
1	国外科研组织，国际会议 PROOFS2020-2021	程序委员会主席	2020-9, 2022-8
2	国外科研组织, 国际 IEEE 电路系统学会 VLSI 系统应用技术委员会委员	委员	2018-1, 至今
3	国外科研组织, 网安 A 类国际会议 CHES2022-2023	程序委员会委员	2021-5, 至今
4	国外科研组织, 网安 A 类国际会议 DAC2020-2022	程序委员会委员	2019-10, 2022-9
5	国外学术期刊, Frontiers of Information Technology & Electronic Engineering	首批通讯专家	2019-7, 至今
6	国外学术期刊, CyberSecurity	青年编委委员	2019-8, 至今
7	国外学术期刊, IEEE Access	副编辑	2019-4, 2022-2
8	国内科研组织，中国密码学会 (CACR) 密码芯片专业委员会	委员	2019-3, 至今
9	国内科研组织，中国密码学会 (CACR) 密码测评专业委员会	委员	2020-12, 至今
10	国内科研组织，中国计算机学会 (CCF) 容错计算专业委员会	委员	2020-8, 至今
11	国内科研组织，中国计算机学会 (CCF) 集成电路专业委员会	委员	2020-8, 至今
12	国内科研组织，中国计算机学会 CCF YOCSEF 杭州	委员	2018-6, 2019-5

13	国内科研组织,中国中文信息学会大数据安全与隐私计算专委会	委员	2022-1,至今			
学术荣誉称号（不超过 5 项）						
序号	称号	授予机构	获得时间			
1	省部级奖励 军队科技进步奖二等奖	中央军委	2020-12-25			
2	其他 2021 年华为杯第四届中国研究生创“芯”大赛全国二等奖	中国学位与研究生教育学会	2021-12-27			
3	其他 2021 年密码测评学术会议（CryptoTE2021）最佳论文奖	中国密码学会 密码测评专业委员会	2021-07-23			
4	其他 2019 年亚洲硬件安全年会（AsianHOST2019）最佳海报论文奖	亚洲硬件安全年会技术委员会	2019-10-22			
5	其 他 2018 年 中 国 密 码 年 会（ChinaCrypt2018）最佳论文奖	中国密码学会	2018-10-28			
在重要国际学术会议报告情况（不超过 5 项）						
序号	报告名称	会议名称	主办方	时间	地点	报告类别
1	Persistent Fault Analysis on Block Ciphers	CHES2018	IACR Transactions on Cryptographic Hardware and	2018-9	荷兰 阿姆斯特丹（线下举行）	大会报告



			Embedd ed System s			
2	Persistent Fault Attack in Practice	CHES2020	IACR Transa ctions on Crypto graphi c Hardwa re and Embedd ed System s	2020-9	中国 北 京（线上 举行）	大会报告
3	Free Fault Leakages for Deep Exploitatio n: Algebraic Persistent Fault Analysis on Lightweight Block Ciphers	CHES2022	IACR Transa ctions on Crypto graphi c Hardwa re and Embedd ed System s	2022-9	比 利 时 鲁汶（线 上 线 下 混合）	大会报告



4	From Homogeneous to Heterogeneous: Leveraging Deep Learning based Power Analysis across Devices	DAC2020	Design Automation Conference	2020-7	美国 旧金山（线上举行）	大会报告
5	DARPT: Defense Against Remote Physical Attack Based on TDC in Multi-tenant Scenario	DAC2022	Design Automation Conference	2022-7	美国 旧金山（线上线下混合）	大会报告



六、申报单位发展需求与推荐人选的相关性及申报单位提供的支持保障措施

1. 申报单位在推荐人选所属学科和科研领域的布局及发展状况（200 字以内）

浙大计算机学院拥有 4 个国家一级学科和 2 个全国重点实验室。近三年获 1 项国家科技奖，8 项省部级科技奖，年均到款经费超 7.5 亿元。浙大网安中心现有固定研究人员近 50 名，含国青千等高层次人才 11 人，承担国家/省部级重大项目 30 余项，获最佳论文奖等重要奖项 21 项。浙大在 CSRankings 安全方向排名全国第一、亚洲第二、全球并列第六。网络空间安全学科布局强调软硬结合，芯片硬件安全是其重要方向。

2. 推荐人选对申报单位科研发展的作用（学科带动、科研水平提升、队伍建设等）（200 字以内）

推荐人是芯片硬件安全方向的带头人，在网络空间安全学科发展中起了重要的作用，主要包括：

（1）学科建设发展：构建了硬件安全本科生基础课程和研究生进阶课程；

（2）科研水平提升：承担了 2122.83 万的科研项目，获 2 项省部级科技奖，使得浙大芯片硬件安全的研究水平上升到一个新的层次和水平；

（3）师资队伍建设：促进建立了一支高水平的密码学和芯片硬件安全学科研究队伍，共计 8 位海内外高层次人才。



3. 申报单位对推荐人选的培养使用所提供的保障措施及落实计划（包括岗位设置、人才培养、科研场所、实验平台、招生计划、资源共享、经费投入、项目倾斜、后勤保障等）（300 字以内）

推荐人张帆教授是浙江省千人计划入选者，具有很强的科研能力和组织协调能力，发展潜力明显，承担了多项国家和省部级重点科研项目。张帆教授在芯片硬件安全方面的研究成果处于国际顶尖水平，符合该人才计划申报条件。如果他的申请获得批准，学校和推荐人所在学院将在以候选人为核心的科研创新团队建设、科研条件保障、科研发展空间等方面为他创造更好的条件。



七、承诺与推荐意见

1. 推荐人选承诺

本人在此郑重承诺：

（一）申报材料符合《中华人民共和国保守国家秘密法》和《科学技术保密规定》等相关法律法规；申报材料所有内容真实有效，不存在任何违背《关于进一步加强科研诚信建设的若干意见》规定的行为。

（二）本人未发生过以下行为：

1. 抄袭、剽窃他人科研成果或者伪造、篡改研究数据、研究结论；
2. 购买、代写、代投论文，虚构同行评议专家及评议意见；
3. 违反论文署名规范，擅自标注或虚假标注获得科技计划等资助；
4. 弄虚作假，骗取科技计划项目、科研经费以及奖励、荣誉等；
5. 违反科学道德，违背职业操守、师德师风规范等；
6. 违反党纪、触犯国家法律法规。

（三）在参与申报、评审和实施全过程中，遵守评审规则和工作纪律，杜绝以下行为：

1. 以任何形式探听尚未公布的评审专家名单及其他评审过程中的保密信息；
2. 本人或委托他人通过各种方式及各种途径联系有关专家进行请托、游说，违规到评审会议驻地游说评审专家和工作人员、询问评审或尚未正式向社会公布的信息等干扰评审或可能影响评审公正性的活动；
3. 向评审工作人员、评审专家等提供任何形式的礼品、礼金、有价证券、支付凭证、商业预付卡、电子红包，或提供宴请、旅游、娱乐健身等任何可能影响评审公正性的活动；



4. 其它违反财经纪律和相关管理规定的行为。

如有违反，本人愿接受项目管理机构和相关部门做出的各项处理决定，包括但不限于取消人才称号，追回支持经费，向社会通报违规情况，取消一定期限人才计划申报资格，记入科研诚信严重失信行为数据库以及接受相应的党纪政纪处理等。

(签字):

年 月 日

2. 申报单位承诺和意见

本单位严格履行法人负责制，已就所申报材料内容的真实性和完整性进行审核，不存在违背《关于进一步加强科研诚信建设的若干意见》规定和其它科研诚信要求的行为，申报材料符合国家保密有关规定，在参与项目申报和审评活动全过程中，遵守有关评审规则和工作纪律，杜绝违法违规行为。

本单位承诺本单位与推荐人选存在实质劳动关系，签订了劳动合同，并支付薪酬或缴纳社会保险。推荐人选已完成内部公示无异议，同意推荐。

本单位严格履行法人负责制，已就所申报材料内容的真实性和完整性进行审核，不存在违背《关于进一步加强科研诚信建设的若干意见》规定和其它科研诚信要求的行为，申报材料符合国家保密有关规定，在参与项目申报和审评活动全过程中，遵守有关评审规则和工作纪律，杜绝违法违规行为。

本单位承诺本单位与推荐人选存在实质劳动关系，签订了劳动合同，并支付薪酬或缴纳社会保险。推荐人选已完成内部公示无异议，同意推荐。

单位法定代表人（签章）:

（公章）

年 月 日



3.推荐单位意见

本单位已就所申报材料内容的真实性和完整性进行审核，不存在违背《关于进一步加强科研诚信建设的若干意见》规定和其它科研诚信要求的行为，申报材料符合国家保密有关规定，在参与项目申报和审评活动全过程中，遵守有关评审规则和工作纪律，杜绝违法违规行为。

推荐人选已通过专家初选，已征求纪检监察部门意见，并经所在地省（自治区、直辖市及计划单列市）党委组织部门或主管部门（单位）审核把关，同意推荐。

本单位已就所申报材料内容的真实性和完整性进行审核，不存在违背《关于进一步加强科研诚信建设的若干意见》规定和其它科研诚信要求的行为，申报材料符合国家保密有关规定，在参与项目申报和审评活动全过程中，遵守有关评审规则和工作纪律，杜绝违法违规行为。

推荐人选已通过专家初选，已征求纪检监察部门意见，并经所在地省（自治区、直辖市及计划单列市）党委组织部门或主管部门（单位）审核把关，同意推荐。

单位（公章）

年 月 日



八、相关材料

- 1. 最高学历毕业证书、学位证书（必须上传）；

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（最高学历毕业证书 、学位证书）

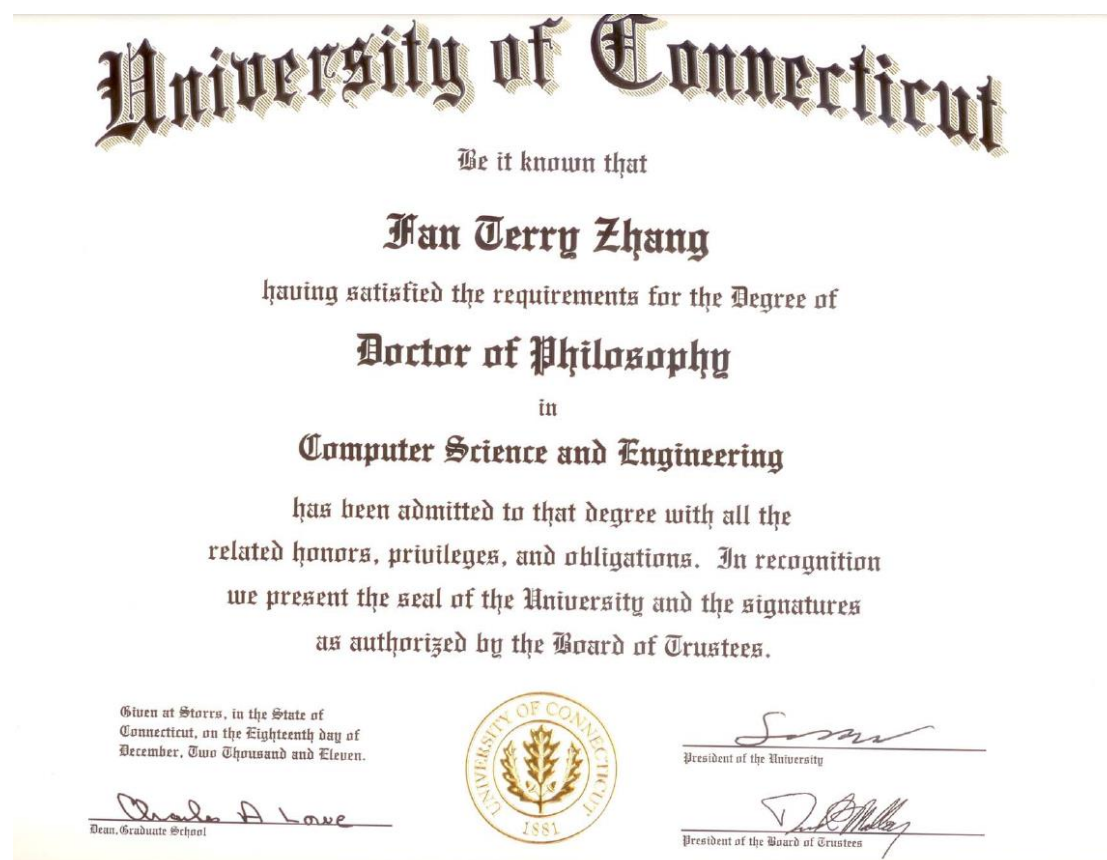
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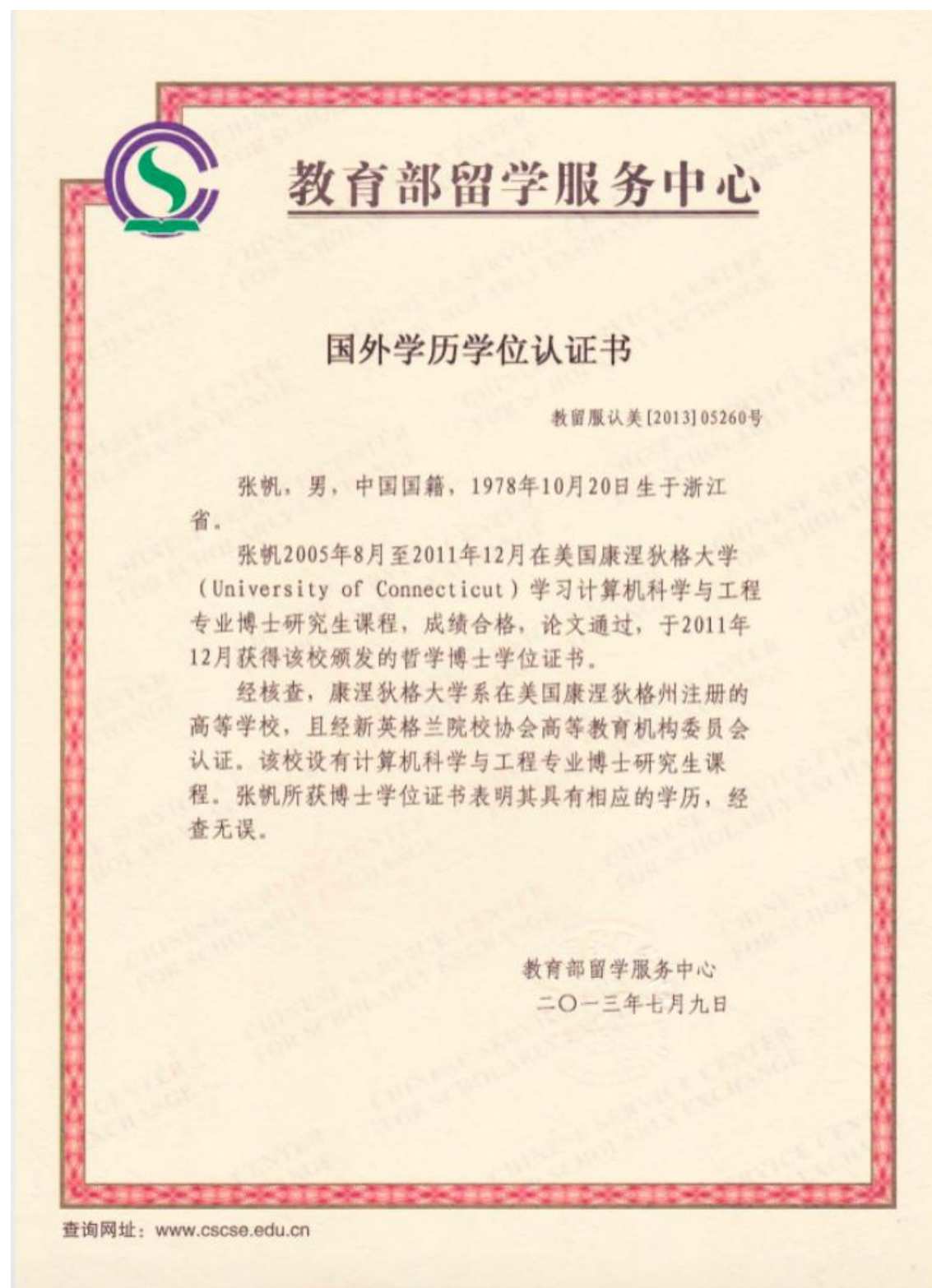
1. 美国康涅狄格大学博士毕业证书



2. 美国康涅狄格大学博士学位证书



3. 国外学历学位认证书



2. 承担任务（注：根据填写的“承担任务情况”依次上传证明材料）；

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（承担主要科研任务情况）

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【系统中已录入的主要科研任务】-共 5 项

1. 省部级：军工 XYY-124002-E52202

关于某部委创新特区项目的证明

兹证明，我校计算机科学与技术学院 张帆教授 主持某
部委创新特区项目 1 项，相关信息如下。

项目名称：XXX 对抗方法验证研究

项目总经费：360 万

项目周期：2022 年 12 月 至 2023 年 12 月

特此证明。

浙江大学先进技术研究院

2023 年 2 月 23 日



2. 省部级：军工 XYY-110202-E62202

关于某部委基础研究重点项目的证明

兹证明，我校计算机科学与技术学院 张帆教授 参研某部委基础研究重点项目 1 项，相关信息如下。

项目名称：XXX 机理及 XX 研究

项目总经费：1935 万

个人承担经费：200 万

项目周期：2022 年 08 月 至 2026 年 08 月

特此证明。

浙江大学先进技术研究院
2023 年 2 月 23 日



3. 国家级：科技部科技创新 2030-人工智能安全理论及验证平台

项目编号：2020AAA0107700

密 级：公开

项目任务书

项目名称： 人工智能安全理论及验证平台

所属重大项目： 科技创新2030-“新一代人工智能”重大项目

指南方向： 2.1人工智能安全理论及验证平台

推荐单位： 浙江省科学技术厅

项目管理专业机构： 科学技术部高技术研究发展中心

项目牵头承担单位： 浙江大学 (公章)

项目负责人： 任奎

执行期限： 2020年11月至2023年10月

中华人民共和国科学技术部制

2020年11月20日

0002AA 2020AAA0107700 2020 2020-11-20 10:07:40



项目基本信息表

项目名称		人工智能安全理论及验证平台			
项目编号		2020AAA0107700			
所属重大项目		科技创新2030-“新一代人工智能”重大项目			
指南方向		2.1人工智能安全理论及验证平台			
密级	公开	单位总数	9	课题数	5
项目类型		重大共性关键技术			
经费预算		总预算5823.00万元，其中中央财政资金1823.00万元，地方财政资金0.00万元，单位自筹资金4000.00万元，其他渠道获得资金 0.00万元			
项目周期节点		起始时间	2020年11月	结束时间	2023年10月
		实施周期	共 36 个月	预计中期时间点	2022年4月
项目 牵头 承担 单位	单位名称	浙江大学		单位性质	大专院校
	单位所在地	浙江省杭州市西湖区		组织机构代码	12100000470095016Q
	通信地址	浙大路38号		邮政编码	310027
	银行账号	1202024609908808891		法定代表人姓名	吴朝晖
	单位开户名称	浙江大学		汇入地点	浙江省杭州市
	开户银行(全称)	中国工商银行杭州浙大支行		银行机构代码	102331002462
推荐 单位	单位名称	浙江省科学技术厅		推荐单位性质	地方



4. 国家级：国家自然科学基金面上项目-基于持久性故障的密码芯片和安全系统的分析机理和方法研究



申请代码	F0206
接收部门	
收件日期	
接收编号	6207070691



6207070691

国家自然科学基金

申 请 书

(2020版)

资助类别： 面上项目

亚类说明：

附注说明：

项目名称： 基于持久性故障的密码芯片和安全系统的分析机理和方法研究

申 请 人： 张帆 电 话： 0571-87952685

依托单位： 浙江大学

通讯地址： 浙江省杭州市浙大路38号浙江大学玉泉校区曹光彪主楼526-3

邮政编码： 310027 单位电话： 0571-88981080

电子邮箱： fanzhang@zju.edu.cn

申报日期： 2020年01月22日

国家自然科学基金委员会



张帆 先生/女士：

根据《国家自然科学基金条例》规定和专家评审意见，国家自然科学基金委员会（以下简称自然科学基金委）决定资助您申请的项目。项目批准号：62072398，项目名称：基于持久性故障的密码芯片和安全系统的分析机理和方法研究，直接费用：58.00万元，项目起止年月：2021年01月至2024年12月，有关项目的评审意见及修改意见附后。

请尽早登录科学基金网络信息系统（<https://isisn.nsf.gov.cn>），获取《国家自然科学基金资助项目计划书》（以下简称计划书）并按要求填写。对于有修改意见的项目，请按修改意见及时调整计划书相关内容；如对修改意见有异议，须在电子版计划书报送截止日期前向相关科学处提出。

电子版计划书通过科学基金网络信息系统（<https://isisn.nsf.gov.cn>）上传，依托单位审核后提交至自然科学基金委进行审核。审核未通过者，返回修改后再行提交；审核通过者，打印纸质版计划书（一式两份，双面打印），依托单位审核并加盖单位公章，将申请书纸质签字盖章页订在其中一份计划书之后，一并将上述材料报送至自然科学基金委项目材料接收工作组。电子版和纸质版计划书内容应当保证一致。**自然科学基金委将对申请书纸质签字盖章页进行审核，对存在问题的，允许依托单位进行一次修改或补齐。**

向自然科学基金委补交申请书纸质签字盖章页，提交和报送计划书截止时间节点如下：

1. **2020年10月14日16点：**提交电子版计划书的截止时间（视为计划书正式提交时间）；
2. **2020年10月21日16点：**提交电子修改版计划书的截止时间；
3. **2020年10月28日16点：**报送纸质版计划书（其中一份包含申请书纸质签字盖章页）的截止时间。
4. **2020年11月18日16点：**报送修改后的申请书纸质签字盖章页的截止时间。





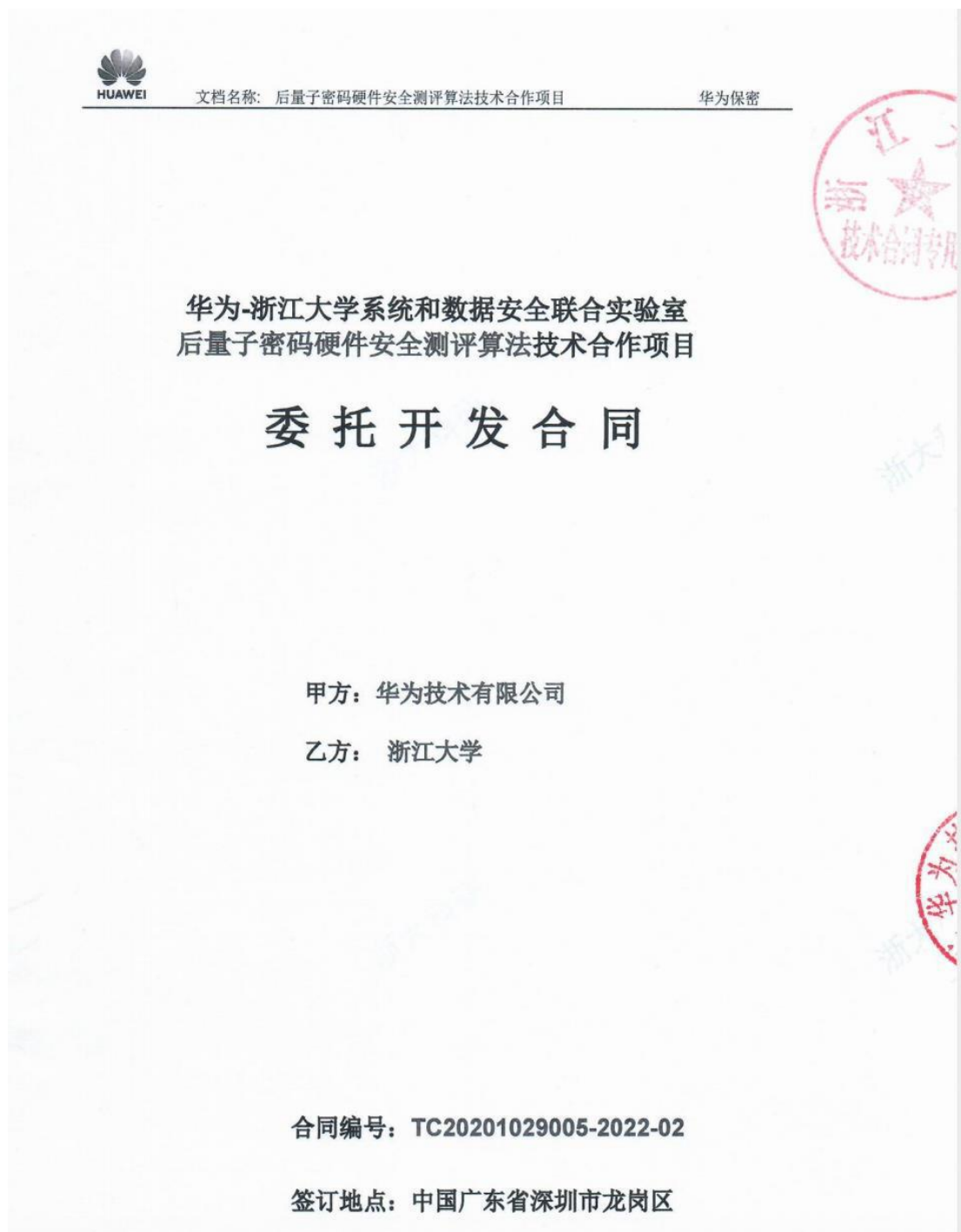
主要参与者（注：主要参与者不包括项目申请人）

编号	姓名	出生年月	性别	职 称	学 位	工作单位	项目分工	电话	证件号码
1	王小娟	1985-11-24	女	副教授	博士	北京邮电大学	项目主研人	010-62282751	1*****2
2	徐洁	1986-12-15	女	讲师	博士	北京邮电大学	项目主研人	010-61198159	3*****4
3	任传伦	1972-06-02	男	研究员	博士	中国电子科技集团公司第三十研究所	项目主研人	010-66306430	3*****2
4	俞赛赛	1982-01-08	男	高级工程师	博士	中国电子科技集团公司第三十研究所	项目主研人	010-66306430	3*****7
5	张帆	1978-10-20	男	教授	博士	浙江大学	项目主研人	0571-87952685	3*****8
6	王伟	1989-02-09	男	无	博士	北京邮电大学	项目主研人	010-61198108	1*****X
7	肖达	1981-06-01	男	讲师	博士	北京邮电大学	项目主研人	010-62283134转0	2*****7
8	满毅	1974-11-01	男	副教授	博士	北京邮电大学	项目主研人	62282751	1*****1
9	黄玮	1980-05-18	男	研究员	博士	北京理工大学	项目主研人	010-68913694	3*****6

总人数	高级	中级	初级	博士后	博士生	硕士生
39	7	2	0	1	11	18



5. 横 向：华为后量子密码硬件安全测评算法技术合作项目等





	2、功能正确,能够在华为实验室进行演示复现; 3、具备清晰的使用指导说明。	的泄露检测方法和实施路径,包含测试向量的设计、有效性分析和测评验证。 软硬件原型验收标准: 1、代码由Python实现,可对NTT(多项式乘法)、随机采样器的测评,具备一定的注释,可移植至华为自研平台; 2、功能正确,能够在华为实验室进行演示复现; 3、具备清晰的使用指导说明。	的泄露检测方法和实施路径,包含测试向量的设计、有效性分析和测评验证。 软硬件原型验收标准: 1、代码由Python实现,可对NTT(多项式乘法)、随机采样器的测评,具备一定的注释,可移植至华为自研平台; 2、功能正确,能够在华为实验室进行演示复现; 3、具备清晰的使用指导说明。
验收方法	案例复现及华为验收小组评审	案例复现及华为验收小组评审	案例复现及华为验收小组评审

5.2 工作任务书履行过程中,乙方应根据甲方的要求按具体的拨付计划由技术委员会与子项目负责人协商后确定。月提交书面的协议工作进度报告,以使甲方及时了解协议工作的真实进度。若乙方预见到协议工作可能延期,应及时书面告知甲方延期的原因及替代方案。若因乙方原因导致乙方未能按本工作任务书5.1款中约定的交付内容、形式、数量和交付时间向甲方如期交付开发成果,甲方有权要求乙方继续履行交付全部开发成果的义务,双方应友好协商解决问题。若非因甲方原因造成的延期超过三十天,甲方有权随时书面通知乙方终止本工作任务书和/或者项目合同。

六、子项目费用

6.1 针对乙方在本合同及本子项目工作任务书下履行的义务,甲方应向乙方支付的费用(包括科研劳务费、差旅费、学校管理费等)总额为人民币80万元(“合同总金额”,含税),不含税金额为人民币80万元。在乙方完全履行本合同及本子项目工作任务书下全部义务的情形下,除非另有书面约定,本子项目合同总金额是甲方应当向乙方支付的全部费用。

6.2 本子项目合同总金额由《华为-浙江大学系统和数据安全联合实验室共建协议》约定的资金池中支付。应按照下表的约定(“拨付计划”)向子项目研究团队拨付各期费用。

阶段	拨付条件	拨付金额(人民币/元,不含税)
1	第1阶段验收通过	24万
2	第2阶段验收通过	24万
3	结项阶段通过验收	32万



密码故障注入技术合作项目

	文档名称: 密码故障注入技术合作项目委托研发合同 合同编号: YBN2020095132	华为保密
密码故障注入技术合作项目 委 托 研 发 合 同		
甲方: 华为技术有限公司 乙方: 浙江大学		
合同编号: YBN2020095132 签订地点: 【中国广东省深圳市龙岗区】		





七、费用与付款计划:

本项目总额为人民币 1,250,000 元 (“合同总金额”, 不含税), 包括科研劳务费、实验设备材料费、差旅费等保证项目按合同要求完成的所有费用。由华为技术有限公司分 4 期支付给浙江大学, 浙江大学开具税率为 0 %的 增值税普通发票。

阶段	支付条件	支付金额 (人民币·元, 不含税)
1	合同签署	375000 (30%)
2	项目第二阶段交付完成, 验收通过	250000 (20%)
3	项目第三阶段交付完成, 验收通过	375000 (30%)
4	全部阶段验收通过	250000 (20%)

- 华为技术有限公司应在收到浙江大学开具的合法有效的税率为 0 %的 增值税普通发票、《技术合同项目验收通知函》(格式符合附件 3 要求) 和《供应商开票清单》(格式符合附件 4 要求) 之日起第 30 日 (即 “到期日”), 以银行转账的方式向浙江大学支付付款计划约定的各阶段支付金额。浙江大学账户信息如下表所述:

开户行	工行杭州市浙大支行
户 名	浙江大学
帐 号	1202024609908808891

八、本研究项目成果及其知识产权所有权分配:

- ☒ 华为技术有限公司 100 %
☐ 浙江大学 %

双方的背景知识产权归各自所有。乙方理解并同意, 甲方及其关联公司有权为使用部分或者全部前述开发成果的目的而免费使用乙方背景知识产权。所述背景知识产权包括但不限于附件 2 所列知识产权。

九、物料返还

不涉及



华为-浙江大学系统和数据安全联合实验室
“Cache旁边分析方法研究”子项目工作任务书

一、子项目名称:

二、子项目合同编号

三、研究子项目类型:

☐ B类项目

四、定义

4.1 不涉及

五、 工作计划及阶段性成果

项目阶段	一	二	三
起止日期	T - T+2	T+2 - T+6	T+6 - T+10
工作描述	开展Cache旁路方法综述调研	基于自研算法, 或者别人开发的算法做分析, 输出旁路demo测评研究报告, 包括具体的方法, 步骤描述, 以及改进研究方向建议	基于市场公开商用产品, 开展Cache旁路分析研究, 输出研究报告
开发成果	《Cache旁路分析方法综述》	《Cache旁路实验平台测评研究报告》, 以及报告所涉及的测评代码和原型	《Cache旁路实际产品测评研究报告》, 以及研究中涉及的测评源码, demo实例

第 1 页, 共 10 页

张君



验收标准	1、文档结构清晰,逻辑性强 2、文档含具体的方法,步骤 3、达成方法复现或者下一步改进研究方向建议 4、代码可读性强,具备一定的注释,实现具体Cache旁路的分析方法,执行步骤,功能正确
验收方法	1、文档经过技术验收小组评审 2、代码以及原型能够在华为实验室进行演示复现

备注T为实际项目开始时间; T是指合同生效日期

5.2 工作任务书履行过程中,乙方应根据甲方的要求按具体的拨付计划由技术委员会与子项目负责人协商后确定。月提交书面的协议工作进度报告,以使甲方及时了解协议工作的真实进度。若乙方预见到协议工作可能延期,应及时书面告知甲方延期的原因及替代方案。若因乙方原因导致乙方未能按本工作任务书5.1款中约定的交付内容、形式、数量和交付时间向甲方如期交付开发成果,甲方有权要求乙方继续履行交付全部开发成果的义务,双方应友好协商解决问题。若非因甲方原因造成的延期超过三十天,甲方有权随时书面通知乙方终止本工作任务书和/或者项目合同。

六、子项目费用

6.1 针对乙方在本合同及本子项目工作任务书下履行的义务,甲方应向乙方支付的费用(包括科研劳务费、差旅费、学校管理等)总额为人民币**73.76**万元(“合同总金额”,含税),不含税金额为人民币**73.76**万元。在乙方完全履行本合同及本子项目工作任务书下全部义务的情形下,除非另有书面约定,本子项目合同总金额是甲方应当向乙方支付的全部费用。

6.2 本子项目合同总金额由《华为-浙江大学系统和数据安全联合实验室共建协议》约定的资金池中支付。应按照下表的约定(“拨付计划”)向子项目研究团队拨付各期费用。

阶段	拨付条件	拨付金额(人民币/万元,含税)
1	本子项目通过实验室技术委员会评审且完成合同签署,预付款	29.504 万元
2	本子项目通过实验室技术委员会中期审视,且甲方收到乙方的科技局备案回执材料	22.128 万元
3	通本子项目过实验室技术委员结题审视	22.128 万元

中期审视是指《华为-浙江大学系统和数据安全联合实验室共建协议》中约定的年中workshop评审。
结题审视是指《华为-浙江大学系统和数据安全联合实验室共建协议》中约定的年度workshop评审。



基于 DVFS 技术的故障注入和密钥恢复研究项目

基于 DVFS 技术的故障注入和密钥恢复研究项目

委托研发合同

甲方：华为技术有限公司

乙方：浙江大学

合同编号：YBN2019085134A2

签订地点：【中国广东省深圳市龙岗区】

张雷



电话	18192592905	13777356409
电邮	maoshaoxu@huawei.com	fanzhang@zju.edu.cn

乙方应向甲方提供上述两列表所列的个人信息并保证其真实性，确保基于合法理由收集个人信息，并授权甲方使用上述个人信息用于本合同履行的目的。甲方应采用适当方式使用乙方提供的个人信息，并根据所适用的法律保护该个人信息。

七、资金投入比例：

☒ 华为技术有限公司 100 %

☐ 其他(如国家、地区政府项目资金) _____ %

八、费用与付款计划：

本项目总额为人民币 55 万 元（“合同总金额”，含税），包括科研劳务费、实验设备材料费、差旅费等保证项目按合同要求完成的所有费用。由华为技术有限公司分 3 期支付给浙江大学，浙江大学开具税率为 0 % 的 增值税普通发票。

阶段	支付条件	支付金额 (人民币·元，含税)
1	合同签订	16.5 万
2	第 2 阶段工作成果交付通过验收	22 万
3	项目结题通过验收	16.5 万

● 华为技术有限公司应在收到浙江大学开具的合法有效的税率为 0 % 的 增值税普通发票、《技术合同项目验收通知函》（格式符合附件 3 要求）和《供应商开票清单》（格式符合附件 4 要求）之日起第 30 日（即“到期日”），以银行转账的方式向浙江大学支付付款计划约定的各阶段支付金额。浙江大学账户信息如下表所述：

开户行	工行杭州市浙大支行
户 名	浙江大学
帐 号	1202024609908808891

九、本研究项目成果及其知识产权所有权分配：

本合同下产生的全部开发成果，包括但不限于乙方向甲方交付的技术方案、技术样品、及其相关的文档、软件代码、建议书或报告等（若有），以及除专利之外的其它知识产权，包括但不限于版权、商业秘密等，归甲方所有，甲方及其关联公司可以使用上述开发成果而不受其他限制。



履行所有适用出口管制法律及相关要求下的义务。

十二、 其他条款

12.1 有关本协议的任何争议应由双方秉承善意友好协商解决。若三十日内协商不成，双方同意将争议提交合同签订地有管辖权的人民法院诉讼解决。

12.2 本协议一式八份，华为公司执二份，浙江大学执六份，均具有同等法律效力。

12.3 保密期截止为合同期结束起1年。

兹证明双方已签立本研究项目表格。

甲方（盖章）：华为技术有限公司

乙方（盖章）：浙江大学

授权代表（签名）：

授权代表（签名）：

姓名：尹辉 (3)

姓名：张明

职务：授权签署人用章

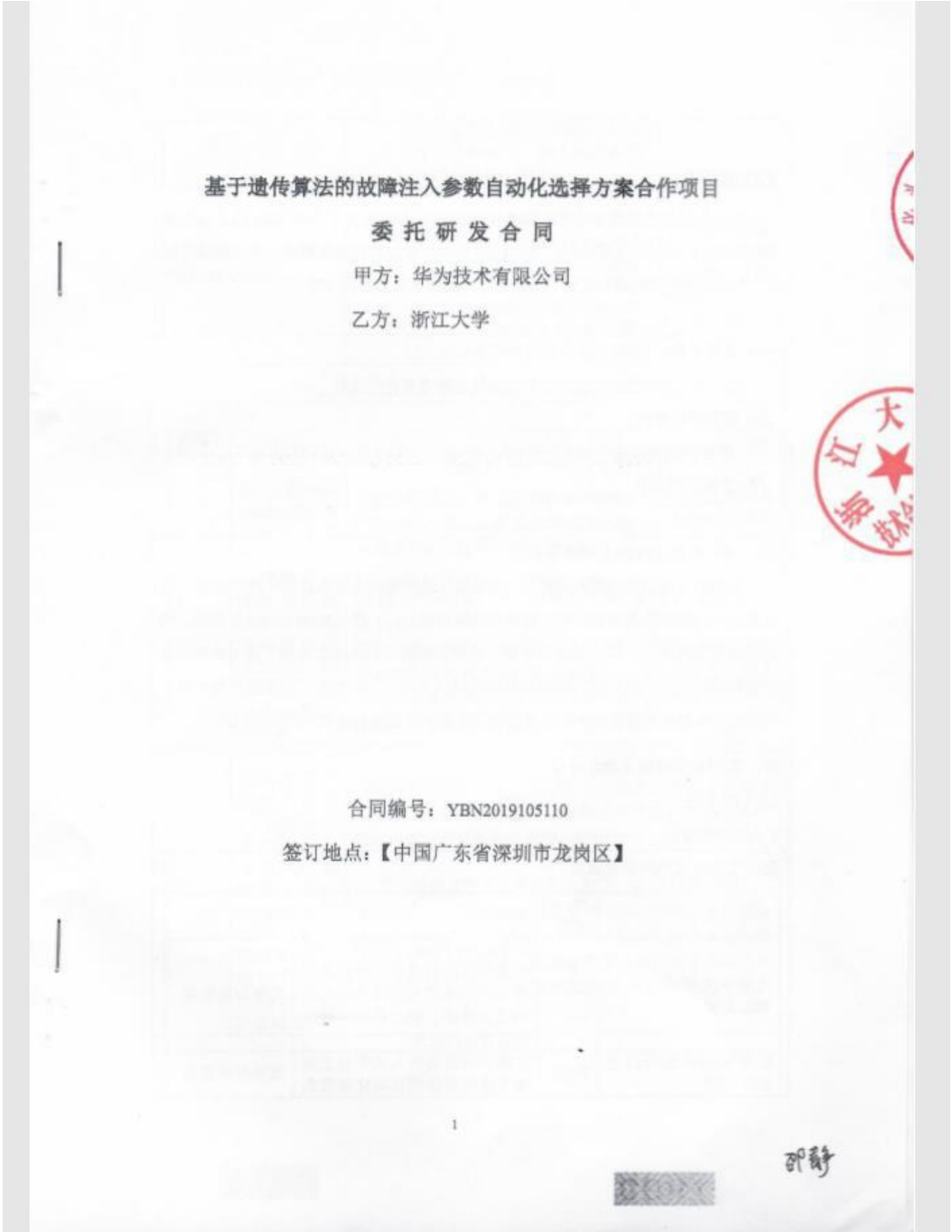
职务：张明

日期：2018.11.6

日期：



基于遗传算法的故障注入参数自动化选择方案合作项目



验证与测试		
● 项目负责人:		
	华为公司项目负责人	浙江大学项目负责人
姓名	章庆隆	张帆
职务	工程师	副教授
地址	杭州华为研究所, 滨江区, 杭州	浙江大学玉泉校区, 西湖区, 杭州
邮编	310051	310027
电话	+86 13811491310	+86 13777356409
电邮	ql.zhang@huawei.com	fanzhang@zju.edu.cn
七、资金投入比例:		
☒ 华为公司 <u>100</u> % ☐ 其他(如国家、地区政府项目资金) _____ %		
八、费用与付款计划:		
本项目总额为人民币 <u>55</u> 万元 (“合同总金额”, 含税), 包括科研劳务费、实验设备材料费、差旅费等保证项目按合同要求完成的所有费用。由华为技术有限公司分 <u>3</u> 期支付给浙江大学, 浙江大学开具税率为 <u>0</u> % 的 <u>增值税普通发票</u> 。		
阶段	支付条件	支付金额 (人民币·万元, 含税)
1	阶段 1 交付件验收通过(合同签署后一周内)	16.5
2	阶段 2 交付件通过甲方验收	16.5
3	项目结题交付件通过甲方验收且甲方收到乙方提供的科技认证登记回执原件或彩色扫描件等相关材料	22
● 华为技术有限公司应在收到浙江大学开具的合法有效的税率为 <u>0</u> % 的 <u>增值税普通发票</u> 、《技术合同项目验收通知函》(格式符合附件 3 要求) 和《供应商开票清单》(格式符合附件 4 要求) 之日起第 <u>30</u> 日 (即 “到期日”), 以银行转账的方式向浙江大学支付付款计划约定的各阶段支付金额。浙江大学账户信息如下表所述:		
开户行	工行杭州市浙大支行	
户名	浙江大学	
帐号	1202024609908808891	
九、本研究项目成果及其知识产权所有权分配:		
☐ 浙江大学 ____ % ☒ 华为技术有限公司 <u>100</u> %		

钓鱼网站品牌仿冒动态检测项目

本项目合同编号：YBN2019085144

保密信息

附件 1：项目任务书

钓鱼网站品牌仿冒动态检测项目

委托研发合同

2 P

甲方：华为技术有限公司

乙方：浙江大学



合同编号：YBN2019085144

签订地点：中国广东省深圳市龙岗区

1



费用与付款计划:

本项目总额为人民币 70 万 元 (“合同总金额”, 不含税), 包括科研劳务费、实验设备材料费、差旅费等保证项目按合同要求完成的所有费用。由甲方分两期支付给乙方, 乙方开具合法有效的发票给甲方。若乙方开具增值税专用发票, 乙方承担税金, 否则由乙方承担税金。

阶段	支付条件	支付比例 (总金额百分比, 不含税)
1	提交项目规划书并验收通过	80%
2	项目全部成果验收通过	20%

甲方应在收到乙方开具的合法有效发票和《供应商开票清单》(格式符合附件 3 要求)之日起第 15 日(即“到期日”), 以银行转账的方式向浙江大学支付付款计划约定的各阶段支付金额。浙江大学账户信息如下表所述:

开户行	工行杭州市浙大支行
户 名	浙江大学
帐 号	1202024609908808891

保证

11.1 未经甲方事先同意, 乙方在本合同下交付的开发成果中不得包含任何导致该开发成果的全部或部分或者使用该开发成果的软件的全部或部分具有对外开源义务的开源软件。

11.2 在受 11.1 条约束的前提下, 乙方向甲方交付任何开发成果的同时, 应按照附件 4 的格式向甲方提供开发成果中包含的所有开源软件清单, 并保证所列信息是准确、详尽的。乙方保证向甲方交付任何开发成果之前, 已全面履行相应开源软件许可证的所有义务。如甲方发现乙方违反此义务的, 乙方在收到甲方书面要求的两个工作日内或双方商定的其他期限内进行整改以达到全面履行, 包括但不限于在相应开源软件许可证允许的范围内, 向甲方免费提供相应的开源软件源代码。

11.3 乙方保证: 参与研究项目的教职员, 在执行研究项目期间及项目结束后三年之



【系统中未录入的补充科研任务】-共 2 项

1. 国家级：国家重大科研仪器研制项目-面向威胁发现的网络流谱分析仪研制



申请代码	F0206
接收部门	
收件日期	
接收编号	6222780010



国家自然科学基金 申 请 书

(2022 版)

资助类别：国家重大科研仪器研制项目

亚类说明：自由申请

附注说明：

项目名称：面向威胁发现的网络流谱分析仪研制

申请人：郭世泽 电话：01066305306

依托单位：北京邮电大学

通讯地址：北京市947信箱

邮政编码：100191 单位电话：62282612

电子邮箱：nsfgsz@126.com

填写日期：2022年02月28日

国家自然科学基金委员会



国家自然科学基金资助项目批准通知

(预算制项目)

郭世泽 先生/女士:

根据《国家自然科学基金条例》、相关项目管理办法规定和专家评审意见,国家自然科学基金委员会(以下简称自然科学基金委)决定资助您申请的项目。项目批准号: 62227805, 项目名称: 面向威胁发现的网络流谱分析仪研制, 直接费用: 772.62万元, 项目起止年月: 2023年01月至 2027年 12月, 有关项目的评审意见及修改意见附后。

请您尽快登录科学基金网络信息系统(<https://grants.nsfc.gov.cn>), 认真阅读《国家自然科学基金资助项目计划书填报说明》并按要求填写《国家自然科学基金资助项目计划书》(以下简称计划书)。对于有修改意见的项目, 请您按修改意见及时调整计划书相关内容; 如您对修改意见有异议, 须在电子版计划书报送截止日期前向相关科学处提出。

请您将电子版计划书通过科学基金网络信息系统(<https://grants.nsfc.gov.cn>)提交, 由依托单位审核后提交至自然科学基金委。自然科学基金委审核未通过者, 将退回的电子版计划书修改后再行提交; 审核通过者, 打印纸质版计划书(一式两份, 双面打印)并在项目负责人承诺栏签字, 由依托单位科研、财务管理等部门审核、签章并在承诺栏加盖依托单位公章, 且将申请书纸质签字盖章页订在其中一份计划书之后, 一并报送至自然科学基金委项目材料接收工作组。纸质版计划书应当保证与审核通过的电子版计划书内容一致。自然科学基金委将对申请书纸质签字盖章页进行审核, 对存在问题的, 允许依托单位进行一次修改或补齐。

向自然科学基金委提交电子版计划书、报送纸质版计划书并补交申请书纸质签字盖章页截止时间节点如下:

1. 年 月 日16点: 提交电子版计划书的截止时间;
2. 年 月 日16点: 提交修改后电子版计划书的截止时间;
3. 年 月 日: 报送纸质版计划书(一式两份, 其中一份包含申请书纸质签字盖章页)的截止时间。





主要参与者 (注: 主要参与者不包括项目申请人)

编号	姓名	出生年月	性别	职 称	学 位	工作单位	项目分工	电话	证件号码
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2	徐洁	1986-12-15	女	讲师	博士	北京邮电大学	项目主研人	010-61198159	3*****4
3	任传伦	1972-06-02	男	研究员	博士	中国电子科技集团公司第三十研究所	项目主研人	010-66306430	3*****2
4	俞赛赛	1982-01-08	男	高级工程师	博士	中国电子科技集团公司第三十研究所	项目主研人	010-66306430	3*****7
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7	肖达	1981-06-01	男	讲师	博士	北京邮电大学	项目主研人	010-62283134转0	2*****7
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9	黄玮	1980-05-18	男	研究员	博士	北京理工大学	项目主研人	010-68913694	3*****6

总人数	高级	中级	初级	博士后	博士生	硕士生
39	7	2	0	1	11	18



2. 省部级：之江实验室重大科研项目-先进工业互联网安全平台

项目编号：



之江实验室重大科研项目课题任务书

项目名称： 先进工业互联网安全平台

课题名称： 工业互联网先进试验场

委托单位（甲方）： 之江实验室 （盖章）

承担单位（乙方）： 浙江大学 （盖章）

起止时间： 2018 年 9 月至 2021 年 8 月

签订日期： 2018 年 8 月

之江实验室制



基本信息	项目预算	合作单位	项目成员	项目经费	经费执行	支出明细	项目状态	经济合同
序号	成员姓名	性别	类别	工作证号	所属单位	成员角色	当前学历	当前职称
1	阮伟	男	科研人员	0016235	控制科学与工程学院--工业控制研究所	负责人	博士研究生	无
2	吴春明	男	科研人员	0095168	计算机科学与技术学院--系统结构与网络安全研究所	参与者	博士研究生	教授
3	王文海	男	科研人员	0096081	控制科学与工程学院--工业控制研究所	子课题负责人	博士研究生	研究员（自然科学）
4	程鹏	男	科研人员	0009848	控制科学与工程学院--工业控制研究所	子课题负责人	博士研究生	教授
5	陈亮	男	科研人员	0917192	先进技术研究院	参与者	博士研究生	无
6	尹中旭	男	其它		国家数字交换系统工程技术研究中心	参与者		
7	杨强	男	科研人员	0010085	电气工程学院--电气自动化研究所	子课题负责人	博士研究生	副教授
8	曹琰	男	其它		国家数字交换系统工程技术研究中心	参与者		
9	张帆	男	科研人员	0013274	计算机科学与技术学院--系统结构与网络安全研究所	子课题负责人	博士研究生	副教授
10	陈铁明	男	其它		浙江工业大学计算机学院	子课题负责人		
11	陈双喜	男	学生		工程师学院	参与者	博士研究生	
12	刘江宣	男	学生		计算机科学与技术学院	参与者	研究生	
13	田逸飞	男	学生		计算机科学与技术学院	参与者	研究生	
14	蔡晶晶	女	学生		工程师学院	参与者	研究生	
15	张继学	男	学生		计算机科学与技术学院	参与者	研究生	
16	吴强	男	其它		中兴通讯股份有限公司	参与者		
17	庞建民	男	其它		国家数字交换系统工程技术研究中心	子课题负责人		
18	陈鸿妮	男	其它		国家数字交换系统工程技术研究中心	子课题负责人		
19	张兴明	男	其它		国家数字交换系统工程技术研究中心	子课题负责人		
20	魏强	男	其它		国家数字交换系统工程技术研究中心	子课题负责人		

返回



3. 成果产出及贡献（注：根据填写的“代表性论著”、“发明专利授权情况”、“软件著作权或动植物新品种”、“标准制定情况”、“主要新产品（含新品种）/新装置（装备）/新工艺/新材料开发/创新服务情况”等依次上传证明材料）；

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一、代表性论著

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论文检索证明汇总

经检索“Engineering Village”，下述文献中 15 篇文献被《Ei Compendex》收录。

经检索《Web of Science》，下述文献中 7 篇文献被《Science Citation Index Expanded (SCI-EXPANDED)》收录。

经检索《Web of Science》，下述文献中 12 篇文献被《Science Citation Index Expanded (SCI-EXPANDED)》引用 82 次，其中自引 23 次，他引 59 次。

(检索时间 2023 年 1 月 19 日)

序号	文献信息	EI 收录	SCI- E 收录	期刊影 响因子 (最新 年 2021)	SCI- E 总 引次 数	SCI- E 他 引次 数
1	Persistent fault analysis on block ciphers Zhang, Fan;Lou, Xiaoxuan;Zhao, Xinjie;Bhasin, Shivam;He, Wei;Ding, Ruyi;Qureshi, Samiya;Ren, Kui IACR Transactions on Cryptographic Hardware and Embedded Systems Year:2018 Volume:2018 Issue:3 Page:150-172 Doi:10.13154/tches.v2018.i3.150-172	是	/		16	9
2	Side-channel analysis and countermeasure design on arm-based quantum-resistant SIKE Zhang, Fan;Yang, Bolin;Dong, Xiaofei;Guilley, Sylvain;Liu, Zhe;He, Wei;Zhang, Fangguo;Ren, Kui IEEE Transactions on Computers Year:2020 Volume:69 Issue:11 Page:1681-1693 Doi:10.1109/TC.2020.3020407	是	是	3.183	2	1
3	DARPT: Defense Against Remote Physical Attack Based on TDC in Multi-tenant Scenario Zhang, Fan;Wang, Zhiyong;Shen, Haoting;Yang, Bolin;Wu, Qianmei;Ren, Kui Proceedings - Design Automation Conference Year:2022 Page:559-564 Doi:10.1145/3489517.3530494	是	/		0	0
4	One Fault is All it Needs: Breaking Higher-Order Masking with Persistent Fault Analysis Pan, Jingyu;Zhang, Fan;Ren, Kui;Bhasin, Shivam Proceedings of the 2019 Design, Automation and Test in Europe Conference and Exhibition, DATE 2019 Year:2019 Page:1-6 Doi:10.23919/DATE.2019.8715260	是	/		11	11
5	From homogeneous to heterogeneous: Leveraging deep learning based power analysis across devices Zhang, Fan;Shao, Bin;Xu, Guorui;Yang, Bolin;Yang, Ziqi;Qin, Zhan;Ren, Kui;Yang, Ziqi Proceedings - Design Automation Conference Year:2020 Volume:2020-July Doi:10.1109/DAC18072.2020.9218693	是	/		0	0



论文检索证明汇总

序号	文献信息	EI 收录	SCI- E 收录	期刊影 响因子 (最新 年2021)	SCI- E 总 引 次 数	SCI- E 他 引 次 数
6	On Efficient and Secure Code-based Masking: A Pragmatic Evaluation Wu, Qianmei;Cheng, Wei;Guilley, Sylvain;Zhang, Fan;Fu, Wei IACR Transactions on Cryptographic Hardware and Embedded Systems Year:2022 Volume:2022 Issue:3 Page:192-222 Doi:10.46586/tches.v2022.i3.192-222	是	/		1	0
7	Free Fault Leakages for Deep Exploitation: Algebraic Persistent Fault Analysis on Lightweight Block Ciphers Zhang, Fan;Feng, Tianxiang;Li, Zhiqi;Ren, Kui;Zhao, Xinjie IACR Transactions on Cryptographic Hardware and Embedded Systems Year:2022 Volume:2022 Issue:2 Page:289-311 Doi:10.46586/tches.v2022.i2.289-311	是	/		1	1
8	Pushing the Limit of PFA: Enhanced Persistent Fault Analysis on Block Ciphers Xu, Guorui;Zhang, Fan;Yang, Bolin;Zhao, Xinjie;He, Wei;Ren, Kui IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems Year:2021 Volume:40 Issue:6 Page:1102-1116 Doi:10.1109/TCAD.2020.3048280	是	是	2.565	1	1
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Persistent Fault Analysis on Block Ciphers

Fan Zhang^{1,2}, Xiaoxuan Lou^{1,2}, Xinjie Zhao⁴, Shivam Bhasin⁵, Wei He⁶,
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Abstract. Persistence is an intrinsic nature for many errors yet has not been caught enough attractions for years. In this paper, the feature of persistence is applied to fault attacks, and the *persistent fault attack* is proposed. Different from traditional fault attacks, adversaries can prepare the fault injection stage before the encryption stage, which relaxes the constraint of the tight-coupled time synchronization. The *persistent fault analysis* (PFA) is elaborated on different implementations of AES-128, specially fault hardened implementations based on Dual Modular Redundancy (DMR). Our experimental results show that PFA is quite simple and efficient in breaking these typical implementations. To show the feasibility and practicability of our attack, a case study is illustrated on the shared library Libgcrypt with rowhammer technique. Approximately 8200 ciphertexts are enough to extract the master key of AES-128 when PFA is applied to Libgcrypt1.6.3 with redundant encryption based DMR. This work puts forward a new direction of fault attacks and can be extended to attack other implementations under more interesting scenarios.

Keywords: Fault Analysis · PFA · DMR · AES · Libgcrypt · Rowhammer.

1 Introduction

Fault attack (FA) is a class of implementation level attacks on embedded systems [Joy12], which is usually used to attack different ciphers such as RSA, AES, PRESENT[BKL⁺07], LED[GPPR11], Piccolo[SIH⁺11]. FA is an active attack that disturbs the operation of target device. The disturbance is realized by forcing the device in a non-nominal operating condition. Common methods include changing the power supply voltage, changing the frequency of the external clock, varying the temperature or exposing the circuits to lasers during the key-dependent computations such as encryptions [BECN⁺06, ACS⁺07, TJ09, Insa, Insb]. The idea of fault attack was first reported on RSA-CRT by Boneh et al. in 1996 [DDL97]. Later, Biham and Shamir proposed the *differential fault analysis* (DFA) attack on the block cipher DES, which combines a fault attack with differential cryptanalysis [BS06]. Since then, DFA has been used to break various block ciphers [BS97, DLV03, TMA11]. Apart from breaking cryptographic systems, FA is also used for other attacks, like bypassing security checks in smartcards [BECN⁺06].

DFA operates in a differential setting, i.e., exploiting the difference of correct and faulty ciphertexts for a fixed input. Later, other fault analysis techniques were introduced. Some analysis techniques exploited the algebraic structure of the algorithm with fault injection

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2. 《Free Fault Leakages for Deep Exploitation: Algebraic Persistent Fault Analysis on Lightweight Block Ciphers》

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Fan Zhang Zhejiang University, Hangzhou, China Tianxiang Feng Zhejiang University, Hangzhou, China Zhiqi Li Zhejiang University, Hangzhou, China Kui Ren Zhejiang University, Hangzhou, China Xinjie Zhao Zhejiang University, Hangzhou, China	PDF Published 2022-02-15 How to Cite Zhang, F., Feng, T., Li, Z., Ren, K., & Zhao, X. (2022). Free Fault Leakages for Deep Exploitation: Algebraic Persistent Fault Analysis on Lightweight Block Ciphers. <i>IACR Transactions on Cryptographic Hardware and Embedded Systems</i>, 2022(2), 289–311. https://doi.org/10.46586/tches.v2022.i2.289-311 More Citation Formats ▾ Issue Volume 2022, Issue 2 Section Articles License Copyright (c) 2022 Fan Zhang, Tianxiang Feng, Zhiqi Li, Kui Ren, Xinjie Zhao This work is licensed under a Creative Commons Attribution 4.0 International License.
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Abstract Persistent Fault Analysis (PFA) is a new fault analysis method for block ciphers proposed in CHES 2018, which utilizes those faults that persist in encryptions. However, one fact that has not been raised enough attention is that: while the fault itself does persist in the entire encryption, the corresponding statistical analysis merely leverages fault leakages in the last one or two rounds, which ignores the valuable leakages in deeper rounds. In this paper, we propose <i>Algebraic Persistent Fault Analysis</i> (APFA), which introduces algebraic analysis to facilitate PFA. APFA tries to make full usage of the free fault leakages in the deeper rounds without incurring additional fault injections. The core idea of APFA is to build similar algebraic constraints for the output of substitution layers and apply the constraints to as many rounds as possible. APFA has many advantages compared to PFA. First, APFA can bypass the manual deductions of round key dependencies along the fault propagation path and transfer the burdens to the computing power of machine solvers such as Crypto-MiniSAT. Second, thanks to the free leakages in the deeper round, APFA requires a much less number of ciphertexts than previous PFA methods, especially for those lightweight block ciphers such as PRESENT, LED, SKINNY, etc. Only 10 faulty ciphertexts are required to recover the master key of SKINNY-64-64, which is about 155 times of reduction as compared to the state-of-the-art result. Third, APFA can be applied to the block ciphers that cannot be analyzed by PFA due to the key size, such as PRESENT-128. Most importantly, APFA replaces statistical analysis with algebraic analysis, which opens a new direction for persistent-fault related researches.	

Free Fault Leakages for Deep Exploitation: Algebraic Persistent Fault Analysis on Lightweight Block Ciphers

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Abstract. *Persistent Fault Analysis* (PFA) is a new fault analysis method for block ciphers proposed in CHES 2018, which utilizes those faults that persist in encryptions. However, one fact that has not been raised enough attention is that: while the fault itself does persist in the entire encryption, the corresponding statistical analysis merely leverages fault leakages in the last one or two rounds, which ignores the valuable leakages in deeper rounds. In this paper, we propose *Algebraic Persistent Fault Analysis* (APFA), which introduces algebraic analysis to facilitate PFA. APFA tries to make full usage of the free fault leakages in the deeper rounds without incurring additional fault injections. The core idea of APFA is to build similar algebraic constraints for the output of substitution layers and apply the constraints to as many rounds as possible. APFA has many advantages compared to PFA. First, APFA can bypass the manual deductions of round key dependencies along the fault propagation path and transfer the burdens to the computing power of machine solvers such as Crypto-MiniSAT. Second, thanks to the free leakages in the deeper round, APFA requires a much less number of ciphertexts than previous PFA methods, especially for those lightweight block ciphers such as PRESENT, LED, SKINNY, etc. Only 10 faulty ciphertexts are required to recover the master key of SKINNY-64-64, which is about 155 times of reduction as compared to the state-of-the-art result. Third, APFA can be applied to the block ciphers that cannot be analyzed by PFA due to the key size, such as PRESENT-128. Most importantly, APFA replaces statistical analysis with algebraic analysis, which opens a new direction for persistent-fault related researches.

Keywords: Algebraic · PFA · Fault Analysis · Fault Attack · PRESENT · LED · SKINNY · LBlock · AES

1 Introduction

Different from *Side Channel Attack* (SCA), *Fault Attack* can be considered as an active attack method that targets physical cryptographic equipment [JT12]. It performs the *fault injection* (FI) on the device, and conducts the *fault analysis* (FA) to recover the key by analyzing those collected faulty ciphertexts [DLV03]. It was first proposed by Boneh *et al.* in 1997 [BDL97]. And then, Biham and Shamir proposed the well-known *Differential Fault*

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3. 《Side-Channel Analysis and Countermeasure Design on ARM-Based Quantum-Resistant SIKE》

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Side-Channel Analysis and Countermeasure Design on ARM-based Quantum-Resistant SIKE

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Fan Zhang ; Bolin Yang ; Xiaofei Dong ; Sylvain Guilley ; Zhe Liu ; Wei He ; Fangguo Zhang ; Kui Ren All Authors

Abstract

Authors

Keywords

Abstract:

The implementations of post-quantum cryptography have been newly explored, whereas, the protection against side-channel attacks shall be considered upfront. In this paper, the security of supersingular isogeny key encapsulation (SIKE), a second-round candidate of NIST's on-going post-quantum standardization process, is thoroughly evaluated under side-channel analysis. First, the vulnerabilities of reference and optimized implementations are thoroughly analyzed both in terms of horizontal and vertical leakages. After the optimized SIKE, which is based on Three-point Montgomery Differential Ladder, is proved to be constant-time and there is no horizontal leakage, a vertical vulnerability is analyzed and a theoretical differential power analysis (DPA) attack is proposed. In order to exploit this vulnerability, the differential electromagnetic attack (DEMA) is put into practice to extract the private key of SIKE based on a real 32-bit ARM platform. To the best of our knowledge, this is the first side-channel attack on SIKE implemented at ARM platform. More importantly, an efficient window-based countermeasure is proposed to eliminate the vertical leakage and prevent side-channel attacks with only a little overhead. Through careful evaluation and comparison with other countermeasures, this method can lead to higher security at a very small price in terms of time and memory aspects.

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Side-Channel Analysis and Countermeasure Design on ARM-based Quantum-Resistant SIKE

Fan Zhang, Member, IEEE, Bolin Yang, Xiaofei Dong, Sylvain Guilley, Member, IEEE, Zhe Liu, Senior Member, IEEE, Wei He, Fanguo Zhang, and Kui Ren, Fellow, IEEE

Abstract—The implementations of post-quantum cryptographic algorithms have been newly explored, whereas, the protection against side-channel attacks shall be considered upfront, since it can have a non-negligible impact on security and performance. In this paper, the security of supersingular isogeny key encapsulation (SIKE), a second-round candidate of NIST's on-going post-quantum standardization process, is thoroughly evaluated under side-channel analysis. First, the vulnerabilities of reference and optimized implementations of SIKE are thoroughly analyzed in terms of both horizontal and vertical side-channel leakage. After the optimized SIKE, which is based on Three-point Montgomery Differential Ladder algorithm, is proved to be constant-time and there is no horizontal leakage, a vertical vulnerability is analyzed based on the source code at the algorithmic level, and a theoretical differential power analysis (DPA) attack is proposed. In order to exploit this vulnerability, the differential electromagnetic attack (DEMA) is put into practice to extract the private key of SIKE based on a 32-bit ARM platform. To the best of our knowledge, this is the first practical side-channel attack at SIKE implemented on real ARM-based devices. Our experiments show that the DEMA needs only hundreds of electromagnetic traces to carry out the attack. More importantly, an efficient window-based countermeasure is proposed to eliminate the vertical leakage and prevent side-channel attacks with only a little overhead. The security of our countermeasure is carefully evaluated against most of well-known power analysis attacks. Through careful evaluation and comparison with other countermeasures, this method can lead to higher security at a very small cost in terms of time and memory.

Index Terms—PQC, SIKE, Vulnerability Detection, DPA, DEMA, Window-based Countermeasure

1 INTRODUCTION

SINCE the Diffie-Hellman (DH) key exchange scheme was first proposed in 1976 [1], the public-key cryptography (PKC) is rapidly developed, which is served as the foundation of modern system and communication security. For example, the widely-used RSA, ElGamal and ECC have evolved as the most popular algorithms to implement public-key encryption and digital signatures. The security of these cryptosystems depends on the hardness of certain problems in number theory such as the factoring problem (FP), the discrete logarithm problem (DLP) and elliptic curve

problem [2]. However, these hard problems are proved to be very fragile when facing to the appearance of physical quantum computers that put Shor's quantum algorithm into reality [3]. Even though the quantum computers have not emerged on a large scale until now, it is urgent to research on new quantum-resistant cryptography algorithms to replace previous PKC, which are commonly named as Post-Quantum Cryptography (PQC).

PQC is usually divided into four classes: lattice-based, hash-based, code-based and multivariate-quadratic-equations-based cryptography [4]. In 2016, National Institute of Standards and Technology (NIST) has initiated a process to solicit, evaluate, and standardize the quantum-resistant public-key cryptography [5]. Among Round 1 submissions, 17 public-key encryption and key-establishment algorithms and 9 digital signature algorithms head to Round 2. The Supersingular Isogeny Key Encapsulation (SIKE) is the only one submission to NIST apart from other four classes, which is based on supersingular isogeny elliptic curves. And it is very promising and unique among all the Round 2 candidates.

For the supersingular isogeny-based cryptography, its debut is years later than those four classes aforementioned. It was first proposed by Jao and De Feo in 2011 [6], which relies on the conjectured hardness of finding isogenies between supersingular elliptic curves. Later, it was continuously optimized to improve its performance. For example, Faz-Hernández et al. proposed several algorithmic optimizations on both elliptic-curve and field arithmetic operations [7]. Most importantly, they optimized the classic three-point ladder algorithm to compute elliptic-curve

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RESEARCH-ARTICLE

DARPT: defense against remote physical attack based on TDC in multi-tenant scenario

Authors: Fan Zhang, Zhiyong Wang, Haoting Shen, Bolin Yang, Qianmei Wu, Kui Ren [Authors Info & Claims](#)

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ABSTRACTReferencesIndex TermsComments

ABSTRACT

With rapidly increasing demands for cloud computing, Field Programmable Gate Array (FPGA) has become popular in cloud datacenters. Although it improves computing performance through flexible hardware acceleration, new security concerns also come along. For example, unavoidable physical leakage from the Power Distribution Network (PDN) can be utilized by attackers to mount remote Side-Channel Attacks (SCA), such as Correlation Power Attacks (CPA). Remote Fault Attacks (FA) can also be successfully presented by malicious tenants in a cloud multi-tenant scenario, posing a significant threat to legal tenants. There are few hardware-based countermeasures to defeat both remote attacks that aforementioned. In this work, we exploit Time-to-Digital Converter (TDC) and propose a novel defense technique called DARPT (Defense Against Remote Physical attack based on TDC) to protect sensitive information from CPA and FA. Specifically, DARPT produces random clock jitters to reduce possible information leakage through the power side-channel and provides an early warning of FA by constantly monitoring the variation of the voltage drop across PDN. In comparison to the fact that 8k traces are enough for a successful CPA on FPGA without DARPT, our experimental results show that up to 800k traces (100 times) are not enough for the same FPGA protected by DARPT. Meanwhile, the TDC-based voltage monitor presents significant readout changes (by 51.82% or larger) under FA with ring oscillators, demonstrating sufficient sensitivities to voltage-drop-based FA.



DARPT: Defense Against Remote Physical Attack Based on TDC in Multi-tenant Scenario

Fan Zhang^{*,}, Zhiyong Wang^{*,}, Haoting Shen^{*,}, Bolin Yang^{*,}, Qianmei Wu[°] and Kui Ren[°]
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ABSTRACT

With rapidly increasing demands for cloud computing, Field Programmable Gate Array (FPGA) has become popular in cloud datacenters. Although it improves computing performance through flexible hardware acceleration, new security concerns also come along. For example, unavoidable physical leakage from the Power Distribution Network (PDN) can be utilized by attackers to mount remote Side-Channel Attacks (SCA), such as Correlation Power Attacks (CPA). Remote Fault Attacks (FA) can also be successfully presented by malicious tenants in a cloud multi-tenant scenario, posing a significant threat to legal tenants. There are few hardware-based countermeasures to defeat both remote attacks that aforementioned. In this work, we exploit Time-to-Digital Converter (TDC) and propose a novel defense technique called DARPT (Defense Against Remote Physical attack based on TDC) to protect sensitive information from CPA and FA. Specifically, DARPT produces random clock jitters to reduce possible information leakage through the power side-channel and provides an early warning of FA by constantly monitoring the variation of the voltage drop across PDN. In comparison to the fact that 8k traces are enough for a successful CPA on FPGA without DARPT, our experimental results show that up to 800k traces (100 times) are not enough for the same FPGA protected by DARPT. Meanwhile, the TDC-based voltage monitor presents significant readout changes (by 51.82% or larger) under FA with ring oscillators, demonstrating sufficient sensitivities to voltage-drop-based FA.

1 INTRODUCTION

As more powerful computation capability is required for machine learning, FPGA is increasingly utilized as accelerators in cloud computing centers due to its high parallelization and flexible implementation. Meanwhile, services such as FPGA-as-a-Service (FaaS) and Acceleration-as-a-Service (AaaS) are also provided on many cloud platforms [7]. With such services, users can quickly develop customized IPs to achieve profitability. To improve the utilization of FPGA resources, the concept of multi-tenant is proposed, which shares resources on one FPGA among multiple users [29]. The multi-tenant virtualization renders the user units to be isolated from each other at the logic level. However, it fails to cut down the physical relevance at the electrical level between different users as shown

in [9]. For example, different values and operations in the Programmable Logic (PL) will affect the instantaneous voltage and current in Power Distribution Network (PDN). That is, all users' logic on the same FPGA will have a joint effect on the voltage fluctuations in the PDN, which might be exploited by malicious tenants.

In typical physical attacks such as Side-Channel Attacks (SCA) and Fault Attacks (FA), an attacker can recover the secret information by collecting and exploiting the leakage of the device or injecting the fault through local devices. In general, those attacks using near-field devices may not exist in cloud conditions due to physical isolation. Unfortunately, the relation between voltage fluctuation and user implemented logic in the PDN makes the physical attacks feasible in the multi-tenant scenario. Through the PDN, both SCA and FA can be implemented remotely, which are called remote SCA and remote FA respectively. A malicious user has the authority to submit a bitstream that influences the PDN and collects the voltage variation during the execution process stealthily. Remote SCA on some encryption algorithms such as AES and RSA have been implemented in the cloud successfully [23, 26]. In addition, it is feasible to conduct fault injection attacks on other users sharing the same FPGA by executing high-power-wasting designs [9, 12].

As the mechanism of SCA and FA are different from each other, it is challenging to defend against both with one solution. Common strategies to resist SCA include *masking* and *hiding*. Proposed *hiding* designs include clock misalignment and noise compensation [6]. At the software level, the clock misalignment is usually implemented by inserting some dummy operations into the algorithm. Considering the noise compensation, this approach usually introduces considerable overhead due to the numerous amounts of power compensation units [13]. The countermeasures against FA are dedicated to error detection and threshold implementations [3]. The traditional way to detect fault injection leverages a redundant computing unit, which consumes double resources.

In this paper, we propose a novel countermeasure called DARPT which can resist both SCA and FA with relatively low overhead. DARPT is composed of only basic primitives and hence it can be migrated to diverse programmable logic parts for security consideration. Our design mainly consists of a generator of clock jitters and a detector of fault injection for defense against SCA and FA, respectively. The threshold detection method against fault injection with TDC was performed by [16]. Subsequently, we propose a doubled clock frequency generator based on TDC which further boosts the SCA security. Our contributions are concluded as follows:

- We propose a new hardware-based countermeasure to defend against both SCA and FA. The proposed defense exploits the

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5. 《Pushing the Limit of PFA: Enhanced Persistent Fault Analysis on Block Ciphers》

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Abstract:

Persistent fault analysis (PFA) is a newly proposed cryptanalysis for block ciphers. Although the injected fault is persistent during the entire encryption, the corresponding analysis is only applied to the last round in the original PFA. In this article, the enhanced PFA (EPFA) is proposed, which can push the limit of PFA by exploiting the fault leakage in deeper rounds and target to reduce the number of required ciphertexts as small as possible. EPFA is first introduced as a general method with a specific application to advanced encryption standard (AES). Then it is extended to other substitution-permutation network (SPN)-based block ciphers, such as LED and SKINNY, both of which have unique features that EPFA fits well. To improve the efficiency of EPFA, a parallel algorithm based on mixed radix numbers is developed, which fully utilizes the power of GPU. Our experimental results show that EPFA can reduce the number of required ciphertexts to be under 1000, which is only about 40% of the 2500 ciphertexts in previous PFA on AES. In contrast to the single-threaded implementation, the parallel EPFA can have a speedup roughly about 200 times.

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I. Introduction

Fault attacks can inject deliberate faults to cause faults in the computation and produce faulty results [1], [2]. Fault analysis is the offline stage of fault attacks, which focuses on analyzing the faulty results to recover some secret information used in the computation [3].

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Pushing the Limit of PFA: Enhanced Persistent Fault Analysis on Block Ciphers

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Abstract—Persistent fault analysis (PFA) is a newly proposed cryptanalysis for block ciphers. Although the injected fault is persistent during the entire encryption, the corresponding analysis is only applied to the last round in the original PFA. In this article, the enhanced PFA (EPFA) is proposed, which can push the limit of PFA by exploiting the fault leakage in deeper rounds and target to reduce the number of required ciphertexts as small as possible. EPFA is first introduced as a general method with a specific application to advanced encryption standard (AES). Then it is extended to other substitution-permutation network (SPN)-based block ciphers, such as LED and SKINNY, both of which have unique features that EPFA fits well. To improve the efficiency of EPFA, a parallel algorithm based on mixed radix numbers is developed, which fully utilizes the power of GPU. Our experimental results show that EPFA can reduce the number of required ciphertexts to be under 1000, which is only about 40% of the 2500 ciphertexts in previous PFA on AES. In contrast to the single-threaded implementation, the parallel EPFA can have a speedup roughly about 200 times.

Index Terms—Advanced encryption standard (AES), fault analysis, fault attack, GPU-accelerated computing, LED, persistent fault analysis (PFA), SKINNY.

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I. INTRODUCTION

FAULT attacks can inject deliberate faults to cause faulty computations and produce faulty results [1], [2]. Fault analysis is the offline stage of fault attacks, which focuses on how to use the faulty results to recover some secret information used in the computation [3].

It has been demonstrated that many block cipher cryptosystems are vulnerable to fault attacks. First in 1997, Biham and Shamir developed differential fault analysis (DFA), showing that DFA is applicable to almost any secret key cryptosystem [1]. The powerful cryptanalysis method is based on the differences between correct and faulty ciphertexts. Under a transient fault model, DFA can extract the full DES key by analyzing 50–200 ciphertexts [1]. Later in 2003, Pierre and Gilles presented DFA on the advanced encryption standard (AES) [3]. AES without faults is not easy to compromise. However, when the adversary can inject faults and collect some faulty ciphertexts, he is able to recover the secret key in a few minutes. This could be a great threat if the injection of faults and the collection of ciphertexts can be done at a low cost. One promising research direction is to make fault attacks more practical, either to simplify the injections with ease efforts and fewer engineering requirements or to push the limit of analysis costs like using as few ciphertexts or resources as possible.

In previous works, most fault models consider the fault to be transient, which means that the fault only appears for a very short period in the process (typically one clock cycle), or permanent, which causes the device to be damaged permanently. Zhang *et al.* [4] proposed a new type of fault named persistent faults in 2018, which can be persistent during many encryptions and disappear when the target gets refreshed (such as rebooted). In Zhang's work [4], the online injection part is done by the RowHammer technique, which injects a persistent bit-flip fault in the S-box. The offline analysis part is named as persistent fault analysis (PFA). Using such faults with persistency, they applied proper statistical methods to extract the secret key with less than 2500 faulty ciphertexts. PFA can be very promising in this way. However, how to reduce the number of required ciphertexts and how to fully utilize the fault leakages, have not been investigated yet.

In this article, we propose a powerful extension of PFA, named as enhanced PFA (EPFA). EPFA naturally inherits all the advantages of PFA, for example, it needs ciphertexts only, it does not damage the physical device and it is able to break some countermeasures. Moreover, EPFA requires much

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【系统中未录入的补充著作】-共 1 篇

1. 《密码故障分析与防护》



【系统中未录入的补充论文】-共 10 篇

1. 《One Fault is All it Needs: Breaking Higher-Order Masking with Persistent Fault Analysis》

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One Fault is All it Needs: Breaking Higher-Order Masking with Persistent Fault Analysis

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Abstract:

Persistent fault analysis (PFA) was proposed at CHES 2018 as a novel fault analysis technique. It was shown to completely defeat standard redundancy based countermeasure against fault analysis. In this work, we investigate the security of masking schemes against PFA. We show that with only one fault injection, masking countermeasures can be broken at any masking order. The study is performed on publicly available implementations of masking.

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One Fault is All it Needs: Breaking Higher-Order Masking with Persistent Fault Analysis

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Abstract—Persistent fault analysis (PFA) was proposed at CHES 2018 as a novel fault analysis technique. It was shown to completely defeat standard redundancy based countermeasure against fault analysis. In this work, we investigate the security of masking schemes against PFA. We show that with only one fault injection, masking countermeasures can be broken at any masking order. The study is performed on publicly available implementations of masking.

Index Terms—fault attacks, masking, persistent

I. INTRODUCTION

Fault attacks [1] are a type of physical attacks which considers an active attacker capable of disturbing the operation of the target system. Fault attacks have been powerful against standard cryptographic schemes, such as AES, RSA, etc.

Most fault attacks assume a *transient* fault model, where the injected disturbance or fault is temporary, and ideally it affects only one instance of the target function call (eg. one encryption). Some attacks also consider a *permanent* fault model which affects all calls to the target function. Such faults often arise from physical defects in the device. Recently at CHES 2018 [2], a new fault model was highlighted which remains between transient and permanent, called as *persistent* fault. Unlike transient fault, it affects several calls of the target function, however, *persistent* fault is not permanent, and disappears with a device reboot.

A specific fault analysis technique to exploit persistent fault on block ciphers was also developed and called as *Persistent Fault Attack (PFA)* [2]. PFA was shown to break fault countermeasures based on module redundancy and comparison. Masking [3] is one of the most-studied countermeasures against side-channel attacks.

In this work, we investigate the security of some popular masking schemes against PFA. Publicly available implementations are used for the analysis. Our results show that masking countermeasures can be easily broken with PFA. We highlight the main advantage of PFA over other fault attacks. PFA needs only one fault injection as compared to typically one fault per encryption for other fault analysis technique. With one fault

injection and multiple encryptions with same fault, PFA can break masking countermeasure at any order. This reduces the practical effort that an attacker should bare to minimum.

The rest of the paper is organized as follows. Section 2 recalls principles of PFA and masking. Section 3 applies PFA on general masking construction. Case study on security of public implementation of masking schemes against PFA is described in Section 4 and final conclusions are drawn in Section 5.

II. BACKGROUND

This section recalls general background concepts about PFA and masking.

A. Persistent Fault Analysis (PFA)

PFA was recently introduced as a novel fault analysis technique in CHES 2018 [2]. Unlike other fault attacks which rely on transient or permanent fault model, PFA exploits persistent fault model. As stated earlier, under persistent fault model, the fault affects several consecutive encryptions. The fault, typically, alters a stored constant (like one or more entry in Sbox look-up) in the target algorithm.

For better comprehension, let us take an example of PRESENT cipher where a random nibble fault alters one Sbox element v to v' . In absence of fault, all elements in 4×4 Sbox including v, v' have an expectation of $\mathbb{E}(v) = \mathbb{E}(v') = \frac{1}{16}$. If a persistent fault is injected to change v to v' , $\mathbb{E}(v) = 0, \mathbb{E}(v') = \frac{2}{16}$, while all other elements still hold the expectation $\frac{1}{16}$. If in a certain Sbox call in any round, the original output is v , it will be replaced by v' , leading to faulty ciphertext. Some encryptions will still be correct as they won't access the element v of the Sbox during the whole encryption. This difference can be detected statistically over a big set of ciphertexts, just by observing the distribution of each nibble, leaking information of the whole last round key k . Fig. 1 illustrates PFA. A fault is injected into the Sbox and turns an element of Sbox from $v = 10$ into $v' = 12$. v, v' are not required to be known to the attacker and can be brute forced. The following statistical tools can be used for key recovery:

- 1) t_{min} : find the missing value in Sbox table. Then $k = t_{min} \oplus v$;

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[†] This research was conducted in parts during author's summer visit to Temasek Laboratories, NTU, Singapore.



2. 《From Homogeneous to Heterogeneous: Leveraging Deep Learning based Power Analysis across Devices》



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Learn From Your Leakage: Attacks and Defenses for Side-Channel Analysis

Chair: Ramesh Karri - *New York Univ, New York, New York*

Today side-channel attacks have emerged as one of the most powerful attacks on protected devices. At the same time, various techniques have been proposed to conduct such attacks. One prominent direction is to employ machine learning to implement side-channel attacks. This session explores the influence of portability on the performance of profiling attacks and the implications of the size of the learning networks on the efficiency of the attack. It further explores how to fuse multiple sources of leakage, and presents a new metric for detecting cache timing attacks.

81.1 Reuse-Trap: Re-Purposing Cache Reuse Distance to Defend Against Side Channel Leakage

Speaker: Hongyu Fang - *George Washington Univ, Washington, DC, DC*

Authors: Hongyu Fang - *George Washington Univ, Washington, DC, DC*
Milos Doroslovacki - *George Washington Univ, Washington, DC*
Guru Venkataramani - *George Washington Univ, Washington, DC*

81.2 Deep Learning Multi-Channel Fusion Attack Against Side-Channel Protected Hardware

Speaker: Benjamin Hettwer - *Robert Bosch GmbH, Stuttgart, Germany*

Authors: Benjamin Hettwer - *Robert Bosch GmbH, Stuttgart, Germany*
Daniel Fennes - *Ruhr Univ. Bochum, Germany*
Sebastien Leger - *Robert Bosch GmbH, Stuttgart, Germany*
Jan Richter-Brockmann - *Ruhr Univ. Bochum, Germany*
Stefan Gehrler - *Robert Bosch GmbH, Stuttgart, Germany*
Tim Güneysu - *Ruhr Univ. Bochum, Germany*

81.3 Learning From A Big Brother - Mimicking Neural Networks in Profiled Side-channel Analysis

Speaker: Stjepan Picek - *Delft Univ. of Technology, Delft, The Netherlands*

Authors: Daan van der Valk - *Delft Univ. of Technology, Delft, The Netherlands*
Marina Krcek - *Delft Univ. of Technology, Delft, The Netherlands*
Stjepan Picek - *Delft Univ. of Technology, Delft, The Netherlands*
Shivam Bhasin - *Nanyang Technological Univ, Singapore, Singapore*

81.4 From Homogeneous to Heterogeneous: Leveraging Deep Learning Based Power Analysis Across Devices

Speaker: Fan Zhang - *Zhejiang Univ. & Alibaba-Zhejiang University Joint Institute of Frontier Technologies, Hangzhou, China*

Authors: Fan Zhang - *Zhejiang Univ. & Alibaba-Zhejiang University Joint Institute of Frontier Technologies, Hangzhou, China*
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Zhan Qin - *Zhejiang Univ. & Alibaba-Zhejiang University Joint Institute of Frontier Technologies, Hangzhou, China*
Kui Ren - *Zhejiang Univ. & Alibaba-Zhejiang University Joint Institute of Frontier Technologies, Hangzhou, China*



From Homogeneous to Heterogeneous: Leveraging Deep Learning based Power Analysis across Devices

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Abstract—In this paper, we raise practical situations in profiling based power analysis when profiling and target devices are quite different at several levels. “Crossed devices” are newly termed, including homogeneous and heterogeneous devices, which have not been carefully investigated. We identify such device variations and take a further step towards leveraging the deep learning based power analysis. Traditional template attacks and straight-forward deep learning based power analysis will fail, when the gap across devices is significantly enlarged. In this paper, we propose a novel frequency and learning based power analysis mechanism, which is able to explore new attacking power of deep learning and address challenges caused by device variations. For the first time, power traces collected from our own PIC devices can be utilized to successfully attack the public dataset in DPAContest v4 which is based on a totally different AVR microcontroller.

I. INTRODUCTION

Profiling based side-channel attack (SCA) is a significant threat to embedded devices. An adversary can recover the secret key with the prior knowledge of the dependency between physical leakages and data being processed. The attack consists of two phases: *profiling* and *attacking*. In the former phase, an adversary collects physical leakages such as power consumptions or Electromagnetic Emissions (EM) from a profiling device which is a controlled copy of the target device, and builds templates or models. In the latter phase, the adversary can extract the secret key of the target device with power/EM traces collected from both devices. The most essential assumption is that the profiling device should be as similar to the target device as possible. Profiling based power attacks mainly include template-based power analysis [1], [2] (denoted as TA), machine learning-based power analysis [3]–[5] (ML-PA) and, recently, deep learning-based ones [6]–[9] (DL-PA).

DL-PA normally does not require pre-processing compared with ML-PA and TA, therefore a number of recent works are devoted to this area. Maghrebi et al [6] first proposed to use deep learning technology to build the classifier and recovered the secret key for an AES algorithm running on a microcontroller. Cagli et al [7] further explored the applicability of the convolutional neural network (CNN) to break the jitter based countermeasures. They combined CNN with the data augmentation technique in the case of over-fitting and trace misalignment, showing that it can effectively improve the efficiency of power analysis. Benadjila et al [8] implemented DL-PA on a masked AES with no prior knowledge of the mask value. Kim et

al [9] showed that even with some artificial noise, CNN can recover the key using only one trace.

Despite of the positive results from profiling based SCA, the concern about *device variations* has not been raised with enough attentions. In many laboratory experiments, power consumptions in both profiling and attacking phases are collected from the same device in one acquisition campaign. However, in practical scenarios, even though data are collected from the same device, they might still differ due to different ambient temperatures, environmental noises or measurement variations. It makes the key recovery still difficult on a single device, not to mention that on two physical copies.

The concept of *cross-device attack* refers to the scenario of using power traces collected from one device to attack the other one. However, in our opinion, the degree of (dis)similarity between profiling and target devices is still very vague. In this paper, we categorize such variation into four types. (1) **Same devices**: The traces used in both profiling and attacking phases are collected from the exact same device. The only difference is the key variation. (2) **Identical devices**: The profiling and target device are two physical copies of the same chip model. All the design and configurations are identical. This scenario is very common in laboratory experiments. Normally, the two identical copies are either purchased as off-the-shelf ones from the same batch, or built from the same circuit design. The slight variation may come from different custom printed circuit boards (PCB) [10] or different RAM size [11]. (3) **Homogeneous devices**: This case actually extends the dissimilarity of crossed devices quite a lot at the chip level. Both devices have chips from the same manufacturer but with different models and structures. (4) **Heterogeneous devices**: This is the ultimate problem in practice which is believed to be the most difficult to solve. The core chips of the two devices to be compared come from different manufacturers, which naturally differ in all aspects, such as models, instruction set architectures (ISA), power dissipations, etc. These four types are newly termed in this paper, which are used to clarify the level of the so-called “crossed devices” and to describe the extent of another attacking capability of DL-PA.

Some researchers became aware of such differences and conducted TA on two identical devices. Renauld et al [12] confirmed the existence of variation between profiling and target device so that TA may not succeed in practice. Lomné et al [13] conducted TA and linear regression analysis on several copies of microcontrollers with different CMOS technologies (350, 130 and 90nm). But they actually performed the attack only on different copies of the same model (i.e., *identical devices* in our terms), while the attack across devices of different models has not been fulfilled yet. Montminy et al [11] analyzed different devices with different RAM sizes in the same series (i.e., *identical devices*). They utilized TA with a specific preprocessing method (zero-mean unit-variance normalization) to eliminate the distribution difference between traces collected from

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3. 《 On Efficient and Secure Code-based Masking: A Pragmatic Evaluation 》

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On Efficient and Secure Code-based Masking: A Pragmatic Evaluation

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Keywords: Side-Channel Analysis, Code-based Masking, Efficient Implementation, Security Evaluation, Inner Product Masking, Redundant Sharing

Abstract

Code-based masking is a highly generalized type of masking schemes, which can be instantiated into specific cases by assigning different encoders. It captivates by its side-channel resistance against higher-order attacks and the potential to withstand fault injection attacks. However, similar to other algebraically-involved masking schemes, code-based masking is also burdened with expensive computational overhead. To mitigate such cost and make it efficient, we contribute to several improvements to the original scheme proposed by Wang et al. in TCHES 2020. Specifically, we devise a computationally friendly encoder and accordingly accelerate masked gadgets to leverage efficient implementations. In addition, we highlight that the amortization technique introduced by Wang et al. does not always lead to efficient implementations as expected, but actually decreases the efficiency in some cases. From the perspective of practical security, we carry out an extensive evaluation of the concrete security of code-based masking in the real world.

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On Efficient and Secure Code-based Masking: A Pragmatic Evaluation

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Abstract. Code-based masking is a highly generalized type of masking schemes, which can be instantiated into specific cases by assigning different encoders. It captivates by its side-channel resistance against higher-order attacks and the potential to withstand fault injection attacks. However, similar to other algebraically-involved masking schemes, code-based masking is also burdened with expensive computational overhead. To mitigate such cost and make it efficient, we contribute to several improvements to the original scheme proposed by Wang et al. in TCHES 2020. Specifically, we devise a computationally friendly encoder and accordingly accelerate masked gadgets to leverage efficient implementations. In addition, we highlight that the amortization technique introduced by Wang et al. does not always lead to efficient implementations as expected, but actually decreases the efficiency in some cases. From the perspective of practical security, we carry out an extensive evaluation of the concrete security of code-based masking in the real world. On one hand, we select three representative variations of code-based masking as targets for an extensive evaluation. On the other hand, we aim at security assessment of both encoding and computations to investigate whether the state-of-the-art computational framework for code-based masking reaches the security of the corresponding encoding. By leveraging both leakage assessment tool and side-channel attacks, we verify the existence of “security order amplification” in practice and validate the reliability of the leakage quantification method proposed by Cheng et al. in TCHES 2021. In addition, we also study the security decrease caused by the “cost amortization” technique and redundancy of code-based masking. We identify a security bottleneck in the gadgets computations which limits the whole masked implementation. To the best of our knowledge, this is the first time that allows us to narrow down the gap between the theoretical security order under the probing model (sometimes with simulation experiments) and the concrete side-channel security level of protected implementations by code-based masking in practice.

Keywords: Side-Channel Analysis · Code-based Masking · Efficient Implementation · Security Evaluation · Inner Product Masking · Redundant Sharing

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4. 《Design and Evaluation of Fluctuating Power Logic to Mitigate Power Analysis at the Cell Level》

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Design and Evaluation of Fluctuating Power Logic to Mitigate Power Analysis at the Cell Level

Publisher: IEEE

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Fan Zhang ; Bolin Yang ; Bojie Yang ; Yiran Zhang ; Xuanle Ren ; Shivam Bhasin ; Kui Ren All Authors

Abstract

Authors

Keywords

Abstract:

In this paper, we design a novel cell-level power-analysis countermeasure, named fluctuating power logic (FPL), which diffuses the correlation between the real power consumption and the fixed data transitions by employing a cascade voltage logic. The countermeasure further acts as a cell-level VDD randomizer, making it a strong candidate for implementing algorithmic countermeasure and exploiting its noise generation capabilities. This proposed scheme is illustrated by a standard flip-flop. HSPICE based simulation results show that the modified flip-flop is resistant against power analysis at the cost of doubled power dissipation. Two illustrative case studies of PRESENT and AES substitutions have been explored. Furthermore, our proposal can be combined with other cell-level countermeasures against power analysis, such as wave dynamic differential logic. The resistance is evaluated by the correlation power analysis and the test vector leakage assessment. The new logic outperforms other counterparts in consideration of both security and cost, which renders it as a practical solution for resource-constrained systems. The proposed cell-level countermeasure can naturally mitigate other side-channel analysis such as electromagnetic analysis.

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Design and Evaluation of Fluctuating Power Logic to Mitigate Power Analysis at the Cell Level

Fan Zhang¹, Bolin Yang, Bojie Yang, Yiran Zhang, Xuanle Ren, Shivam Bhasin, Kui Ren

Abstract—In this paper, we design a novel cell-level power-analysis countermeasure, named *fluctuating power logic (FPL)*, which diffuses the correlation between the real power consumption and the fixed data transitions by employing a *cascade voltage logic*. The countermeasure further acts as a cell-level V_{DD} randomizer, making it a strong candidate for implementing algorithmic countermeasure and exploiting its noise generation capabilities. This proposed scheme is illustrated by a standard flip-flop. HSPICE based simulation results show that the modified flip-flop is resistant against power analysis at the cost of doubled power dissipation. Two illustrative case studies of PRESENT and AES substitutions have been explored. Furthermore, our proposal can be combined with other cell-level countermeasures against power analysis, such as wave dynamic differential logic. The resistance is evaluated by the correlation power analysis and the test vector leakage assessment. The new logic outperforms other counterparts in consideration of both security and cost, which renders it as a practical solution for resource-constrained systems. The proposed cell-level countermeasure can naturally mitigate other side-channel analysis such as electromagnetic analysis.

Index Terms—fluctuating power logic, side channel analysis, dual-rail pre-charge, wave dynamic differential logic, flip-flop.

I. INTRODUCTION

Cryptographic technologies and secure implementations of cryptographic algorithms have been developed and widely used in electronic banking, virtual private networks, online payment and so on. The Internet of Things (IoT) and Cyber-Physical Systems (CPS) are the primary contributors to the revolution of the way that humans and intelligent systems interact. With the rapid development, the security and privacy of sensitive information handled by such systems are emerging as a serious concern [1], [2]. While these technologies offer a lot of new possibilities, the increasing complexities of hardware and software also increase the vulnerability to security attacks.

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One severe security vulnerability of embedded devices is side-channel analysis (SCA) [3], which aims to extract the secrets using unintentional physical leakages from underlying logic elements. Power analysis (PA) is one of the most classical attack approaches, which includes simple power analysis (SPA), differential power analysis (DPA) [3], correlation power analysis (CPA) [4] and more. These attacks exploit the fact that the power dissipation of the implemented cryptographic modules inherently correlates to their switching operations.

The CMOS is the basic building block of modern circuits. Its dynamic power consumption is caused by charging and discharging the capacitive loads when internal and output nodes perform transitions, which accounts for a large portion of the total power in the circuits [5], [6]. Variant data transitions ($0 \rightarrow 1$ and $1 \rightarrow 0$) consume more distinguishable power than invariant ones ($0 \rightarrow 0$ and $1 \rightarrow 1$). Such power activity is consequently associated with the key-dependent variables being processed by the algorithm in a non-invasive manner [7]. The dependency can be approximately described by a power model such as the Hamming weight (HW) or the Hamming distance (HD) model.

Since the power analysis brings severe security threats to modern circuits, effective SCA countermeasures are in high demand. The two mainstream SCA countermeasures are hiding and masking [8]. Both techniques make it difficult to deduce the key-dependent data from observable power dissipations, specifically, in two distinct fashions as shown in Fig. 1. Examples of hiding countermeasures are noise generators [9], clock randomizers [9] and dual-rail precharge logics such as the sense-amplifier based logic (SABL) [10] at the transistor level, and the wave dynamic differential logic (WDDL) [10] or the balanced cell-based dual-rail Logic (BCDL) [11] at the gate level. The first two countermeasures have limited impacts on the attacking difficulty and can be eventually exploited. BCDL eliminates the early propagation effect (EPE) from WDDL by synchronizing pairs of inputs in a compound N-input gate. However, the logic styles such as SABL, WDDL and BCDL require a very strict complementary capacitive balance, making them quite difficult to implement in practice. Further, some researchers have explored quasi-delay insensitive (QDI) asynchronous circuits coupled with 1-to-N encoding to achieve side-channel resistance [12].

Masking [13] is an algorithm-level countermeasure which attempts to *de-correlate the dependency between the actual data and the power model*. Even with enough knowledge of power models, the adversary still cannot extract the secret because the key-dependent variables are actually masked with random and unknown values. Masking is one of the most

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hPRESS: A Hardware-enhanced Proxy Re-encryption Scheme using Secure Enclave

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Fan Zhang ; Ziyuan Liang ; Cong Zuo ; Jun Shao ; Jianting Ning ; Jun Sun ; Joseph K. Liu ; Yibao Bao All Authors

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Abstract

Authors

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Abstract:

Proxy re-encryption (PRE) allows a proxy to transform one ciphertext to another under different encryption keys while keeping the underlying plaintext secret. Because of the ciphertext transformability of PRE, there are many potential private communicating applications of this feature. However, existing PRE schemes are not as full-fledged as expected. The lack of necessary features makes them hard to apply in real-world scenarios. So far, there does not exist a unidirectional multi-hop PRE scheme with constant decryption efficiency and constant ciphertext size without extensions. Impractical performance and weak scalability also hinder PRE from most real-world applications. In this work, we present a new PRE scheme with secure hardware enclave named hPRESS (hardware-enhanced proxy re-encryption scheme using secure enclave). To the best of our knowledge, hPRESS is the first unidirectional multi-hop PRE scheme which achieves both constant decryption efficiency and constant ciphertext size without extensions. A detailed security analysis demonstrates that our proposal is CCA secure based on the security of the underlying encryption schemes and the secure enclave. We also implement a prototype based on Intel SGX, one of the most popular secure enclave techniques in recent years, and evaluate its performance. The experimental results show that, compared with previous PRE schemes, our hPRESS is almost one order of magnitude faster in terms of the decryption and transformation.

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hPRESS: A Hardware-enhanced Proxy Re-encryption Scheme using Secure Enclave

Fan Zhang, Member, IEEE, Ziyuan Liang, Cong Zuo,
Jun Shao, Jianting Ning, Jun Sun, Joseph K. Liu, and Yibao Bao

Abstract—Proxy re-encryption (PRE) allows a proxy to transform one ciphertext to another under different encryption keys while keeping the underlying plaintext secret. Because of the ciphertext transformability of PRE, there are many potential private communicating applications of this feature. However, existing PRE schemes are not as full-fledged as expected. The lack of necessary features makes them hard to apply in real-world scenarios. So far, there does not exist a unidirectional multi-hop PRE scheme with constant decryption efficiency and constant ciphertext size without extensions. Impractical performance and weak scalability also hinder PRE from most real-world applications. In this work, we present a new PRE scheme with secure hardware enclave named hPRESS (hardware-enhanced proxy re-encryption scheme using secure enclave). To the best of our knowledge, hPRESS is the first unidirectional multi-hop PRE scheme which achieves both constant decryption efficiency and constant ciphertext size without extensions. A detailed security analysis demonstrates that our proposal is CCA secure based on the security of the underlying encryption schemes and the secure enclave. We also implement a prototype based on Intel SGX, one of the most popular secure enclave techniques in recent years, and evaluate its performance. The experimental results show that, compared with previous PRE schemes, our hPRESS is almost one order of magnitude faster in terms of the decryption and transformation.

Index Terms—Proxy Re-encryption, Secure Enclave, Trusted Computing, Intel SGX

I. INTRODUCTION

PROXY re-encryption (PRE) is a powerful cryptographic primitive, which allows a semi-trusted proxy to transform a ciphertext of user A (called delegator) to another ciphertext of user B (called delegatee). The two plaintexts of the original

and transformed ciphertexts are the same, and they are never revealed (even to the proxy) during the transformation. If the transformed ciphertext can be further transformed, we call the underlying scheme as multi-hop PRE; otherwise, single-hop PRE. Moreover, if the transformation can happen in two directions between the delegator and the delegatee, we call the underlying scheme as bidirectional PRE; if it can only happen from the delegator to the delegatee, we call it as unidirectional PRE.

A. Motivation

Most existing PRE schemes are far from real-world applications, which is our main motivation to propose a new practical PRE scheme. To illustrate the deficiencies of current PRE schemes, we consider several potential use cases of PRE, analyze features in need of a practical PRE scheme.

Digital right management (DRM) provides licensing restrictions to digital content for authenticated users. It takes advantage of the online content distribution while preventing illegal redistribution of the content. Taban et al. [1] first proposed a secure and interoperable DRM architecture with PRE and re-signature, which is similar with PRE. They introduced an intermediate module called the Domain Interoperability Manager (DIM) to deal with the problem of content and license translation across different DRM regimes. DIM operators are regarded as semi-trusted proxies in their design. DIM converts a ciphertext computed under one key to another, without the DIM learning any information about the plaintext message or the secret keys of the content providers or other authenticated parties. Although PRE primitives perfectly fit in the scenario of DRM and DIM, there is still no DRM systems based on PRE in real-world applications. The reason is that existing PRE schemes do not provide all features demanded by a practical DRM system. First, DRM architecture needs a unidirectional PRE scheme. The content providers only allow DIM to re-encrypt from higher-level authenticated parties to lower-level local groups or individual devices, and forbid the inverse transformation. Content providers should not need to trust authenticated users in order for them to be able to forward the license back. The PRE scheme adopted by DRM also needs to be a multi-hop one. In most cases, the digital contents are not directly authenticated from the content providers to the final user devices. There are several proxy nodes in the authentication chain, and the contents are always authenticated from high-level parties to low-level ones. Thus the PRE scheme for a practical DRM system is a unidirectional multi-hop one, and it is also expected to support constant ciphertext

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6. 《Persistent Fault Attack in Practice》

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Persistent Fault Attack in Practice

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Abstract. *Persistence fault analysis (PFA)* is a novel fault analysis technique proposed in CHES 2018 and demonstrated with rowhammer-based fault injections. However, whether such analysis can be applied to traditional fault attack scenario, together with its difficulty in practice, has not been carefully investigated. For the first time, a persistent fault attack is conducted on an unprotected AES implemented on ATmega163L microcontroller in this paper. Several critical challenges are solved with our new improvements, including (1) how to decide whether the fault is injected in SBox; (2) how to use the maximum likelihood estimation to pursue the minimum number of ciphertexts; (3) how to utilize the unknown fault in SBox to extract the key. Our experiments show that: to break AES with physical laser injections despite all these challenges, the minimum and average number of required ciphertexts are 926 and 1641, respectively. It is about 38% and 28% reductions of the ciphertexts required in comparison to 1493 and 2273 in previous work where both fault value and location have to be known. Furthermore, our analysis is extended to the PRESENT cipher. By applying the persistent fault analysis to the penultimate round, the full PRESENT key of 80 bits can be recovered. Eventually, an experimental validation is performed to confirm the accuracy of our attack with more insights. This paper solves the challenges in most aspects of practice and also demonstrates the feasibility and universality of PFA on SPN block ciphers.

Keywords: PFA · Fault Injection · SRAM · MLE · AES · PRESENT

1 Introduction

Fault attack (FA) on cryptographic systems is an active attack with two phases. In Phase one, the adversary disturbs the operation of the target device, which is known as *fault injection (FI)*. In Phase two, he analyzes the faulty ciphertexts to recover the key, which is known as *fault analysis*.

Fault injection techniques can include changing the power supply voltage, the external clock, or the temperature, etc, which are low-cost but low-precision. Laser is one of the most popular methods due to its precision. Laser injection was first developed in the field of radiation resistance. The specific manifestation of *Single Event Effect (SEE)* [WM62] is that the incident high-energy particles can generate a large number of ionization charges inside the chip and then generate current pulses, which may lead to changes of storage

7. 《Terminator on SkyNet: A Practical DVFS Attack on DNN Hardware IP for UAV Object Detection》

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RESEARCH-ARTICLE



Terminator on SkyNet: a practical DVFS attack on DNN hardware IP for UAV object detection

Authors: Junge Xu, Bohan Xuan, Anlin Liu, Mo Sun, Fan Zhang, Zeke Wang, Kui Ren [Authors Info & Claims](#)

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ABSTRACT

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Comments



ABSTRACT

With increasing computation of various applications, dynamic voltage and frequency scaling (DVFS) is gradually deployed on FPGAs. However, its reliability and security haven't been sufficiently evaluated. In this paper, we present a practical DVFS fault attack targeting at the SkyNet accelerator IP and successfully destroy the detection accuracy. With no knowledge about the internal accelerator structure, our attack can achieve more than 98% detection accuracy loss under ten vulnerable operating point pairs (OPPs). Meanwhile, we explore the local injection with 1 ms duration and next double the intensity which can achieve more than 50% and 74% average accuracy loss respectively.

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Terminator on SkyNet: A Practical DVFS Attack on DNN Hardware IP for UAV Object Detection

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ABSTRACT

With increasing computation of various applications, dynamic voltage and frequency scaling (DVFS) is gradually deployed on FPGAs. However, its reliability and security haven't been sufficiently evaluated. In this paper, we present a practical DVFS fault attack targeting at the SkyNet accelerator IP and successfully destroy the detection accuracy. With no knowledge about the internal accelerator structure, our attack can achieve more than 98% detection accuracy loss under ten vulnerable operating point pairs (OPPs). Meanwhile, we explore the local injection with 1 ms duration and next double the intensity which can achieve more than 50% and 74% average accuracy loss respectively.

1 INTRODUCTION

Given their high flexibility and energy efficiency, FPGAs have become one of the most promising edge computing platforms and have been widely deployed as hardware accelerators for various neural networks. Therefore, neural network implementation on FPGA has drawn close attentions in recent years. Some researches have shown that FPGA-based hardware accelerator can be a superior solution in terms of both throughput and energy efficiency when convolutional neural networks (CNN) are trained in particular circumstances [10]. Although the implementation of neural networks is achieving better performance, the raising concerns on its robustness and security have not been adequately summarized and fully addressed.

The need for better performance, power- and energy-efficiency drives the development of modern hardware-software energy management technology. DVFS mechanism is one of the most commonly used technologies. In order to maximize the utility of it, hardware and software operate cooperatively at very fine granularities to regulate the frequency and voltage of processors. The author [8] combined the fine-grained DVFS framework with CNN-based object detection accelerator in FPGA. However, considering the complexity of software-hardware interoperability requirements and the pressure of cost and time limit to the market, designers of DVFS may do not fully consider the security of it, which makes it possible to break through the safe defense-line of DVFS with only software methods.

Fault injection attack is an important kind of method to destroy the accuracy of neural networks through modifying electronic circuits' behaviors. Various fault attacks on FPGAs such as glitch disturbance [1], rowhammering [9] and glitch injection [11] have

been demonstrated previously. Most of recent fault attacks targeting at neural networks focus on model weight manipulations [6], which will leave traces in memory no matter how much data has been manipulated. Such persistent fault injection into the networks can not avoid detection and can be bypassed by parameter reloading. Inconspicuous fault attack targeting at neural networks' accelerators has been paid close attentions on recently and has been proposed in several papers, such as glitch injection [11]. However, there is no specific work on DVFS fault attacks on FPGAs.

In this paper, we summarize the research gap in related fields and proposed the first stealthy and non-invasive fault attack targeting at neural network hardware IP based on DVFS mechanism. In specific work, we combine the DVFS framework with the SkyNet accelerator. And the DVFS framework consists of DVS and DFS modules implemented by some hardware components which mapped to corresponding FPGA drivers. Therefore, hardware fault can be injected through software interfaces to corresponding FPGA drivers. Through these software interfaces, we can dynamically control the voltage and frequency out of specification and thus induce abnormal behaviors of electronic circuits. When the voltage/frequency pair is in different value areas of the two-dimensional coordinates, the induced fault will cause three different results: silent data corruptions (SDCs), accelerator hangs, or system crashes. SDCs are the basic of our inconspicuous attack while an accelerator hang or system crash will result in function disruption and arouse vigilance. Our attack, like a terminator on the detection accuracy of SkyNet, aims at inducing temporal faults into the intermediate results of the network which will propagate to the inference stage and lead to accuracy loss.

The main contributions of our work are summarized as follows:

- We present the first DVFS fault attack on FPGA, which is stealthy and does not require any physical access to hardware.
- We design the architecture of our evaluation system with the MMCM module and the PMBus compliant power supply system which consists of a DVFS framework and the SkyNet accelerator. And we show how to induce SDC fault into the accelerator.
- We evaluate our attack on the DAC-SDC2019 champion SkyNet which aims at real-world UVA single object detection. And we investigate both global and local fault injection. For global fault injection, our attack can achieve more than 98% detection accuracy loss. For local fault injection, we can achieve more than 74% average accuracy loss with 95.45% fault probabilities.

The remaining part of this paper will be organized as follows. In Section 2, we talk about some previous related works on DVFS methodology, neural network accelerator and fault attack. In Section 3, we describe our threat model, the violation of timing constraint

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8. 《A Systematic Evaluation of Wavelet-Based Attack Framework on Random Delay Countermeasures》

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Abstract

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IV. Wavelet-Based Attack Framework Against Random Delay Countermeasures

V. Evaluation of Proposed Attack Framework

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Abstract:

Random delay countermeasure is a commonly used defense against side-channel attacks, which brings certain interference and disturbance to those calculation sequences in the time domain. Data alignment and frequency attack are considered as typical techniques to counteract the random delay countermeasure. However, these attacks have limitations from the perspectives of both efficiency and performance. In comparison, facing those delays, wavelet analysis is considered as a more efficient technique due to its detailed and comprehensive interpretation of a signal. This paper applies different wavelet techniques to three attack components: noise reduction, trace alignment and key extraction. For the first time, the unified wavelet-based attack framework against random delays is proposed where wavelet analysis is fully applied in the entire attack life cycle. In particular, a novel method of trace alignment at the wavelet level is proposed in this framework, which is based on wavelet pattern detection to synchronize the misaligned power traces. Most importantly, the overall wavelet-based attack framework is systematically evaluated over three random delay strategies, after the respective contribution of each component is investigated through a series of comparative experiments. Experimental results show that the performance of the wavelet-based attack framework is significantly improved compared to standard attack procedures and frequency ones, which can be regarded as a unified and effective solution to conquer random delay countermeasures.

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I. Introduction

Side channel attack (SCA) tends to extract the secret keys using physical leakages such as power, electromagnetic emissions, time, and more. Among them, power analysis [1] is one of the most powerful means which exploits the power consumption leaked through cryptographic devices. The instantaneous power consumption is closely related to the executed operations and processed intermediate data. The secret keys could be extracted by means of analyzing the power features. Differential power analysis (DPA) [2] and Correlation Power Analysis (CPA) [3] are two foremost and fundamental power analysis methods to recover the keys based on statistical methodology. To exploit the dependence of the leaked power characteristics and executed operations as well as the processed data, the target parts of those power measurements have to be aligned to the same position. In order to mitigate SCAs, different countermeasures

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A Systematic Evaluation of Wavelet-Based Attack Framework on Random Delay Countermeasures

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Abstract—Random delay countermeasure is a commonly used defense against side-channel attacks, which brings certain interference and disturbance to those calculation sequences in the time domain. Data alignment and frequency attack are considered as typical techniques to counteract the random delay countermeasure. However, these attacks have limitations from the perspectives of both efficiency and performance. In comparison, facing those delays, wavelet analysis is considered as a more efficient technique due to its detailed and comprehensive interpretation of a signal. This paper applies different wavelet techniques to three attack components: noise reduction, trace alignment and key extraction. For the first time, the unified wavelet-based attack framework against random delays is proposed where wavelet analysis is fully applied in the entire attack life cycle. In particular, a novel method of trace alignment at the wavelet level is proposed in this framework, which is based on wavelet pattern detection to synchronize the misaligned power traces. Most importantly, the overall wavelet-based attack framework is systematically evaluated over three random delay strategies, after the respective contribution of each component is investigated through a series of comparative experiments. Experimental results show that the performance of the wavelet-based attack framework is significantly improved compared to standard attack procedures and frequency ones, which can be regarded as a unified and effective solution to conquer random delay countermeasures.

Index Terms—Random delay countermeasures, side-channel analysis, wavelet framework, performance evaluation.

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I. INTRODUCTION

SIDE channel attack (SCA) tends to extract the secret keys using physical leakages such as power, electromagnetic emissions, time, and more. Among them, *power analysis* [1] is one of the most powerful means which exploits the power consumption leaked through cryptographic devices. The instantaneous power consumption is closely related to the executed operations and processed intermediate data, so the secret keys could be extracted by means of analyzing the power features. Differential power analysis (DPA) and correlation power analysis (CPA) [2] are two foremost and fundamental power analysis methods to recover the keys based on statistical methodology. To exploit the dependence of the leaked power characteristics and executed operations as well as the processed data, the target parts of those power measurements have to be aligned to the same position. In order to mitigate SCAs, different countermeasures are proposed to increase the difficulty for adversaries.

The concept of *random delay countermeasure* against SCA was proposed in [3]. By making power traces out of synchronization, random delays reduce the correlation of the consumed power and executed operations along with processed data. Compared with random masking, it improves the attack complexity as a type of hiding manner. The reason why random delay countermeasure is still acknowledged and widely used is that its implementation strategy is quite flexible and easy to adopt. In a nutshell, strategies to implement random delay countermeasures can be divided into three categories.

From the aspect of measurements, it is quite normal that the power traces are triggered and then collected in an oscilloscope controlled by a host computer. If the trigger signal arrives at a random interval, the starting moment of encryptions varies every time, which is called *random propagation delay*.

From the aspect of the algorithm itself, the delays are often realized through dummy operations such as *NOP* instructions, which is called *random delays insertion*. At first, a gate-level random delay insertion against DPA was proposed in [4] where the instantaneous power consumption and total charge quantity are randomized. Besides the hardware solution, Tunstall et al. proposed a different method of random delays in embedded software, which increases the desynchronization compared to those whose lengths are uniformly distributed with less time consumption [5]. As an improvement of Benoit and Tunstall's work, Coron et al. proposed effective countermeasures of random delays in CHES 2009 and 2010 [6], [7], which are also implemented at the software level.



9. 《A Lightweight Detection Algorithm For Collision-Optimized Divide-and-Conquer Attacks》

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A Lightweight Detection Algorithm For Collision-Optimized Divide-and-Conquer Attacks

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Changhai Ou ; Siew Kei Lam ; Chengju Zhou ; Guiyuan Jiang ; Fan Zhang All Authors

17 Full Text Views

Abstract

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Keywords

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Abstract:

By introducing collision information into divide-and-conquer attacks, several existing works transform the original candidate space, which may be too large to enumerate, into a significantly smaller collision space, making key recovery possible. However, the use of inefficient collision detection algorithms and fault tolerance mechanisms make them time-consuming and their success rate low. Moreover, they may still leave very huge chain spaces that makes it difficult for key recovery. In this paper, we exploit Correlation-enhanced Collision Attack (CCA) to optimize Template Attack (TA), and propose a Lightweight Collision Detection (LCD) algorithm. The proposed method exploits a jump detection mechanism to efficiently reduce the repetitive collision detections on chains with the same prefix sub-chains. We then introduce guessing theory to reorder the collision detection of the sub-keys according to their guessing lengths, and provide us with an evaluation tool. Finally, we design a highly efficient fault tolerance mechanism for our LCD to allow flexible thresholds adjustment, and further optimize sieving mechanism to efficiently extract the best chains with the largest number of collisions. Experimental results fully demonstrate LCD's superiority.

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A Lightweight Detection Algorithm For Collision-Optimized Divide-and-Conquer Attacks

Changhai Ou, Siew-Kei Lam, Chengju Zhou, Guiyuan Jiang and Fan Zhang

Abstract—By introducing collision information into divide-and-conquer attacks, several existing works transform the original candidate space, which may be too large to enumerate, into a significantly smaller collision space, thus making key recovery possible. However, the inefficient collision detection algorithms and fault tolerance mechanisms make them time-consuming and their success rate low. Moreover, they may still leave very huge chain spaces that makes it difficult for key recovery. In this paper, we exploit collision attack to optimize Template Attack (TA), and propose a Lightweight Collision Detection (LCD) algorithm. The proposed method exploits a jump detection mechanism to efficiently reduce the repetitive collision detections on chains with the same prefix sub-chains. We then introduce guessing theory to reorder the collision detection of the sub-keys according to their guessing lengths, and provide us with an evaluation tool. Finally, we design a highly efficient fault tolerance mechanism for our LCD to allow flexible thresholds adjustment, and further optimize sieving mechanism to efficiently extract the best chains with the largest number of collisions. Experimental results fully demonstrate LCD's superiority.

Index Terms—collision detection, LCD, collision attack, template attack, fault tolerance, key enumeration, side-channel attack

1 INTRODUCTION

SECRET information will leak through side-channels such as timing [1], power consumption [2], electromagnetic [3], cache patterns [4] and acoustic [5] when cryptographic algorithms are implemented on devices. Side-channel attacks (SCAs) can be launched in two ways: divide-and-conquer and analytical. The former like Correlation Power Analysis (CPA) [6] and Template Attack (TA) [7], is performed on each sub-key like AES-128 and conquer them one by one. The latter like collision attacks [8], [21], conquers the key by solving a system of equations. It exploits more information compared to the divide-and-conquer manner, but is more complex. Key enumeration strategies [12], [20], which enumerate the full-key candidates from the most possible one to the least possible one, are exploited to optimize the key recovery if the divide-and-conquer attacks do not rank the sub-keys as their best candidates. However, they are limited by the computing power of the attacker, and can only be exploited in scenarios where cryptographic implementations are “practically insecure” (for which the leakage allows for key enumeration).

In this paper, our goal is neither to improve the collision attack like [8], [10], [26], nor to enhance the key enumeration schemes. Instead, we aim to exploit collision information from collision attacks like Correlation-enhanced

Collision Attacks (CCA) [14] to optimize the divide-and-conquer attacks like TA, and transform their huge candidate space to collision chain space. In this case, the candidates dissatisfying the collision conditions are discarded, thus key recovery can still be feasibly applied to the remaining chain space that is much smaller than the one provided by divide-and-conquer attack (see experiments in Section 3). We called these combined collision attacks, Collision-Optimized Divided-and-Conquer Attacks (CODCAs) in [16]. They are independent of specific distinguishers used, and their collision detection algorithms play a very important role in performance. Related works will be described in the next subsection before introducing our contributions.

1.1 Related Works

The idea of exploiting collision information to optimize the divide-and-conquer attacks was first given in [3]. For example, the divide-and-conquer attack TA can be optimized by the collision attack like CCA. The combined attack exploits key-related information from the outputs of two attacks, and aims to recover the full key from them under given thresholds. This means that only a very small part of candidates within thresholds are considered. The proposed CODCAs in this paper can exploit collision information to further reduce the candidate space. Let κ_j denote the j^{th} sub-key, and τ_κ denote the threshold, which means only the τ_κ best candidates of each sub-key will be considered. Similarly, we also set a threshold τ_d for CCA, which means only the τ_d best candidates for each XOR value $\delta_{j_1, j_2} = \kappa_{j_1} \oplus \kappa_{j_2}$ of two sub-keys κ_{j_1} and κ_{j_2} are considered. A collision happens if a pair of candidates of two sub-keys and their corresponding δ candidate are within thresholds simultaneously, and several collisions form a chain. For example, a chain consisting of 3 sub-keys κ_1 , κ_2 and κ_3 is formed if a candidate of κ_2 simultaneously collides with a candidate of

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10. 《Passive Attacks Against Searchable Encryption》

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Passive Attacks Against Searchable Encryption

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Jianting Ning | Jia Xu; Kaitai Liang; Fan Zhang; Ee-Chien Chang | All Authors

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Abstract

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II. Preliminaries and Problem Formulation

III. Overview of Assumptions

IV. Passive Attack Under Type I Assumption

V. Enhanced Passive Attack Under Type II Assumption

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Abstract:

Searchable encryption (SE) provides a privacy-preserving mechanism for data users to search over encrypted data stored on a remote server. Researchers have designed a number of SE schemes with high efficiency yet allowing some degree of leakage profile to the remote server. The leakage, however, should be further measured to allow us to understand what types of attacks an SE scheme would encounter. This paper considers passive attacks that make inferences based on prior knowledge and observations on queries issued by users. This is in contrast to previously studied active attacks that adaptively inject files and queries. We consider several assumptions on the types or prior knowledge the attacker possessed and propose a few passive attacks. In particular, under the “full-fledged” assumption, the keyword recovery rate of our attack is optimal in the sense that it is equal to the theoretical upper bound. We further present several enhanced attacks under other weaker assumptions on various levels of the prior knowledge that the attacker can obtain, in which the keyword recovery rates are optimal or nearly optimal (i.e., approaching the theoretical upper bound). In addition, we provide extensive experiments to show the “power” of our passive attacks. This paper highlights the importance of minimizing the prior knowledge of a server and the leakage of search queries. It also shows that simply distorting the frequency of the keyword to hold against our passive attacks may not scale well.

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I. Introduction

To date the advanced cloud-based technologies provide flexible data outsourcing service for the Internet users. The service, however, may incur the concern on the leakage of personal sensitive data, due to the fact that the data is out of data owner’s premises. To guarantee the confidentiality of the data, one of the choices is using encryption techniques. Although protecting personal data in a cloud server, traditional encryption mechanisms limit further operations over encrypted data. To address this problem, searchable encryption (SE) [28] has been introduced to allow a valid user to issue a search query, so that the server can

Authors

Figures

References

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Passive Attacks Against Searchable Encryption

Jianting Ning[✉], Jia Xu, Kaitai Liang, *Member, IEEE*, Fan Zhang[✉], *Member, IEEE*, and Ee-Chien Chang

Abstract—Searchable encryption (SE) provides a privacy-preserving mechanism for data users to search over encrypted data stored on a remote server. Researchers have designed a number of SE schemes with high efficiency yet allowing some degree of leakage profile to the remote server. The leakage, however, should be further measured to allow us to understand what types of attacks an SE scheme would encounter. This paper considers passive attacks that make inferences based on prior knowledge and observations on queries issued by users. This is in contrast to previously studied active attacks that adaptively inject files and queries. We consider several assumptions on the types or prior knowledge the attacker possessed and propose a few passive attacks. In particular, under the “full-fledged” assumption, the keyword recovery rate of our attack is optimal in the sense that it is equal to the theoretical upper bound. We further present several enhanced attacks under other weaker assumptions on various levels of the prior knowledge that the attacker can obtain, in which the keyword recovery rates are optimal or nearly optimal (i.e., approaching the theoretical upper bound). In addition, we provide extensive experiments to show the “power” of our passive attacks. This paper highlights the importance of minimizing the prior knowledge of a server and the leakage of search queries. It also shows that simply distorting the frequency of the keyword to hold against our passive attacks may not scale well.

Index Terms—Searchable symmetric encryption, passive attacks, search query privacy, leakage of file-access pattern.

I. INTRODUCTION

TO DATE the advanced cloud-based technologies provide flexible data outsourcing service for the Internet users. The service, however, may incur the concern on the leakage of personal sensitive data, due to the fact that the data is out of data owner’s premises. To guarantee the confidentiality of the data, one of the choices is using encryption techniques.

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Although protecting personal data from being read by malicious cloud server, traditional encryption mechanisms limit further operations over encrypted data. In particular, if all data are encrypted and remotely stored in a cloud server, how can the server successfully respond a search query over the encryption? To address the problem, *searchable encryption* (SE) [28] has been introduced to allow a valid user to issue a search query, so that the server can return the corresponding encrypted data without knowing “what the query is” and “what the underlying data is”.

In theory, SE schemes with no information leakage can be built on top of several well studied cryptographic primitives such as oblivious RAM, fully-homomorphic encryption, and secure two-party computation [32]. However, such constructions require either heavy computation complexity or large storage cost so that they may be inadequately employed in practice [6], [24]. Researchers have focused on designing searchable symmetric encryption (SSE) schemes with better efficiency (e.g., [10], [28]), which can be deployed in practice, but being subject to some information leak. Cash *et al.* [10] claimed that “in order to provide truly practical SSE solutions one needs to accept a certain level of leakage”. One of the most recent concerns on SSE is the *leakage* of query [9]. One may aim to understand how much information an SSE scheme does truly leak in reality and meanwhile, what the information does mean for the scheme. Informally, there are two main types of information that may be leaked in practice. One is *search pattern* [9], [32] that can be used to determine if two queries correspond to the same keyword(s), and the other is *file-access pattern* [9], [32] referring to the information that is implied by the query results.

Recently, researchers have attempted to launch attacks against SSE schemes (those with some level of information leakage). Islam *et al.* [19] demonstrated that a server can identify the queries of a user from the leakage of search and file-access patterns if the server is with knowledge of all the contents of the user’s files. Cash *et al.* [9] later presented an enhanced attack for larger keyword universe even if the server has less knowledge of the user’s files. Liu *et al.* [23] proposed a new attack on keyword recovery, assuming the attacker has knowledge of the distribution on keyword(s) searched by user. Zhang *et al.* [32] introduced an “active” attack, called file-injection attack, in which the server is able to recover all the keywords from the given queries if it is allowed to actively inject some deliberately chosen files into the encrypted database. In this paper, we ask:

How effective are passive attacks against SSE schemes?

A. Our Contributions

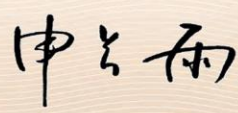
In this paper, we study the prior knowledge and the search query leakage to quantitatively analyze the practical security



二、发明专利授权情况

【系统中已录入的授权专利】-共 5 项

1. 《基于 SIKE 算法的旁路防御方法、装置、电子设备及可读介质》

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发 明 名 称：基于 SIKE 算法的旁路防御方法、装置、电子设备及可读介质		
发 明 人：张帆;杨博麟		
专 利 号：ZL 2020 1 0553818.1		
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2. 《基于 TDC 模块的硬件攻击的防御方法及装置、电子设备》

证书号第 5257150 号





发明专利证书

发明名称：基于 TDC 模块的硬件攻击的防御方法及装置、电子设备

发明人：张帆;王智勇

专利号：ZL 2022 1 0035523.4

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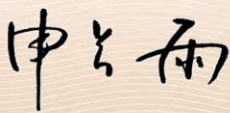
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3. 《一种基于稳态故障的芯片安全性的评估方法》

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4. 《基于线性码掩码和位切片技术的抗旁路攻击的防御方法》

证书号 第 5258219 号





发 明 专 利 证 书

发 明 名 称：基于线性码掩码和位切片技术的抗旁路攻击的防御方法

发 明 人：张帆;吴茜眉

专 利 号：ZL 2022 1 0047078.3

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专 利 权 人：浙江大学

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5. 《一种适用于硬件实现的轻量级认证加密解密实现方法》

证书号第 4930485 号





发明专利证书

发明名称：一种适用于硬件实现的轻量级认证加密解密实现方法

发明人：马孝宇;韩雁;张帆

专利号：ZL 2021 1 0000330.0

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【系统中未录入的补充授权专利】-共 2 项

1. 《一种针对分组密码的代数持久性故障分析方法及装置》

证书号第 5257021 号





发明专利证书

发明名称：一种针对分组密码的代数持久性故障分析方法及装置

发明人：张帆;冯天祥;黎治圻;任奎

专利号：ZL 2022 1 0038526.3

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专利权人：浙江大学

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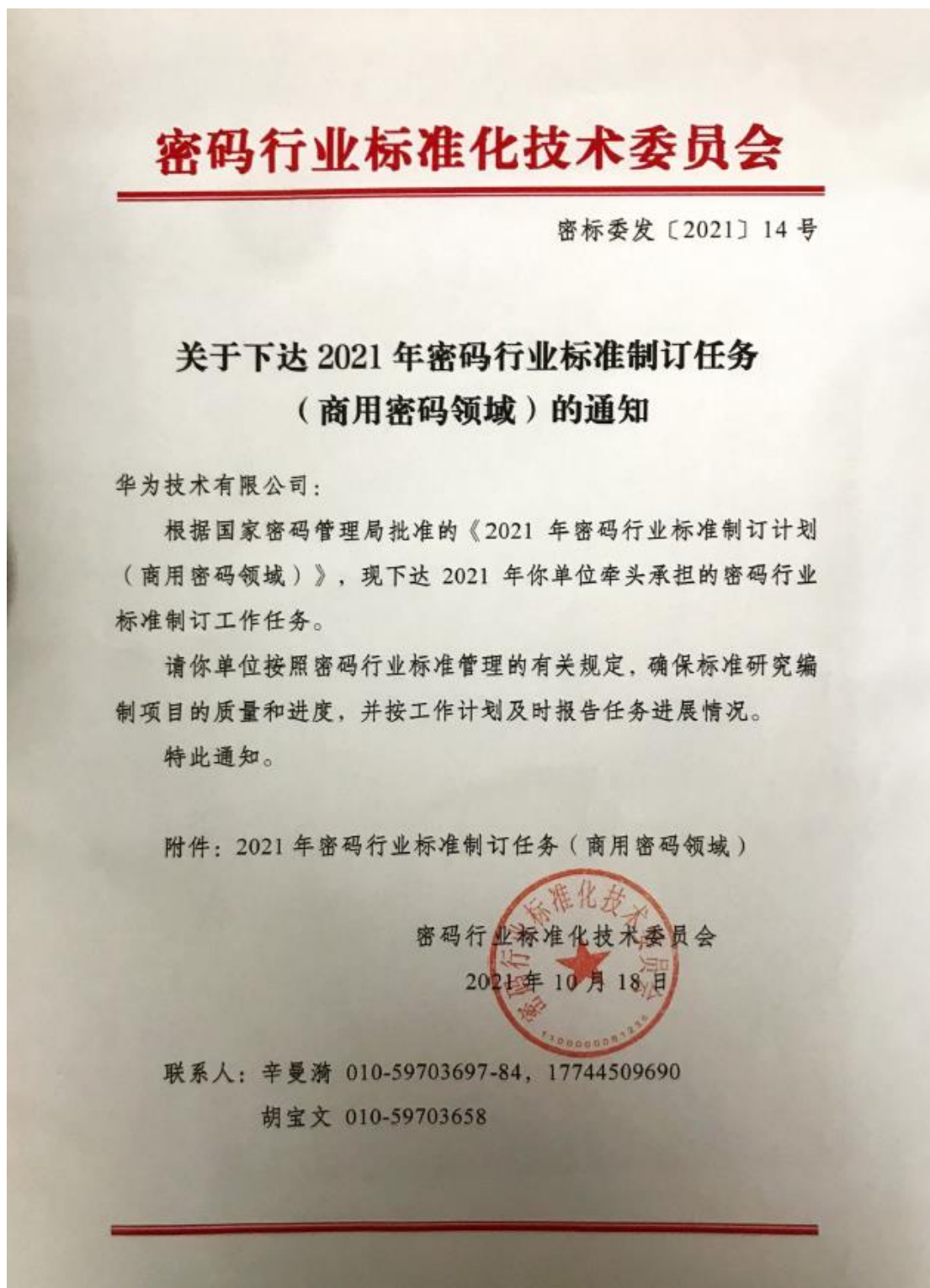
2. 《一种基于 U2F 物理令牌的物联网设备的集中式认证系统》

证书号第 4530993 号		
发明专利证书		
发明名称：一种基于 U2F 物理令牌的物联网设备的集中式认证系统		
发明人：林峰;王超;罗浩;张帆;韩劲松;许文曜;任奎		
专利号：ZL 2020 1 0428749.1		
专利申请日：2020 年 05 月 20 日		
专利权人：浙江大学		
地址：310058 浙江省杭州市西湖区余杭塘路 866 号		
授权公告日：2021 年 07 月 06 日 授权公告号：CN 111585771 B		
国家知识产权局依照中华人民共和国专利法进行审查，决定授予专利权，颁发发明专利证书并在专利登记簿上予以登记。专利权自授权公告之日起生效。专利权期限为二十年，自申请日起算。 专利证书记载专利权登记时的法律状况。专利权的转移、质押、无效、终止、恢复和专利权人的姓名或名称、国籍、地址变更等事项记载在专利登记簿上。		
		
局长 申长雨		2021 年 07 月 06 日
第 1 页 (共 2 页)		
其他事项参见续页		



三、标准制定情况

1. 《汽车智能处理器硬件可信根和安全启动密码应用技术研究》



附件

2021 年密码行业标准制订任务（商用密码领域）

承担单位：华为技术有限公司

序号	项目名称	类型	时间安排	工作组
1	数据库管理系统密码应用技术要求	研究	2022.8 前完成 研究报告	应用工作 一组
2	操作系统密码应用技术要求	研究	2022.8 前完成 研究报告	应用工作 一组
3	汽车智能处理器硬件可信根和安全启动密码应用技术研究	研究	2022.8 完成研 究报告	应用工作 一组
4	车载领域 SoC 密码安全子系统保护轮廓与测评要求研究	研究	2022.8 完成研 究报告	测评工作 一组



标准进展：密标委标准通知已下达，结项时间定在22年9月

应用组：
汽车智能处理器硬件可信根和安全启动密码应用技术研究

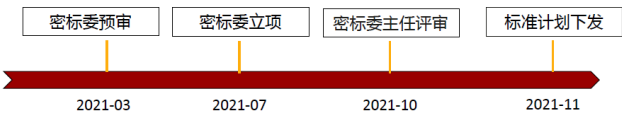
目的：
基于密码技术的信任机制，解决汽车智能处理器从电到进入系统期间的安全可信，防止篡改和替换，实现安全启动

交付件：
研究报告：
《汽车智能处理器硬件可信根和安全启动密码应用技术研究》
附件带《汽车智能处理器硬件可信根和安全启动密码应用指南》框架草案

- 课题组成员单位：排名不分先后
 - 兴唐通信科技
 - 浙江大学
 - 华中科技大学
 - 中科院信工所
 - 武汉大学
 - 深圳纽创信安
 - 大疆创新
 - 电子科技大学
 - 清华大学
 - 成都信息工程大学
 - 北京理工大学
 - 华为技术有限公司
 - 北京数字认证
- 期待更多企业参与！

- 密标委责任专家
 - 夏鲁宁 北京数字认证
 - 袁峰 北京国脉信安

- 秘书处老师
 - 刘中 北京数字认证



**密标委通知原件
3 Huawei Confidential



课题组：参编单位&专家职责，阶段计划

角色安排&专家阵型

角色	单位（主要人员）	职责
牵头	华为（张小虎）	牵头标准撰写、负责评审汇报&结项
参与	兴唐通信科技（梁承志、吕晓晨）	-参与标准撰写、评审
参与	大疆（张翌维）	
参与	北京数字认证（李向峰、李文超、刘晨光）	
参与	武大（涂航）	
参与	清华（贾珂婷）	
参与	北理（王安）	
参与	浙大（张帆）	
参与	信工所（马原）	
参与	OSR（张浩）	
参与	华科（刘振林）	
参与	成都电子科大（肖莹、魏宁）	
参与	成都信息工程大学（吴震、杜之波、王敏）	

5 Huawei Confidential

阶段计划及风险

阶段	输入	输出	完成度	风险
阶段一 (22.02.25)	开工会材料	开工会确定的研究报告目录和任务分工	未开始	时间较紧， 交稿时间<1个月
阶段二 (22.03.15)	任务分工	初稿	未开始	暂无
阶段三 (22.03.25)	初稿	评审意见	未开始	暂无
阶段四 (22.04.15)	刷新后的初稿	评审意见&初稿 ready	未开始	暂无



2. 《车载领域 SoC 密码安全子系统保护轮廓与测评要求研究》

课题组：参编单位&专家职责，阶段计划

角色安排&专家阵型		
角色	单位（主要人员）	职责
牵头	华为（张小虎）	牵头标准撰写、负责评审汇报&结项
参与	商密检测中心（雷银花）	-参与标准撰写、评审 -提供专家信息和引荐
参与	武大（涂航）	
参与	清华（贾珂婷）	
参与	北理（王安）	
参与	浙大（张帆）	
参与	信工所（马原）	
参与	OSR（张浩）	
参与	华科（刘振林）	
参与	成都电子科大（肖望、魏宁）	
参与	成都信息工程大学（吴震、杜之波、王敏）	

阶段计划及风险				
阶段	输入	输出	完成度	风险
阶段一 (22.03.25)	开工会材料	开工会确定的研究报告目录和任务分工	未开始	时间较紧，交稿时间1个月
阶段二 (22.04.20)	任务分工	初稿	未开始	暂无
阶段三 (22.04.30)	初稿	评审意见&初稿 ready	未开始	暂无



四、主要新产品（含新品种）/新装置（装备）情况

1. 《国家重大科研仪器研制项目-面向威胁发现的网络流谱分析仪研制》



申请代码	F0206
接收部门	
收件日期	
接收编号	6222780010



国家自然科学基金
申 请 书

(2022版)

资助类别：国家重大科研仪器研制项目

亚类说明：自由申请

附注说明：

项目名称：面向威胁发现的网络流谱分析仪研制

申请人：郭世泽 电 话：01066305306

依托单位：北京邮电大学

通讯地址：北京市947信箱

邮政编码：100191 单位电话：62282612

电子邮箱：nsfgsz@126.com

填写日期：2022年02月28日

国家自然科学基金委员会



国家自然科学基金资助项目批准通知

(预算制项目)

郭世泽 先生/女士:

根据《国家自然科学基金条例》、相关项目管理办法规定和专家评审意见,国家自然科学基金委员会(以下简称自然科学基金委)决定资助您申请的项目。项目批准号: 62227805, 项目名称: 面向威胁发现的网络流谱分析仪研制, 直接费用: 772.62万元, 项目起止年月: 2023年01月至 2027年12月, 有关项目的评审意见及修改意见附后。

请您尽快登录科学基金网络信息系统(<https://grants.nsfc.gov.cn>), 认真阅读《国家自然科学基金资助项目计划书填报说明》并按要求填写《国家自然科学基金资助项目计划书》(以下简称计划书)。对于有修改意见的项目, 请您按修改意见及时调整计划书相关内容; 如您对修改意见有异议, 须在电子版计划书报送截止日期前向相关科学处提出。

请您将电子版计划书通过科学基金网络信息系统(<https://grants.nsfc.gov.cn>)提交, 由依托单位审核后提交至自然科学基金委。自然科学基金委审核未通过者, 将退回的电子版计划书修改后再行提交; 审核通过者, 打印纸质版计划书(一式两份, 双面打印)并在项目负责人承诺栏签字, 由依托单位科研、财务管理等部门审核、签章并在承诺栏加盖依托单位公章, 且将申请书纸质签字盖章页订在其中一份计划书之后, 一并报送至自然科学基金委项目材料接收工作组。纸质版计划书应当保证与审核通过的电子版计划书内容一致。自然科学基金委将对申请书纸质签字盖章页进行审核, 对存在问题的, 允许依托单位进行一次修改或补齐。

向自然科学基金委提交电子版计划书、报送纸质版计划书并补交申请书纸质签字盖章页截止时间节点如下:

1. 年 月 日16点: 提交电子版计划书的截止时间;
2. 年 月 日16点: 提交修改后电子版计划书的截止时间;
3. 年 月 日: 报送纸质版计划书(一式两份, 其中一份包含申请书纸质签字盖章页)的截止时间。





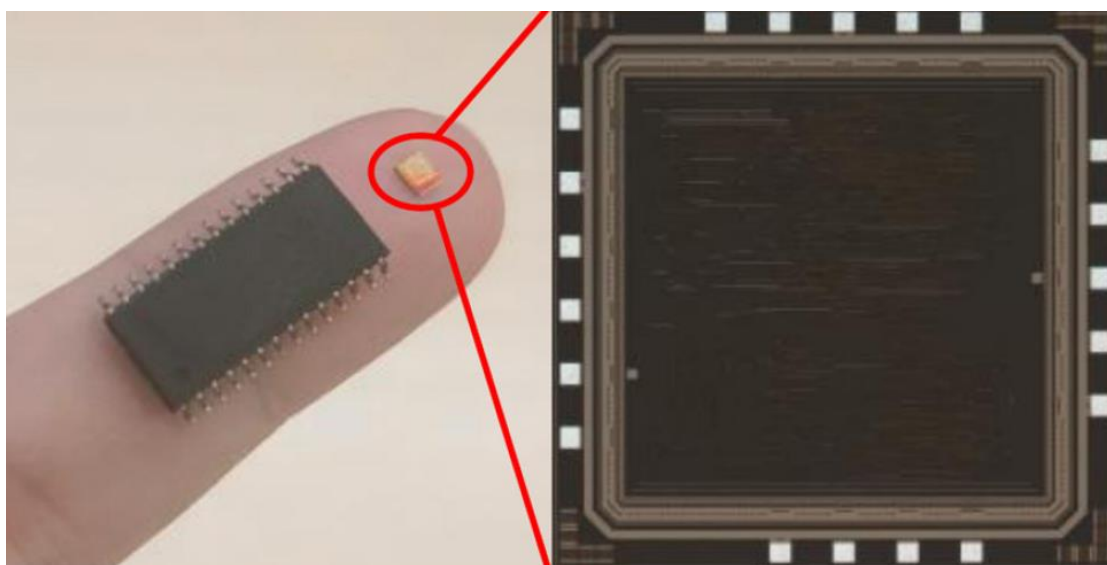
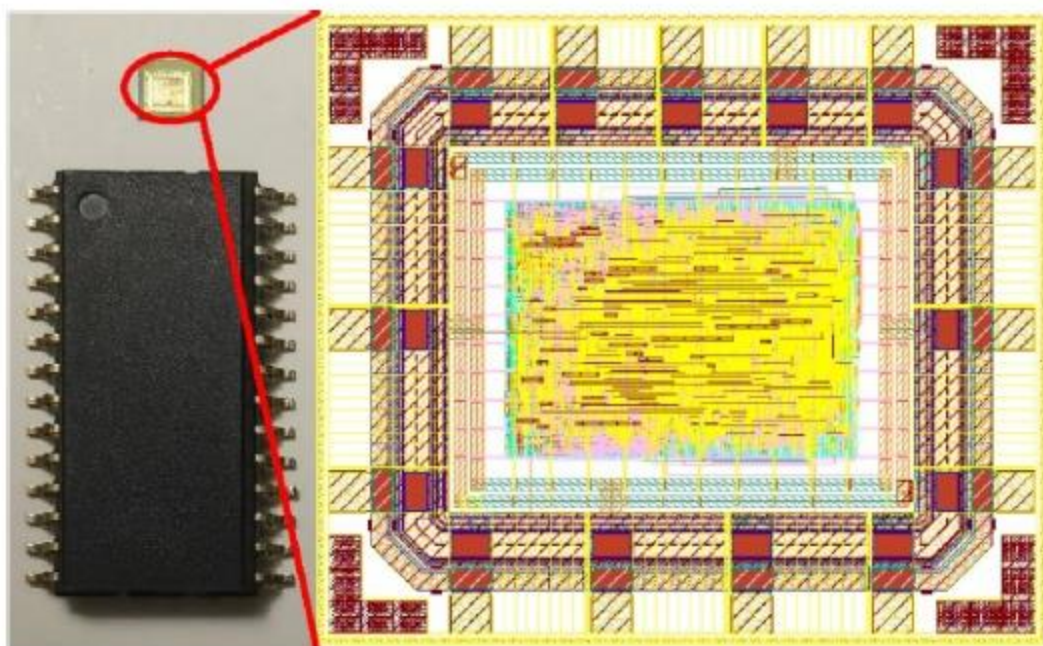
主要参与者 (注: 主要参与者不包括项目申请人)

编号	姓名	出生年月	性别	职 称	学 位	工作单位	项目分工	电话	证件号码
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2	徐洁	1986-12-15	女	讲师	博士	北京邮电大学	项目主研人	010-61198159	3*****4
3	任传伦	1972-06-02	男	研究员	博士	中国电子科技集团公司第三十研究所	项目主研人	010-66306430	3*****2
4	俞赛赛	1982-01-08	男	高级工程师	博士	中国电子科技集团公司第三十研究所	项目主研人	010-66306430	3*****7
5	张帆	1978-10-20	男	教授	博士	浙江大学	项目主研人	0571-87952685	3*****8
6	王伟	1989-02-09	男	无	博士	北京邮电大学	项目主研人	010-61198108	1*****X
7	肖达	1981-06-01	男	讲师	博士	北京邮电大学	项目主研人	010-62283134转0	2*****7
8	满毅	1974-11-01	男	副教授	博士	北京邮电大学	项目主研人	62282751	1*****1
9	黄玮	1980-05-18	男	研究员	博士	北京理工大学	项目主研人	010-68913694	3*****6

总人数	高级	中级	初级	博士后	博士生	硕士生
39	7	2	0	1	11	18



2. 《NIST 认证加密算法 ASCON 芯片-SMIC 180nm 工艺》



Authenticated encryption chip implementation against side-channel attack

Xiaoyu Ma^{1,a}, Fan Zhang^{2,b*} and Yan Han^{1,c}

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²College of Computer Science and Technology, Zhejiang University, Hangzhou, Zhejiang, 310027, PRC

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^{b*}Corresponding author's e-mail: fanzhang@zju.edu.cn

Abstract. An authenticated encryption chip with novel nonce generation circuit was developed. This circuit associates the nonce generation with the contents and receiving time of plaintext. It will also generate overlapping power compensation for the whole chip. The integrated chip was fabricated under SMIC 180nm technology. Under 1.8V V_{DD} , 100MHz global clock, the power consumption is about 14mW within 50k gates. The test result exhibits more than 10 times the strength in resistance to side-channel attack than the unprotected version without increasing hardware cost.

1. Introduction

Side-Channel Attacks (SCA) is one of the most important security problems faced by authenticated encryption (AE) hardware implementation. When the cryptographic implementation works it is unavoidable that information leaks. An attacker can then exploit the leakage and destroy the confidentiality of data. Side-channels that can leak information for instance time, electromagnetic, and power. Among them, power analysis and electromagnetic analysis are most commonly used. Their principles are the same, so it is enough to analyse one of them. To reveal the secret key of hardware implementation, they both analyse a large number of traces that have been recorded while the device is processing[1]. They are universally applicable methods since they have the advantages that no in-depth knowledge of the device is required. When the implementation is functioning, the internal information processing will have a certain connection with the power consumption. The attacker can deduce the hardware internal data information by analysing the power or electromagnetic changes of the implementation, then crack the secret key[2]. The two most common way of power analysis are Differential Power Analysis (DPA) and Correlation Power Analysis (CPA). The former analyses the rationality of guessing the key by calculating the differential value. The latter cracks the key by calculating the correlation coefficient.

In this article, we chose the unprotected version of ASCON as authenticated encryption prototype. An authenticated encryption chip with power compensation and novel nonce generation circuit was generated. Two methods, "Reversed Grouping Padding (RGP)" and "Appended Random Delay (ARD)" were proposed to improve hardware resistance to SCA. These methods were realized by modifying the nonce generation mechanism without adding complex peripheral circuits. Our integrated chip was



Low power & high throughput authenticated encryption chip for radio frequency applications

Xiaoyu Ma^{1,a}, Yan Han^{1,b*} and Fan Zhang^{2,c}

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²College of Computer Science and Technology, Zhejiang University, Hangzhou, Zhejiang, 310027, PRC

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^{b*}Corresponding author's e-mail: hany@zju.edu.cn

ABSTRACT

A novel authenticated encryption chip for radio frequency communication application was developed. A range of low power techniques were applied to reduce the power consumption. The proposed authenticated encryption chip exhibits ultra low power dissipation of 139.2 μ W/MHz @1.8V V_{DD} and ultra high throughput of 1.07Gbps with a reasonable core area of 0.461mm² (46.8k gates) under SMIC 180nm technology.

Keywords: Authenticated Encryption, Radio Frequency, low power, chip design

1. INTRODUCTION

Radio Frequency (RF) communication is a technology that uses radio signals for information exchange. Due to the fast transmission speed and wireless, RF technology has now played an important role in Internet of Things (IoT)[1], military security, and so on. However, as this technology widely applied, many security issues are increasingly prominent. Information leakage, property misappropriation and other issues occurred frequently, which restricted the development of RF communication.

Authenticated Encryption (AE) algorithm is a good way to ensure information security. It not only provides data encryption, but data source authentication as well, which can simultaneously guarantee the confidentiality, integrity and authentication of data. In addition, compared with the combination of encryption algorithm and message authentication code, AE algorithm can reduce the potential information leakage caused by data convergence between different algorithms effectively. The problem is that low area and low power are particularly important for hardware implementations.

The researches on AE Algorithm is Developing Worldwide. In order to solicit outstanding AE algorithms around the world, the Competition for Authenticated Encryption: Security, Applicability, and Robustness (CAESAR) came into being. After carefully assessing by cryptographers around the world, the candidate algorithms of the competition have been proved to have good security and reliability. Among all final round candidates, ASCON and ACORN were proved to be suitable for lightweight cases. AEGIS and OCB were designed for high-performance applications. Deoxys and COLM were good at defence in depth cases. A year after the final round of the CAESAR competition was announced, the National Institute of Standards and Technology (NIST) launched the Lightweight Cryptography Algorithms Competition. ASCON, as one of the candidate algorithms in the competition, became one of the winning algorithms in the final round in March 2021. After the evaluation and consideration from cryptographers worldwide, it is highly possible for the excellent algorithms to be promoted and applied as standards in industry.

In this article, we proposed a chip implementation of a lightweight AE algorithm ASCON in the RF field. It combines both encryption, decryption and authentication functions. The chip can adaptively adjust the encryption or decryption function according to the received data frame. It turned out to achieve a low core area of 0.461mm² (1.4mm² with I/Os) under SMIC 180nm technology with a considerable core power consumption of 46.4 μ W/MHz. At the operating frequency of 100MHz, our chip achieves a high throughput of 1.07Gbps.





4. 其他材料（注：根据填写的“国内外科研组织及重要学术期刊任职情况”、“学术荣誉称号”、“在重要国际学术会议报告情况”等依次上传证明材料）；

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一、国内外科研组织及重要学术期刊任职情况

【系统中已录入的国外科研组织任职情况】-共 4 项

1. 国际会议 PROOFS2020 和 2021 程序委员会主席

PROOFS 2021

10TH INTERNATIONAL WORKSHOP ON SECURITY PROOFS FOR EMBEDDED SYSTEMS
BEIJING, SEPTEMBER 17, 2021

[HOMEPAGE](#) [CFP](#) [COMMITTEES](#) [SUBMISSION](#) [REGISTRATION](#) [PROGRAM](#) [VENUE](#)



03.09.2021

The final program is available.

31.08.2021

The registration is open.

19.07.2021

Deadline extension until August 9

17.05.2021

Paper submission is open.

18.04.2021

The call for papers has been published.

08.02.2021

The new website for PROOFS 2021 is going online.

COMMITTEES

PROGRAM COMMITTEE

- Ludovic Aprville, Telecom ParisTech.
- Shivam Bhasin, Temasek Labs@NTU.
- Olivier Bronchain, UCL.
- Lei Bu, Nanjing University.
- Guo Chun, IIE,CAS.
- Naofumi Homma, Tohoku University.
- Cetin Koc, University of California Santa Barbara.
- Quentin Meunier, Université Pierre et Marie Curie.
- Debdeep Mukhopadhyay, IIT Kharagpur, India.
- Changhai Ou, Nanyang Technological University.
- Guilherme Perin, UFRMM.
- Francesco Regazzoni, ALaRI - USI (Switzerland).
- Fu Song, ShanghaiTech University.
- Gilles Van Assche, STMicroelectronics.
- An Wang, Beijing Institute of Technology.
- Xiaofei Xie, Kyushu University.
- Yongwang Zhao, Zhejiang University.

STEERING COMMITTEE

- Jean-Luc Danger, Télécom ParisTech, France.
- Sylvain Guilley, GET/ENST, CNRS/LTCl, France.
- Çetin Kaya Koc, UCSB, USA.
- David Naccache, ENS, France.
- Akashi Satoh, UEC, Japan.
- Werner Schindler, BSI, Germany.

ORGANIZING COMMITTEE

- General chair: Ulrich Kühne, Télécom Paris, France.
- Program chair: Fan Terry Zhang, Zhejiang University, China.

IMPORTANT DATES

- ✓ Submission (abstract):
Monday, August 2
- ✓ Submission (full paper):
Monday, August 9
- ✓ Notification:
Thursday, August 26
- ✓ Final version due:
Tuesday, September 14, 2021
- ✓ PROOFS workshop:
Friday September 17, 2021



PROOFS 2020 CFP (Published Version)

This CFP will expire on Tuesday, October 13th. To extend this CFP [click here](#).

To edit this CFP use 'Edit' in the main menu.

To view other versions of the current CFP, use the context menu in the upper right corner.

The CFP is in your watchlist.

[View publishing history](#)
[Remove from watchlist](#)

PROOFS 2020: 9th International Workshop on Security Proofs for Embedded Systems

World Wide Web

September 13, 2020

Conference website	http://www.proofs-workshop.org/2020/
Submission link	https://easychair.org/conferences/?conf=proofs2020
Submission deadline	July 19, 2020
Author notification date	August 19, 2020
Camera Ready deadline	September 1, 2020

Topics: [hardware security](#) [security proofs](#) [embedded systems](#) [cryptography](#)

Due to the current situation of COVID-19, the workshop of PROOFS 2020 will go online together with CHES 2020 this year. The workshop organizers are waiting for the official technical solution for IACR.

PROOFS 2020

The goal of the PROOFS workshop is to promote methodologies that increase the confidence in the security of embedded systems, especially those which contain cryptographic algorithms. Concretely, the PROOFS workshop seeks contributions in both theory and practice of methods and tools applied to the security of embedded systems. Examples include (semi-)formal methods, simulation-based leakage evaluation and security checks, protocol verification techniques, test and verification of secure embedded systems (software and hardware), and provable security for physical attacks.

Submission Guidelines

Submitted papers are limited to 15 pages in [EPIC style](#). The papers can be submitted via [easychair](#).

List of Topics

Topics of interest include, but are not limited to, the following:

- (Automated) security proofs
- Applications of formal methods in security
- Protocol verification
- Security evaluation of real-world systems
- Leakage-resilient cryptography
- Side-channel analysis and countermeasures
- Fault attacks and defenses
- Information leakage models
- Tamper-resistant hardware

2. 国际组织 IEEE 电路系统学会 VLSI 系统应用技术委员会委员

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[Technical Committee](#) [VLSI Systems and Applications \(VSA\)](#)

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- VSA Reports

VLSI Systems and Applications (VSA)

Scope

The goal of the VSA-TC is to promote research and education in the field of digital VLSI systems and its applications in electrical engineering, computer science and engineering, and related fields. System design aspects include the design of architectures, algorithms, developing theory, design methodologies, and implementation, as they relate to the design of VLSI systems and system-on-chips. Application areas include, but are not limited to, signal processing, communications, wireless, networking systems, video, multimedia, radar, optical, nanotechnologies, biotechnology, and other emerging technologies.

Future research directions: where the field is going

The field in which the VSA-TC is currently active is certainly broad, and is expected to further broaden in the future. In particular, the VSA-TC field is expected to *expand both at the bottom* (wider range of potential technologies that will complement or eventually replace CMOS) *and the top* (more complex system). This is a consequence of expected technology breakthrough and exponentially increasing complexity at the same time.

Chair(s)

Tony Tae-Hyoung Kim



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Mona E. Zaghloul | The George Washington University - USA

Email: [\(Click to show email\)](#)

Chuan Zhang | Southeast University - China

Email: [\(Click to show email\)](#)

Fan (Terry) Zhang | ZheJiang University - China

Email: [\(Click to show email\)](#)

Tong Zhang | Rensselaer Polytechnic Institute - USA

Email: [\(Click to show email\)](#); [\(Click to show email\)](#)

Xinmiao Zhang | The Ohio State University, Columbus - USA

Email: [\(Click to show email\)](#)

Zhengya Zhang | University of Michigan - USA

Email: [\(Click to show email\)](#)

Danella Zhao | Old Dominion University - USA

Email: [\(Click to show email\)](#); [\(Click to show email\)](#)

Jun Zhou | University of Science and Technology

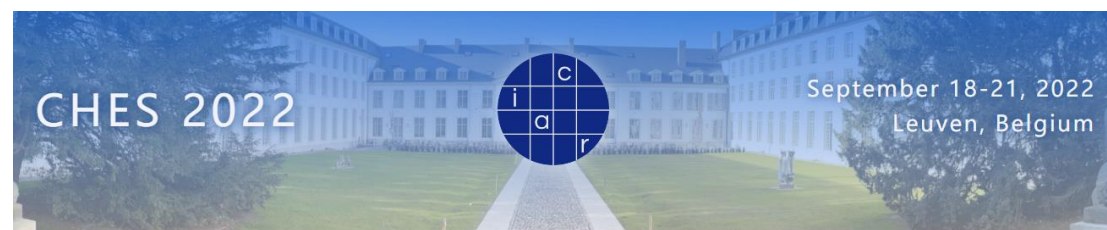
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115

-2023/2/28

3. 国际会议（网安 A 类）CHES2022-2023 程序委员会委员

CHES 2022



The banner features a blue-tinted photograph of a large, historic building with a central courtyard. Overlaid on the image is a circular logo with a grid pattern containing the letters 'c', 'i', 'a', and 'r'. The text 'CHES 2022' is on the left, and 'September 18-21, 2022 Leuven, Belgium' is on the right.

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Call for Papers

[Call for Papers as PDF](#)

General Information

Having been established in 1999, the Cryptographic Hardware and Embedded Systems (CHES) conference is the premier venue for research on both design and analysis of cryptographic hardware and software implementations. As an area conference of the International Association for Cryptologic Research (IACR), CHES bridges the cryptographic research and engineering communities, and attracts participants from academia, industry, government and beyond. CHES 2022 will take place in Leuven, Belgium in September 18-21, 2022. The conference website is accessible at <https://ches.iacr.org/2022>.

The scope of CHES is intentionally diverse, meaning we solicit submission of papers on topics including, but not limited to, the following (with new topics for CHES 2022 **in boldface**):

Georg Sigl TU Munich & Fraunhofer AISEC	François-Xavier Standaert UCLouvain	Rainer Steinwandt University of Alabama in Huntsville	Takeshi Sugawara The University of Electro-Communications
Petr Svenda Masaryk University	Jakub Szefer Yale	Adrian Thillard Ledger	Yosuke Todo NTT Corporation
Meltem Sönmez Turan NIST	Alexandre Venelli NXP Semiconductors	Christine van Vredendaal NXP Semiconductors	Junwei Wang
Xavier Bonnetain University of Waterloo	Wenjie Xiong Virginia Tech	Bo-Yin Yang Academia Sinica	Bohan Yang Tsinghua University
Yuval Yarom The University of Adelaide	Yu Yu Shanghai Jiao Tong University	Fan Zhang Zhejiang University	



CHES 2023



CHES 2023

September 10-14, 2023
Prague, Czech Republic

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- Sponsors
- Contact

Call for Papers

[Call for Papers as PDF](#) [Paper Submission](#)

General Information

Having been established in 1999, the Cryptographic Hardware and Embedded Systems (CHES) conference is the premier venue for research on both design and analysis of cryptographic hardware and software implementations. As an area conference of the International Association for Cryptologic Research (IACR), CHES bridges the cryptographic research and engineering communities, and attracts participants from academia, industry, government and beyond. CHES 2023 will take place in Prague, Czech Republic in September 10-14, 2023. The conference website is accessible at

<https://ches.iacr.org/2023>

The scope of CHES is intentionally diverse, meaning we solicit submission of papers on topics including, but not limited to, the following (with new topics for CHES 2023 highlighted in **bold blue**):

Cryptographic implementations:

- Hardware architectures
- Cryptographic processors and coprocessors
- True and pseudorandom number generators

Interactions between cryptographic theory and implementation issues:

- Quantum cryptanalysis
- Algorithm subversion and subversion prevention

Pascal Sasdrich Ruhr-University Bochum	Patrick Schaumont Worcester Polytechnic Institute	Georg Sigl TU Munich & Fraunhofer AISEC	Marcos Simplicio Jr. University of São Paulo
François-Xavier Standaert UCLouvain	Rainer Steinwandt University of Alabama in Huntsville	Takeshi Sugawara The University of Electro-Communications	Petr Svenda Masaryk University
Junko Takahashi NTT Corporation	Russell Tessier University of Massachusetts Amherst	Nicolas Thériault Universidad de Santiago de Chile	Alexandre Venelli NXP Semiconductors
Wenjie Xiong Virginia Tech	Yu Yu Shanghai Jiao Tong University	Fan Zhang Zhejiang University	



4. 国际会议（网安 A 类）DAC2020-2022 技术委员会委员

DAC 2020



DAC 2021

DAC 2021 - SEC2 TPC Member Invitation

Alexis Bauer Kolak
发给 fanzhang

发件人: Alexis Bauer Kolak <do_not_reply@linklings.com>
收件人: fanzhang <fanzhang@zju.edu.cn>
时间: 2020年10月20日 (周二) 23:59
大小: 10 KB



Dear Fan,

On behalf of DAC 2021 (www.dac.com), I would like to invite you to serve as a TPC member in track SEC2. Hardware Security II: Attack and Defense.

PLEASE READ CAREFULLY

*** A TPC member is required to attend the virtual TPC meeting tentatively scheduled for February 9, 2021.

*** Furthermore, we expect all TPC members to attend DAC July 11-15, 2021.

A TPC member should be deeply involved in DAC and should meet other DAC attendees, attend technical presentations etc. and as such should be aware of the new enable him/her to provide excellent reviews in subsequent years as well.

There are some submission constraints for TPC members:

- A TPC member can submit maximum 5 papers to DAC
- A TPC member can submit maximum 3 papers to own track
- No paper can be submitted to own track by Track Co-Chair

An important goal for DAC 2021 is to receive IN-DEPTH REVIEWS i.e. reviews that are close to the depth/thoroughness typical for top journals. This will require more review period will be between November 17, 2020 and late January 2021. More details on the review process will follow as we get closer to the submission deadline

Please accept or decline this invitation by Thursday, October 22.

To accept or decline this invitation, click one of the following links or carefully paste it into your browser:

Accept: https://ssl.linklings.net/organizations/dac/?page=AcceptInvitation&id=2AmUEBSzoN&new_year=dac2021

Decline: https://ssl.linklings.net/organizations/dac/?page=DeclineInvitation&id=2AmUEBSzoN&new_year=dac2021

The DAC 2021 Executive Committee and I appreciate your time and consideration. I look forward to your positive reply and working with you.

Best regards,
Vivek De



DAC 2022

59 DAC: Invitation to serve as a TPC member ★

Alexis Kolak

发给 fanzhang

发件人: Alexis Kolak <noreply@softconf.com>

收件人: fanzhang <fanzhang@zju.edu.cn>

时间: 2021年10月5日 (周二) 05:28 📧

大小: 7 KB

Dear Fan,

On behalf of DAC 2022 (www.dac.com), we would like to invite you to serve as a TPC Member for track SEC3. Embedded and Cross-Layer Security.

PLEASE READ CAREFULLY

*** A TPC member is required to attend the virtual TPC meeting tentatively scheduled for February 9, 2022

*** Furthermore, we expect all TPC members to attend DAC July 10-14, 2022.

A TPC member should be deeply involved in DAC and should meet other DAC attendees, attend technical presentations etc. and as such should be able to provide excellent reviews in subsequent years as well.

There are some submission constraints for TPC members:

- A TPC member can submit maximum 5 papers to DAC
- A TPC member can submit maximum 3 papers to own track
- No paper can be submitted to own track by Track Chair

An important goal for DAC 2022 is to receive IN-DEPTH REVIEWS i.e. reviews that are close to the depth/thoroughness typical for top journals. This

Review period will be between December 13, 2021 and late January 18, 2022. More details on the review process will follow as we get closer to the start.

Please reply by October 8.

PLEASE REPLY HERE:

<https://www.softconf.com/dac2022/super/sl.cgi?sl=483686-D4E8>

Helen Li, DAC 2022 Technical Program Chair

Renu Mehra, DAC 2022 Technical Program Co-Chair

--

DAC 2022 - <https://www.softconf.com/dac2022/research>



【系统中已录入的国外期刊任职情况】-共 3 项

1. 国际 SCI 期刊 FITEE 首批通讯专家



2. 国际 SCI 期刊 CyberSecurity 青年编委委员

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3. 国际 SCI 期刊 IEEE Access 副编辑



Multidisciplinary | Rapid Review | Open Access Journal

8 April 2019

Dear Dr. Zhang,

I would like to recommend you as a potential Associate Editor for IEEE Access. We would like to ask you at this point if you would be interested in this position.

As an Associate Editor, you will be expected to:

- Coordinate the careful and critical review of a minimum of 2 articles per month (on average) by inviting and receiving a minimum of 2 independent reviews for every article.
 - Briefly evaluate each article assigned to you to confirm it is suitable for peer review (in scope, has technical merit, is not simplistic, grammar, etc.).
 - Facilitate a rapid review process that takes less than two weeks, with an expedited timeline.
 - Invite and assign willing reviewers for each paper within 2 days of receiving a paper.
 - Remind reviewers if they have not submitted their review within 1 week of accepting the invitation to review.
 - If a reviewer does not respond to reminder emails, then invite and assign new reviewers.
 - Render a final decision of Accept or Reject within 1 day of receiving the reviewer comments.
 - Provide brief comments on Accept and Reject decisions.
- Contact the Managing Editor if you notice author (or reviewer) misconduct. Plagiarism or suggesting one's own references as a reviewer (or editor) is not appropriate and would warrant a detailed investigation.
- In cases of article discrepancies, evaluate changes to accepted articles to confirm the changes are acceptable (this is rare).
- Serve a 3-year term.
- Maintain an account in the IEEE Access ScholarOne Manuscript website with a valid ORCID (<https://orcid.org/>). This website will be used for the review process, including assigning reviewers and rendering publication decisions.

Additionally, if you are engaged in a trending and interesting topic, and can compile a full list of Guest Editors (a minimum of 4, that are geographically diverse) you may propose a Special Section for IEEE Access that focuses on topics of research that belong within the IEEE, but would be considered out of scope for most of the IEEE journals (e.g. interdisciplinary or boundary-pushing topics). Kindly note that this is not mandatory. Each Special Section will include an Editorial that introduces the section and at least 4-8 published articles on the topic. The proposals will be reviewed by an Editorial Committee, and then the Special Sections will be announced. If your proposal is selected, you will be the lead Editor for that Special Section.

If you are still interested in this position, please confirm at this time.

Thank you very much for your interest in IEEE Access,

Prof. Derek Abbott
Editor-in-Chief, IEEE Access

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【系统中已录入的国内科研组织任职情况】-共 6 项

1. 中国密码学会（CACR）密码芯片专业委员会委员



中国密码学会

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密码芯片专业委员会第三届成员名单

时间: 2019年03月19日 来源: 中国密码学会

分享:

(2018年10月26日中国密码学会三届四次理事会审议通过)

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5	白国强	清华大学微电子学研究所	副教授	委员
6	包仕波	兴唐通信科技有限公司	副总工/高级工程师	委员
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中国密码学会2013年年会

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中国密码学会2016年年会

中国密码学会2014年年会

中国密码学会2012年年会

中国密码学会分支机构评估管理办法



中国密码学会

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2. 中国密码学会（CACR）密码测评专业委员会委员

中国密码学会密码测评专业委员会第二届成员名单

时间：2022年11月22日 来源：中国密码学会

分享： 

2022年11月12日（第四届十一次常务理事会）

序号	姓名	机构职务	工作单位	职务/职称
1	罗 鹏	主任委员	国家密码管理局商用密码检测中心	正高级工程师
2	李建华	副主任委员	上海交通大学网络安全技术研究院	教授
3	樊俊锋	副主任委员	深圳市组创信安科技开发有限公司	高级工程师
4	陈 华	副主任委员	中国科学院软件研究所	正高级工程师
5	李大为	委员	哈尔滨工业大学（深圳）	研究员
6	顾 健	委员	国家网络与信息系统安全产品检验检测中心 （公安部第三研究所）	研究员
7	潘小明	委员	浙江省电子信息产品检验研究院	正高级工程师
8	陈清明	委员	上海市信息安全测评认证中心	正高级工程师
9	韩正甫	委员	中国科学技术大学	教授
10	郭山清	委员	山东大学网络空间安全学院	教授
11	何道敬	委员	哈尔滨工业大学（深圳）	教授
12	韦永壮	委员	桂林电子科技大学	教授
13	吴 震	委员	成都信息工程大学	教授
14	林璟锵	委员	中国科学技术大学	教授
15	乌力吉	委员	清华大学	研究员
16	沈海斌	委员	浙江大学	教授
17	樊 凯	委员	西安电子科技大学	教授
18	张 帆	委员	浙江大学	教授
19	欧长海	委员	武汉大学	教授



3. 中国计算机学会（CCF）容错计算专业委员会委员



4. 中国计算机学会（CCF）集成电路专业委员会委员



5. 中国计算机学会 CCF YOCSEF 杭州委员



6. 中国中文信息学会大数据安全与隐私计算专委会委员



二、学术荣誉称号

【系统中已录入的学术荣誉称号】-共 5 项

1. 2020 年军队科技进步二等奖



2. 2021 年华为杯第四届中国研究生创“芯”大赛全国二等奖



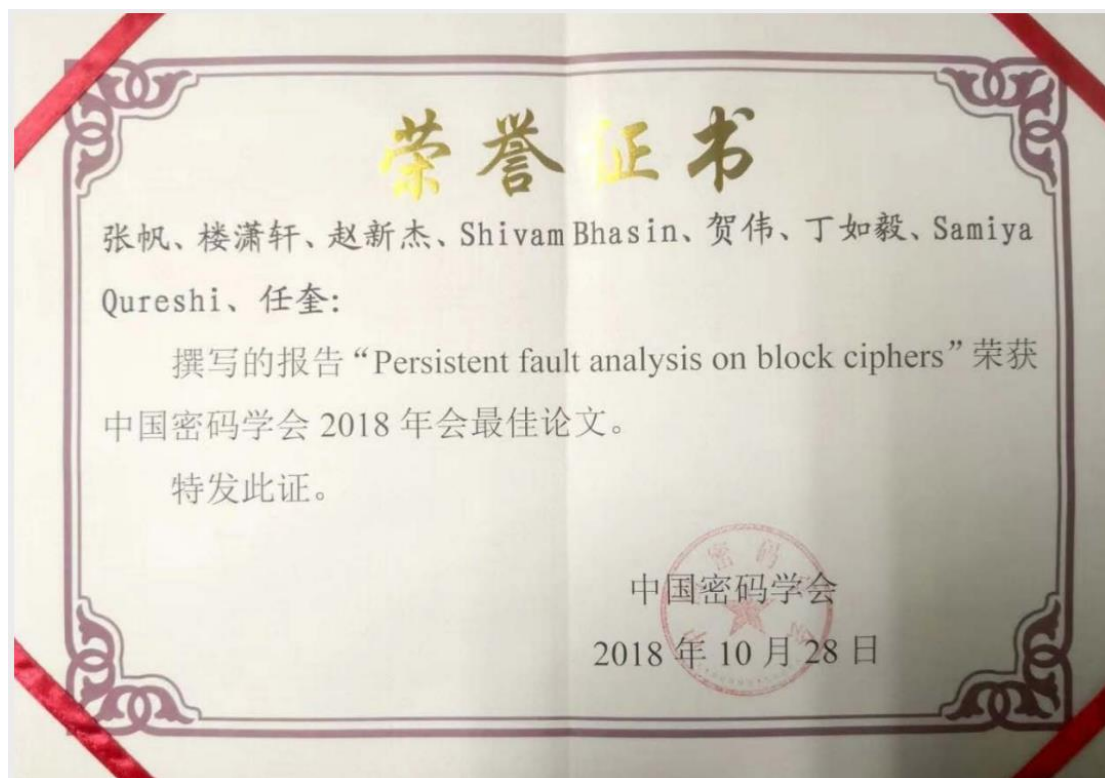
3. 2021 年密码测评学术会议 (CryptoTE2021) 最佳论文奖



4. 2019 年亚洲硬件安全年会 (AsianHOST2019) 最佳海报论文奖



5. 2018 年中国密码年会 (ChinaCrypt2018) 最佳论文奖

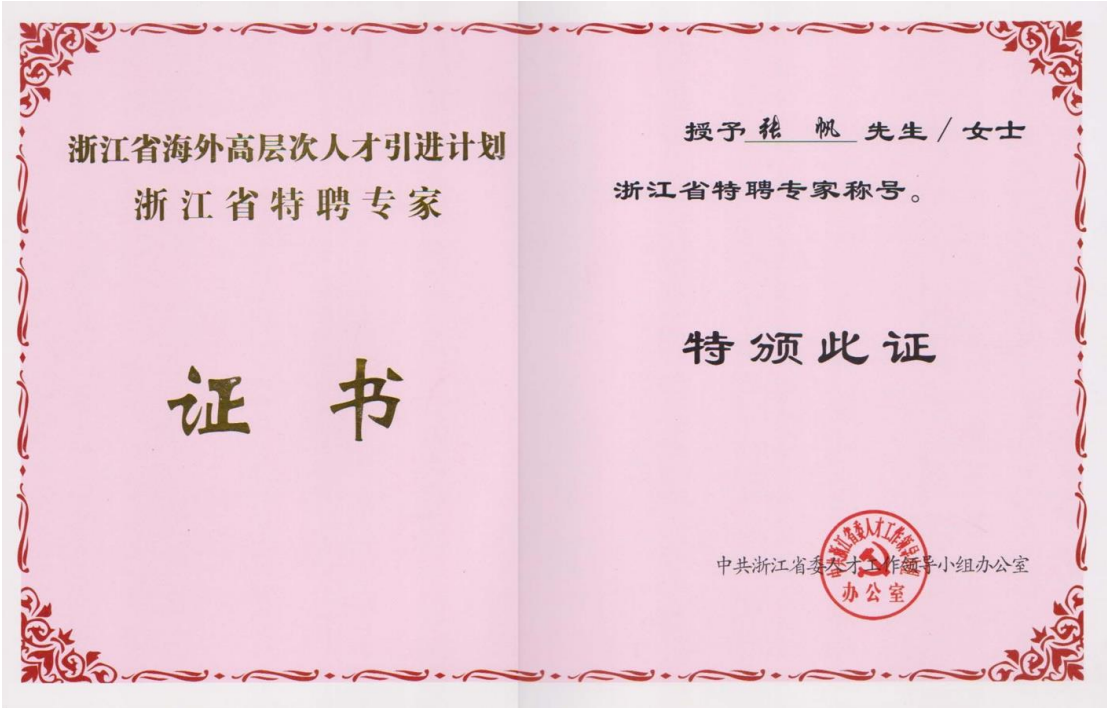


【系统中未录入的补充学术荣誉称号】-共 6 项

1. 2015 年军队科技进步奖二等奖



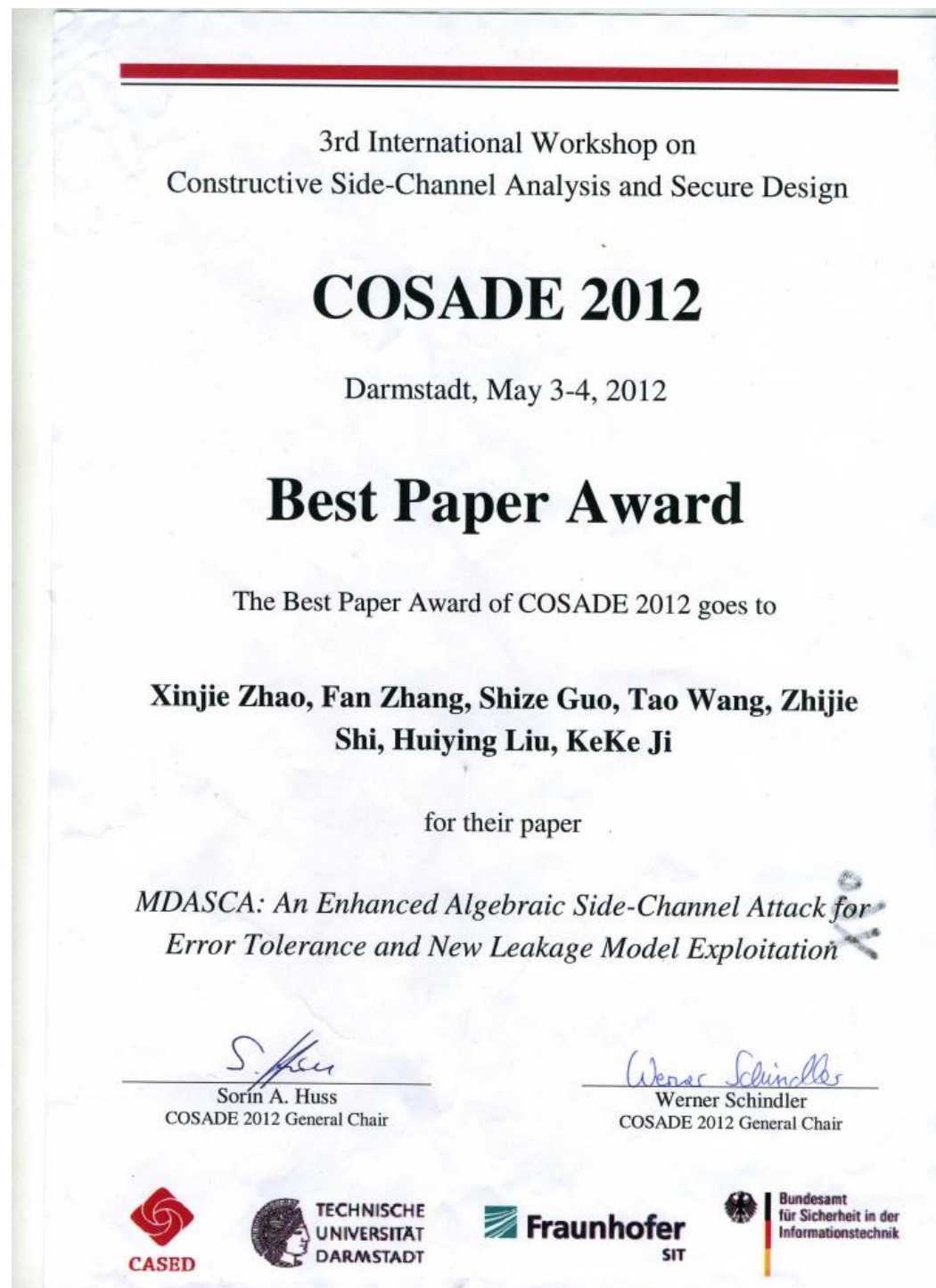
2. 2014 年浙江省千人计划



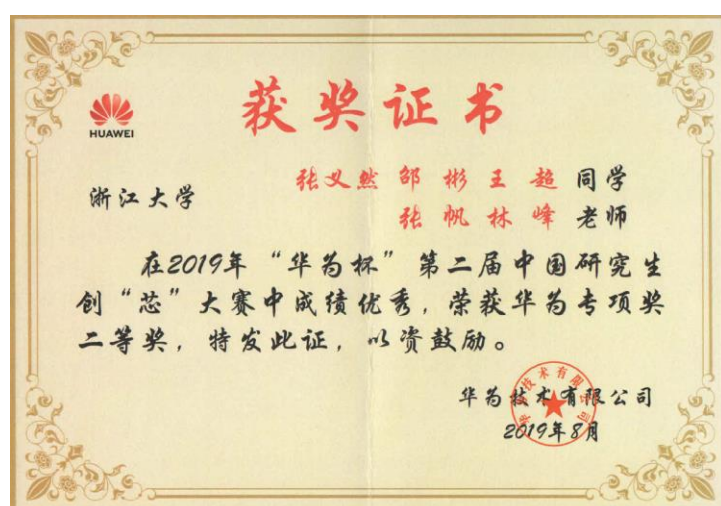
3. 2021 年 SAFE 会议最佳论文奖



4. 2012 年 COSADE 国际会议最佳论文奖



5. 2019 年华为杯第四届中国研究生创“芯”大赛全国一等奖、
专项二等奖、优秀指导教师



6. 2021 年第五届强网杯全国网络安全挑战赛人工智能挑战赛全国一等奖、优秀指导老师



7. 2020 年华为杯第二届中国研究生人工智能创新大赛全国二等奖、专项一等奖、优秀指导教师



三、重要国际学术会议报告情况

【系统中已录入的会议报告情况】-共 5 项

1. 国际会议 CHES2018 大会报告: Persistent Fault Analysis on Block Ciphers



浙江大学 信息与电子工程学院
College of Information Science & Electronic Engineering, Zhejiang University

Persistent Fault Analysis on Block Ciphers

Fan (Terry) Zhang, Xiaoxuan Lou, Xinjie Zhao, Shivam Bhasin,
Wei He, Ruyi Ding, Samiya Qureshi and Kui Ren
Zhejiang University

CHES2018, Amsterdam, The Netherlands, 09/12/2018



15:20 – 15:50	Coffee break
Session 15: Fault Attacks II (chair: Debdeep Mukhopadhyay)	
15:50 – 16:10	New Bleichenbacher Records: Fault Attacks on gDSA Signatures , Akira Takahashi (Kyoto University), Mehdi Tibouchi (Kyoto University, NTT Secure Platform Laboratories), Masayuki Abe (Kyoto University, NTT Secure Platform Laboratories) [slides]
16:10 – 16:30	Persistent Fault Analysis on Block Ciphers , Fan Zhang (Zhejiang University), Xiaoxuan Lou (Zhejiang University), Xinjie Zhao (The Institute of North Electronic Equipment), Shivam Bhasin (Nanyang Technological University), Wei He (Huawei Ltd.), Ruyi Ding (Zhejiang University, Georgia Institute of Technology), Samiya Qureshi (Zhejiang University), Kui Ren (Zhejiang University) [slides]
16:30 – 16:50	SIFA: Exploiting Ineffective Fault Inductions on Symmetric Cryptography , Christoph Dobraunig (TU Graz), Maria Eichlseder (TU Graz), Thomas Korak (Infineon Technologies AG), Stefan Mangard (TU Graz), Florian Mendel (Infineon Technologies AG), Robert Primas (TU Graz) [slides]
16:50 – 17:05	Closing remarks

Design 2016 by [Peter Schwabe](#), last updated Apr 27, 2020
Title photo by [Jan Klomp](#) (CC BY-SA 2.0)



2. 国际会议 CHES2020 大会报告: Persistent Fault Attack in Practice

Persistent Fault Attack in Practice

Fan Zhang, Yiran Zhang, Huilong Jiang, Xiang Zhu, Shivam Bhasin, Zhe Liu, Dawu Gu and Kui Ren

CHES 2020



Fri 18 Sep 22:20
to
Fri 18 Sep 23:00
(UTC: 14:20-15:00)

Session 12: Faults ⓘ 📅

YouTube

Chat

Session chairs: Adrian Thillard, Tobias Schneider and Xiaolu Hou

JackHammer: Efficient Rowhammer on Heterogeneous FPGA-CPU Platforms

Zane Weissman, Thore Tiemann, Daniel Moghimi, Evan Custodio, Thomas Eisenbarth, Berk Sunar

Speaker: Zane Weissman, Thore Tiemann

Media: 📄 🎥

Protecting against Statistical Ineffective Fault Attacks

Joan Daemen, Christoph Dobraunig, Maria Eichsleder, Hannes Gross, Florian Mendel, Robert Primas

Speaker: Robert Primas

Media: 📄 🎥

FEDS: Comprehensive Fault Attack Exploitability Detection for Software Implementations of Block Ciphers

Keerthi K, Indrani Roy, Chester Rebeiro, Aritra Hazra, Swarup Bhunia

Speaker: KEERTHI K

Media: 📄 🖥️ 🎥

Persistent Fault Attack in Practice

Fan Zhang, Yiran Zhang, Huilong Jiang, Xiang Zhu, Shivam Bhasin, Xinjie Zhao, Zhe Liu, Dawu Gu, Kui Ren

Speaker: Fan Zhang

Media: 📄 🎥

Fri 18 Sep 23:00
to
Fri 18 Sep 23:10
(UTC: 15:00-15:10)

Closing Remarks 📅

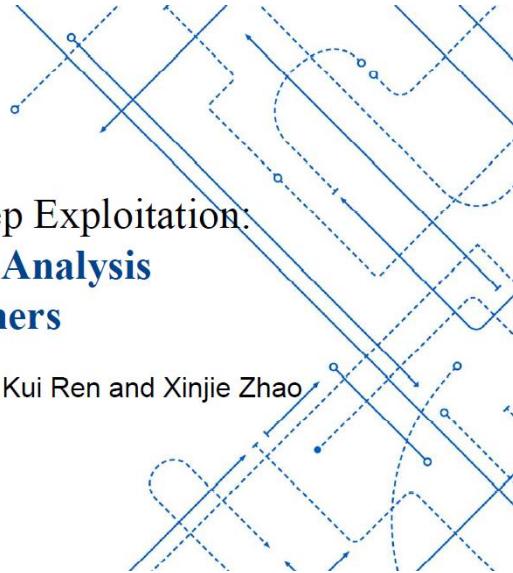


3. 国际会议 CHES2022 大会报告: Free Fault Leakages for Deep Exploitation: Algebraic Persistent Fault Analysis on Lightweight Block Ciphers

Free Fault Leakages for Deep Exploitation: Algebraic Persistent Fault Analysis on Lightweight Block Ciphers

Fan Zhang, Tianxiang Feng, Zhiqi Li, Kui Ren and Xinjie Zhao

CHES 2022



Leuven: 15:50-16:20
Tue Sep 20 21:50
to
Tue Sep 20 22:20

Leuven: 16:20-17:40
Tue Sep 20 22:20
to
Tue Sep 20 23:40

Coffee Break

Location: Pieter de Somer and Grote Aula

Side-Channel Attacks on Post-Quantum Implementations 1

Location: Pieter de Somer

Session Chair: Christine van Vredendaal

[YouTube](#)

Curse of Re-encryption: A Generic Power/EM Analysis on Post-Quantum KEMs

Rei Ueno, Keita Xagawa, Yutaro Tanaka, Akira Ito, Junko Takahashi, Naofumi Homma

Speaker: Rei Ueno (in person)

(paper #1_24)

[Show abstract >](#)

Media:

The Hidden Parallelepiped Is Back Again: Power Analysis Attacks on Falcon

Morgane Guerreau, Ange Martinelli, Thomas Ricosset, Mélissa Rossi

Speaker: Morgane Guerreau (in person)

(paper #3_23)

[Show abstract >](#)

Media:

Don't Reject This: Key-Recovery Timing Attacks Due to Rejection-Sampling in HQC and BIKE

Norman Lahr, Qian Guo, Clemens Hlauschek, Robin Leander Schröder, Thomas Johansson, Alexander Nilsson

Speaker: Robin Leander Schröder (in person)

Fault Analysis, Fault Protection Frameworks and SCA Countermeasures

Location: Grote Aula

Session Chair: Adrian Thillard

[YouTube](#)

Redundancy AES Masking Basis for Attack Mitigation (RAMBAM)

Yaacov Belenky, Vadim Bugaenko, Leonid Azriel, Hennadii Chernyshchyk, Ira Dushar, Oleg Karavaev, Oleh Maksimenko, Yulia Ruda, Valery Teper, Yury Kreimer

Speaker: Yaacov Belenky (in person)

(paper #2_10)

[Show abstract >](#)

Media:

Free Fault Leakages for Deep Exploitation: Algebraic Persistent Fault Analysis on Lightweight Block Ciphers

Fan Zhang, Tianxiang Feng, Zhiqi Li, Kui Ren, Xinjie Zhao

Speaker: Fan Zhang (online)

(paper #2_56)

[Show abstract >](#)

Media:

SIPFA: Statistical Ineffective Persistent Faults Analysis on Feistel Ciphers



4. 国际会议 DAC2018 大会报告：From Homogeneous to Heterogeneous: Leveraging Deep Learning based Power Analysis across Devices

From Homogeneous to Heterogeneous: Leveraging Deep Learning based Power Analysis across Devices

Fan Zhang¹, Bin Shao¹, Guorui Xu¹, Bolin Yang¹,
Ziqi Yang^{2*}, Zhan Qin¹ and Kui Ren¹

¹Zhejiang University, China

²National University of Singapore, Singapore



DAC 2020 | San Francisco, CA | July 20-24

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TUESDAY July 21, 3:30pm - 5:00pm

TOPIC AREA: SECURITY, MACHINE LEARNING/AI

KEYWORD: HARDWARE SECURITY

EVENT TYPE: RESEARCH REVIEWED

Learn From Your Leakage: Attacks and Defenses for Side-Channel Analysis

Chair: Ramesh Karri - *New York Univ, New York, New York*

Today side-channel attacks have emerged as one of the most powerful attacks on protected devices. At the same time, various techniques have been proposed to conduct such attacks. One prominent direction is to employ machine learning to implement side-channel attacks. This session explores the influence of portability on the performance of profiling attacks and the implications of the size of the learning networks on the efficiency of the attack. It further explores how to fuse multiple sources of leakage, and presents a new metric for detecting cache timing attacks.

31.1 Reuse-Trap: Re-Purposing Cache Reuse Distance to Defend Against Side Channel Leakage

Speaker: Hongyu Fang - *George Washington Univ, Washington, DC, DC*

Authors: Hongyu Fang - *George Washington Univ, Washington, DC, DC*

Milos Doroslovacki - *George Washington Univ, Washington, DC*

Guru Venkataramani - *George Washington Univ, Washington, DC*

31.2 Deep Learning Multi-Channel Fusion Attack Against Side-Channel Protected Hardware

Speaker: Benjamin Hettwer - *Robert Bosch GmbH, Stuttgart, Germany*

Authors: Benjamin Hettwer - *Robert Bosch GmbH, Stuttgart, Germany*

Daniel Fennes - *Ruhr Univ. Bochum, Germany*

Sebastian Leger - *Robert Bosch GmbH, Stuttgart, Germany*

Jan Richter-Brockmann - *Ruhr Univ. Bochum, Germany*

Stefan Gehrler - *Robert Bosch GmbH, Stuttgart, Germany*

Tim Güneysu - *Ruhr Univ. Bochum, Germany*

31.3 Learning From A Big Brother - Mimicking Neural Networks in Profiled Side-channel Analysis

Speaker: Stjepan Picek - *Delft Univ. of Technology, Delft, The Netherlands*

Authors: Daan van der Valk - *Delft Univ. of Technology, Delft, The Netherlands*

Marina Krcek - *Delft Univ. of Technology, Delft, The Netherlands*

Stjepan Picek - *Delft Univ. of Technology, Delft, The Netherlands*

Shivam Bhasin - *Nanyang Technological Univ, Singapore, Singapore*

31.4 From Homogeneous to Heterogeneous: Leveraging Deep Learning Based Power Analysis Across Devices

Speaker: Fan Zhang - *Zhejiang Univ. & Alibaba-Zhejiang University Joint Institute of Frontier Technologies, Hangzhou, China*

Authors: Fan Zhang - *Zhejiang Univ. & Alibaba-Zhejiang University Joint Institute of Frontier Technologies, Hangzhou, China*

Bin Shao - *Zhejiang Univ, Hangzhou, China*

Guorui Xu - *Zhejiang Univ, Hangzhou, China*

Bolin Yang - *Zhejiang Univ, Hangzhou, China*

Ziqi Yang - *National Univ. of Singapore, Singapore*

Zhan Qin - *Zhejiang Univ. & Alibaba-Zhejiang University Joint Institute of Frontier Technologies, Hangzhou, China*

Kui Ren - *Zhejiang Univ. & Alibaba-Zhejiang University Joint Institute of Frontier Technologies, Hangzhou, China*



5. 国际会议 DAC2022 大会报告: DARPT: Defense Against Remote Physical Attack Based on TDC in Multi-tenant Scenario



DARPT: Defense Against Remote Physical Attack Based on TDC in Multi-tenant Scenario

Fan Zhang^{1,2}, Zhiyong Wang^{1,2,3}, Haoting Shen^{1,*}, Bolin Yang^{1,2}, Qianmei Wu¹ and Kui Ren¹

¹ School of Cyber Science and Technology, College of Computer Science and Technology, Zhejiang University

² Alibaba-Zhejiang University Joint Research Institute of Frontier Technologies

³ Key Laboratory of Blockchain and Cyberspace Governance of Zhejiang Province

* Corresponding authors: Haoting Shen (Email: htshen@zju.edu.cn)

This work is partly supported by Alibaba-Zhejiang University Joint Institute of Frontier Technologies.



PROGRAM

Wednesday, July 13, 2022

ROBUST QUANTUM COMPUTING: WILD GOOSE CHASE?

Time: 10:30 AM – 12:00 PM

Room: 3005, Level 3

Event Type: Research Manuscript

Topics Area(s): Quantum Computing, Design

Chair/Co-Chair: Yufei Ding, University of California, Santa Barbara; Rolf Drechsler, University of Bremen, DFKI

In this session, we have 4 papers covering various aspects of robust quantum computing flow. There are two presentations that address scalable verification approaches by equivalence checking and BDD-based unitary operator manipulation, respectively. Another paper presents the case of handling non-unitaries in equivalence checking of quantum circuits. The last paper of the session proposes a compression technique for reducing resource requirements for topological quantum error correction circuits.

EQUIVALENCE CHECKING PARADIGMS IN QUANTUM CIRCUIT DESIGN: A CASE STUDY

Tom Peham, Lukas Burgholzer, Robert Wille, Johannes Kepler University Linz, Linz, Austria

ACCURATE BDD-BASED UNITARY OPERATOR MANIPULATION FOR SCALABLE AND ROBUST QUANTUM CIRCUIT VERIFICATION

Chun-Yu Wei, Yuan-Hung Tsai, Chiao-Shan Jhang, Jie-Hong Roland Jiang, National Taiwan University, Taipei, Taiwan

HANDLING NON-UNITARIES IN QUANTUM CIRCUIT EQUIVALENCE CHECKING

Lukas Burgholzer, Robert Wille, Johannes Kepler University Linz, Linz, Austria

A BRIDGE-BASED ALGORITHM FOR SIMULTANEOUS PRIMAL AND DUAL DEFECTS COMPRESSION ON TOPOLOGICALLY QUANTUM-ERROR-CORRECTED CIRCUITS

Wei-Hsiang Tseng, Yao-Wen Chang, National Taiwan University, Taipei, Taiwan

STAY (FAR) AWAY FROM MY DEVICE: REMOTE ATTACKS AND DEFENSES

Time: 10:30 AM – 12:00 PM

Room: 3006, Level 3

Event Type: Research Manuscript

Topics Area(s): Hardware Security: Attack and Defense, Security

Chair/Co-Chair: Seetal Potluri, North Carolina State University; TBA

Side-channel analysis exploits information that originate from side-channels such as the power consumption or the timing behavior of a system. This session deals with side-channel attacks and defenses that are exploited remotely. Two papers concentrate on defense mechanisms against timing side-channel attacks in cache memories. One paper exploits speculative execution to bypass existing countermeasures against cache timing side-channel attacks. The fourth paper proposes a defense against remote power side-channel analysis attacks in multi-tenant FPGAs.

FASE: FAST SELECTIVE FLUSHING TO MITIGATE CONTENTION-BASED CACHE TIMING ATTACKS

Tuo Li, Sri Parameswaran, University of New South Wales, Sydney, Australia

CONDITIONAL ADDRESS PROPAGATION: AN EFFICIENT DEFENSE MECHANISM AGAINST TRANSIENT EXECUTION ATTACKS

Peinan Li, Rui Hou, Lutan Zhao, Yifan Zhu, Dan Meng, Chinese Academy of Sciences, Beijing, China

TIMED SPECULATIVE ATTACKS EXPLOITING STORE TO LOAD FORWARDING BYPASSING CACHE-BASED COUNTERMEASURES

Anirban Chakraborty, Debdeep Mukhopadhyay, Indian Institute of Technology, Kharagpur, Kharagpur, India; Nikhilesh Singh, Chester Rebeiro, Indian Institute of Technology, Madras, Madras, India; Sarani Bhattacharya, KU Leuven, Leuven, Belgium

DARPT: DEFENSE AGAINST REMOTE PHYSICAL ATTACK BASED ON TDC IN MULTI-TENANT SCENARIO

Fan Zhang, Zhiyong Wang, Haoting Shen, Bolin Yang, Qianmei Wu, Kui Ren, Zhejiang University, Hangzhou, China



【系统中未录入的补充会议报告情况】-共 3 项

1. 国际会议 HUWWEI STW 2021 特邀报告

Huawei STW 2021 邀请信

尊敬的张帆先生：

您好！

Huawei STW (Strategy and Technology Workshop)是华为公司与业界共同探讨未来行业发展及技术演进的研讨会，由华为公司技术战略规划部和与海外研究所承办，自 2008 年开始已成功举办 11 届。大会将汇集 200 多位行业顶级技术专家，深入探讨行业趋势和热点话题，携手推动行业技术创新及发展。

STW 2021 将于 2021 年 10 月 14-16 日在深圳华为总部召开。我们诚挚地邀请您作为行业的一名资深专家参加这次大会，并进行专题演讲，向行业专家分享您对硬件可信领域的看法。

衷心感谢您接受我们的邀请！



张帆

华为公司技术战略规划部部长

二〇二一年九月



2. 国内会议 CryptoIC2018 特邀报告

中国密码学会密码芯片专业委员会

Jun. 25th, 2018

Professor Fan Zhang
College of Information Science & Electronic Engineering,
Zhejiang University
Administration Building, #38 Zheda Road,
Hangzhou, 310027, P.R.China
Tel: +86571-87951720
Email: fanzhang@zju.edu.cn

Dear Professor Zhang,

We are pleased to invite you to CryptoIC 2018 to give an invited speech. The conference is scheduled to be held between August 24th and August 25th, 2018 in Beijing, China. CryptoIC is a conference held annually by Cryptographic IC Technical Committee, Chinese Association for Cryptologic Research, and attended by more than 200 researchers from academia, industry and research institutes.

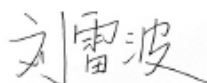
Your expertise and experience in this field of work will be an excellent addition to our program as many of our participants will look forward to hearing and learning from your work. It is an honor to be able to invite you to be an invited speaker as we are aware of your great contribution to this field of work.

CryptoIC 2018 will cover the flight ticket and accommodation expenses and the registration fees.

If you have any additional question or require assistance, please contact the CryptoIC Registration Office via email at wanghn@tsinghua.edu.cn.

We are looking forward to meeting you in Beijing, China!

Best regards,



Dr. Leibo Liu
TPC Chair of CryptoIC 2018
Secretary-General of Cryptographic IC Technical Committee,
Chinese Association for Cryptologic Research
Professor, Institute of Microelectronics, Tsinghua University,
Beijing, China
Tel: +8610-62799016
Fax: +8610-62781553
Email: liulb@tsinghua.edu.cn



3. 国内会议 CryptoIC2022 特邀报告



5. 海归人才回国工作证明材料（与用人单位签署的工作协议）；

目 录

（海归人才回国工作证明材料）

1.	2014 年浙大信电学院聘用合同	151
2.	2017 年浙大信电学院续签聘用合同书	153
3.	2018 年浙大信电学院续签聘用合同书	154
4.	2021 年浙大计算机学院网安中心续签聘用合同书	155



1. 2014 年浙大信电学院聘用合同

聘用合同

甲方(用人单位)名称: 信息与电子工程学院
法定代表人(或主要负责人): 章献民
注册地址: 浙大路38号
经营地址: _____

乙方(劳动者)姓名: 张帆
性别: 男 居民身份证号码: 33025197810201818
居住地联系电话: _____ 手机号码: 13967175420
现居住地址: 余杭塘栖镇西小河31幢 邮编: 311106
户口所在地: 上海奉贤区西渡路1500号 邮编: 200030

根据相关法律、法规的规定,经甲乙双方平等自愿、协商一致,共同签订并履行本合同所列条款。

第一条 聘用合同类型及期限

一、本聘用合同类型及期限按下列第 1 项确定。

1、固定期限:自 2014 年 2 月 16 日起至 2016 年 12 月 31 日止。

2、无固定期限:自 _____ 年 _____ 月 _____ 日起至法定的解除或终止合同的条件出现时止。

3、以完成一定工作任务为期限:自 _____ 年 _____ 月 _____ 日起至 _____ 终止。

二、本合同约定试用期的,试用期自 _____ 年 _____ 月 _____ 日起至 _____ 年 _____ 月 _____ 日止。

第二条 工作内容和工作地点

一、乙方同意根据甲方生产(工作)需要,从事 _____ 工作。甲乙双方

可另行约定岗位具体职责和要求。乙方应按照甲方制定的规章制度和岗位职责要求认真履行义务,按时、按质、按量完成工作任务。未经甲方书面许可不得同时与其他用人单位建立劳动聘用关系。

二、乙方的工作地点为: _____

一、甲、乙任何一方违反本合同，给对方造成损失的，应当赔偿损失。

二、乙方有下列情形之一的，应当依法赔偿甲方损失：

1、因乙方隐瞒与其他用人单位尚未解除或者终止劳动合同事实的，给其他用人单位造成损失的，最终导致甲方承担连带赔偿责任，乙方应当赔偿由此造成甲方的损失。

2、乙方违法解除聘用合同、违反保密义务或者竞业限制约定，给甲方造成损失的，应当承担赔偿责任。

3、乙方对外违法招用劳动者，导致甲方承担连带责任的，乙方应当赔偿由此造成甲方的损失。

4、因乙方过错致使本合同被确认无效，给甲方造成损害的，乙方应当承担赔偿责任。

5、由甲方出资对乙方进行的专业技术培训，乙方违反服务期约定的，应当向甲方支付违约金，违约金数额为服务期尚未履行部分所应分摊的培训费用。详见《专业技术培训协议》。

6、竞业限制违约及赔偿详见保密协议或竞业限制条款。

7、上述赔偿损失的范围包括但不限于：生产经营损失；招聘费用；培训费用；第三人索赔；行政处罚等。

8、如果乙方欠付甲方任何款项或应承担依照法律和本合同约定的赔偿责任，甲方有权在不违反法律法规的前提下，从乙方的奖金或工资中做相应的扣除，不足部分，甲方有权向乙方追偿。

第十二条 争议处理及其他

一、甲乙双方因履行聘用合同发生争议，应协商解决，协商不成或不愿协商的，可以向仲裁机构申请仲裁。对仲裁裁决不服的，可以向人民法院起诉。

二、乙方确认本合同所列现居住地址为本合同相关文件、文书的送达和通知地址。如该地址变更，乙方应及时书面告知甲方，甲方将需送达的资料邮寄至该地址，即视为送达。

三、本合同未尽事项，按国家有关法律法规执行。

四、本合同条款如与今后国家颁布的法律法规相抵触时，按国家新的法律法规执行。

五、本合同依法订立，双方签字盖章后生效，双方必须严格履行。

六、本合同一式 2 份，甲乙双方各执一份。

甲方（盖章）：

乙方（签字）：

甲方法定代表人（主要负责人）

或委托代理人（签字或盖章）

签约日期：

年 月 日

签约日期：

年 月 日

2. 2017 年浙大信电学院续签聘用合同书

续 签 聘 用 合 同 书

浙江大学（甲方）委托其下属 信息与电子工程学院 与乙方 张帆 同志（职工号：0013274）续签聘用合同，原聘用合同书内容不变。续聘合同期限从 2017 年 1 月 1 日至 2018 年 3 月 31 日。原聘用合同未尽事宜或变更原聘用合同内容，双方约定如下：

1、合同期内未晋升学校高级专业技术职务不再续聘。

2、

甲方委托单位公章：

甲方委托单位代表签字：

乙方（签字）：

校人事处合同验章：



2016 年 6 月 22 日

备注：本合同一式四份，学院（或单位）、人事处、本人各执一份，本人档案一份。

3. 2018 年浙大信电学院续签聘用合同书

续 签 聘 用 合 同 书

浙江大学（甲方）委托其下属 信息与电子工程学院（单位）与乙方 张帆 同志（职工号：0013274）续签聘用合同，原聘用合同书内容不变。续聘合同期限从 2018 年 4 月 1 日 至 2020 年 12 月 31 日。原聘用合同未尽事宜或变更原聘用合同内容，双方约定如下：

1、无

甲方委托单位公章：

甲方委托单位代表签字：

乙方（签字）：

校人事处合同验章：



2018 年 3 月 8 日



备注：本合同一式四份，学院（或单位）、人事处、本人各执一份，本人档案一份。



4. 2021 年浙大计算机学院网安中心续签聘用合同书

合同编号：2020-RSC-PYZXXP-0527

续 签 聘 用 合 同 书

浙江大学（甲方）委托其下属计算机科学与技术学院（单位）与乙方张帆同志（职工号：0013274）续签聘用合同，原聘用合同书内容不变。续聘合同期限从2021年01月01日至2025年12月31日。原聘用合同未尽事宜或变更原聘用合同内容，双方约定如下：

无

甲方委托单位公章：

甲方委托单位代表签字：

乙方（签字）：

校合同专用章：



备注：本合同一式四份，学院（或单位）、本人各执一份，本人档案一份，文书档案一份。



6. 外国人永久居留身份证（外籍专家必须上传）；

7. 电子版 2 寸近期免冠证件照片（蓝底、JPG 格式，按“姓名（单位）.JPG”规则命名，分辨率 413*626 以上，文件大小 2M 以下）。

